AD 2. AERODROMES

VIDP AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VIDP - INDIRA GANDHI INTERNATIONAL AIRPORT,DELHI

VIDP AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

1	Aerodrome reference point coordinates and its site	283407N 0770644E 117 DEG /321M from intersection RWY 09/27 and TWY D3	
2	Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves	BRG 228DEG /15Km from Delhi Railway Station.	
3	Aerodrome elevation and reference temperature	778 FT / 41.0 DEG C	
4	Magnetic variation, date of information and annual change	0.75 DEG E (2010) /0.033 DEG E	
5	Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available)	Chief Operations Officer, Delhi International Airport Ltd, New Udaan Bhawan, Opp. Terminal-3 Indira Gandhi International Airport, New Delhi – 110037.,	
		Telephone:	(91) 1147197886 (91) 1147197589
		Fax:	(91) 1147197181
		AFS:	VIDPDIAL
		Email:	Dial.CMS@gmrgroup.in Dial.AOCC@gmrgroup.in
6	Types of traffic permitted (IFR/VFR)	IFR/VFR	
7	Remarks	website: www.newdelhiairport.in	

VIDP AD 2.3 OPERATIONAL HOURS

1	Aerodrome Operator	MON - SAT: 0400 - 1230 UTC (0930 - 1800 IST) (Except 2nd, 4th and 5th SAT) 2nd, 4th and 5th SAT of a month & SUN: NIL
2	Custom and immigration	H24
3	Health and sanitation	H24
4	AIS briefing office	H24
5	ATS reporting office (ARO)	H24
6	MET Briefing office	H24
7	Air Traffic Service	H24
8	Fuelling	H24
9	Handling	H24
10	Security	H24
11	De-icing	NIL

12	Remarks	1. Non-Sked flights not permitted to operate at IGI Airport without a valid Delhi APCH ARR CLR NR (DACN) OR DELHI APCH DEP CLR NR (DDCN) as the case may be, issued by IGIA-AOCC. Non-sked FLT operators, shall apply for DACN/DDCN to IGIA-AOCC at email ids: flight.data@gmrgroup.in and dial.aocc@gmrgroup.in . The detailed procedure for obtaining CLR AVBL at http:// www.newdelhiairport.in/general-aviation.aspx . 2.All international Non-Sked flights shall obtain night parking approval from flight data unit, DIAL (flight.data@gmrgroup.in) at least 5 days in advance and shall arrange Tow Bar themselves. 3. Only VVIP aircraft (President, Vice President, Prime Minister and SPG Protectees), all state aircraft, defence aircraft and SKED civil flight operated by SKED airlines are permitted to operate on RWY 09/27. 4.All Flights departing from uncontrolled airfield within Delhi control zone shall take ATC clearance on telephone no.011-25653454 before departure. 5.The approved hourly RWY traffic handling capacity is as follows: The peak traffic handling capacity is 86 air traffic movement per hour. a. Max DEP. Rate = 51 PER HR b. Max ARR Rate = 42 PER HR

VIDP AD 2.4 HANDLING SERVICES AND FACILITIES

1	Cargo-handling facilities	3 Cargo terminals equipped with advanced storage stacker, material and pallet container handling systems, computerized cargo information, data and documentation systems.
2	Fuel and Oil types	JET A1 All types

3	Fuelling facilities and capacity	IOC: Terminal - 1 Refuellers Capacity:11KL,16KL and 27KL Discharge rate: 400-700 Litres/Min IOST: Terminal - 3 and Terminal- 2 Total No. of hydrant dispensers-28 Discharge rate: 3800 Litres/Min. BPCL: Terminal - 1 Refuellers Capacity: 36KL, 25KL and 11KL Total Capacity: 133 Kilolitres (6- Refuellers). Discharge rate: 1000 Litres/Min. BSSPL: Terminal - 3 and Terminal- 2 Total No. of dispensers-16. Discharge rate:3500 litres/Min HPCL: Terminal - 1 Refuellers Capacity: 16KL and 6KL Total Capacity: 54Kilotitres (4 Refuellers). Discharge rate: 1300 and 500 Litres/Min respectively.
4	De-icing facilities	NIL
5	Hangar space for visiting aircraft	NIL
6	Repair facilities for visiting aircraft	NIL
7	Remarks	Fixed ground power supply and pre-conditioned air are available at every contact stand at Apron-1 and Terminal-3.

VIDP AD 2.5 PASSENGER FACILITIES

1	Hotel(s) at or in the vicinity of aerodrome	93 Rooms hotel at Terminal-3 and in the city.
2	Restaurant(s) at or in the vicinity of aerodrome	At AD and in the city
3	Transportation possibilities	Buses,metro rail,taxis and car rental service from AD
4	Medical Facilities	First aid and ambulance at AD. Hospital in the city.
5	Bank and post office at or in the vicinity of aerodrome	Banks: At AD H-24 Post office: At AD H-24
6	Tourist office	At AD and in the city
7	Remarks	NIL

VIDP AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

1	Aerodrome category for fire fighting	Within ATS HR: CAT-10
2	Rescue equipment	Available as per category. Additional: 1) One portable lighting unit for night operations. 2) One Mobile Command Post (Bus) to deal with aircraft emergencies.

3	Capability for removal of disabled aircraft	1. Primary responsibility for removal of disabled aircraft rests with the concerned airline. 2. IGIA is equipped with disabled aircraft recovery kit for complete narrow body aircraft recovery and partial recovery of all wide body aircraft (except A 380). 3. The Disabled Aircraft Removal Plan (DARP) for IGIA includes other support to be provided for recovery of the aircraft 4. Contact Details of Aerodrome Coordinator for Disabled Aircraft Removal Operations: Aerodrome coordinator: Head ARFF Contact Details: 011-61235004 DRG Office HR APN Control: 011-61234750 (24x7) Dial.aproncontrol@gmrgroup.in Dial.AOCC@gmrgroup.in Dial.CMS@gmrgroup.in
4	Remarks	Nil

VIDP AD 2.7 SEASONAL AVAILABILITY CLEARING

1	Type(s) of clearing equipment	NIL
2	Clearance priorities	NIL
3	Remarks	NIL

VIDP AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

1	Designation, surface and strength of aprons	Refer Aircraft Parking/Docking Chart
2	Designation, width, surface and strength of taxiways	Refer VIDP AD 2.23
3	Location and elevation of altimeter checkpoints	Location: All Aprons Elevation: Apron I: 751 FT General Aviation Apron: 745 FT Apron II: 732 FT Cargo Apron: 737 FT Apron 31: 723 FT Apron 32: 723 FT Apron 33: 726 FT Apron 34: 729 FT Apron 35: 737 FT
4	Location of VOR checkpoints	ON TWYs: TWYs H6 and K6 for RWY 10; TWYs D1, D2 and D3 for RWY 27; TWYs D4, D7, D8 and D9 for RWY 09 TWY K, J5 and G5 for RWY 28 TWYs R7 and R6 for RWY 11L; TWY P7 and P8 for RWY 29R
5	Position of INS checkpoints	
6	Remarks	

VIDP AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

1	Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands	1. Continuous aircraft stand lead-in lines available at MARS (Multiple Aircraft Ramp System) Centre stands for use of wide body aircraft and in single aircraft stands. Broken aircraft stand lead-in lines available at left & right MARS stand for narrow-bodied aircraft. 2.Stand identification signs are provided at all contact stands in Terminal-3 Aprons,Apron 1 and Terminal-2 Apron. On remote stands pole mounted stand identification sign has been provided in Apron I, Terminal- 2 Apron (except Stands 203 to 207), Cargo Apron & Terminal- 3 Apron. 3. Nose-wheel stop lines available at stands. Taxing guidance signs are provided on all TWY intersections, straight section and holding positions as per DGCA CAR. 4. Advanced Visual Docking Guidance System(AVDGS) are provided: i) At T3: A06, A08, A10, A12, A14, A13, A11, A11L, A11R, A09, A09L, A09R, A07, A07L, A07R, A03, A01, B15, B15L, B15R, B17, B17L,B17R, B19, B19L, B19R, B21, B18, B20,B20L, B20R, B22, B22L, B22R, B24, B24L,B24R, B26, B26L, B26R, C28L, C28R,C30L, C30R, C32L, C32R, C34L, C34R, C36, C33, C31L, C31R, C29L, C29R, C27L,C27R, D37, D37L, D37R, D39, D39L, D39R,D41, D43, D45, D47, D49, D51, D53, D55,D46, D48, D50, D52, D54, D56, D58, D60,D62, R02 & R03. ii) At T1: On all Code C Stands. 5. Docking of aircraft is done with the help of marshaller at remote parking stands and other contact stands where AVDGS is not provided or has become unserviceable.
2	Runway and taxiway markings and lights	RWY: 1. Markings: Designation, THR, TDZ, Centreline, Aiming point,Runway side stripe. 2. Lights: THR, TDZ (RWY 11L, RWY 29R, RWY 11R, RWY 29L,RWY 10, RWY 28), Centreline, Edge, End, Wing Bars (RWY 11L, RWY 29R, RWY 11R, RWY29L, RWY 28, RWY 27), RETILs (TWY Y2, TWY Y4, TWY Y1, TWY Z2, TWY Z3, TWY H1, TWY K1, TWY K2, TWY G1, TWY G2, TWY J1,TWY D5. TWY D6, TWY R2, TWY R3, TWY P1, TWY P3), Stop way (RWY 11R, RWY 11L), Approach Lights. TWY: 1. Marking: Centreline,Side stripe, Intermediate Holding Position, Runway holding position, Enhanced taxiway centreline marking 2. Lights: Centre line lights, Intermediate Holding Position lights, Runway crossing lights, taxiway edge lights are on curves only. 3. Portion of TWY LINK 32 south of TWY C, Portion of TWY LINK 30 south of TWY M, TWY LINK 16 (Between TWY C up to GA Apron), TWY LINK 13 and Portion of Taxilane F6 inside apron are having reflective edge marker.

3	Stop bars (if any)	At RHPs: TWY A2, D1,D2, D3, D4,D7, D8, D9,G3, G4, G5, H2, H3, H6, J7, J5,J6, K2, K3,K5, K6, Y8, Y7, Y6, Y5, Z4, Z5, Z6, Z7, P8, P7, P6, P5, P4, P2, R1, R4, R5, R6, R7, T7, T6, T5, T4, S1, S2, S5, S6, S7 and S8. At TWYs Junctions: F5 (south of D/F5 intersection), F5 (north of G/F5 intersection), F4 (south of D/F4 intersection), F4 (north of G/F4 intersection), F3 (south of D/F3 intersection), F3 (north of G/F3 intersection), A (east of LINK 5/A intersection), R (west of A/R intersection), C (east of LINK 5/C intersection), C(south of N/C intersection), A (south of N/ A intersection), A (north of A/ LINK 4 intersection), C (North of P6/ P9 intersection), A (North of A/P9 intersection), C (South of M/C intersection), A (South of M/A intersection).

4	Remarks	1. RWY guard lights are provided at each RWY and TWY intersection. 2. An illuminated wind direction indicator in the form of truncated cone made of fabric of red and white colour in bands have been installed at under mentioned 13 locations: Location — Coordinates i. NORTH OF RWY 09, ABEAM RWY 09 PAPI — 283417.1N 0770532.9E ii. NORTH OF RWY 27 AND WEST OF TWY A2 — 283415.6N 0770628.2E iii. NORTH OF RWY 10 AND EAST OF TWY H3 — 283401.7N 0770529.8E iv. SOUTH OF RWY 28 AND EAST OF TWY J5 — 283329.3N 0770706.4E v. NORTH OF RWY 11R AND EAST OF TWY Z6 — 283252.8N 0770400.6E vi. NORTH OF RWY 11R AND EAST OF TWY Z3 — 283248.6N 0770420.7E vii NORTH OF RWY 29L AND WEST OF TWY Y2 — 283232.1N 0770537.9E viii. SOUTH OF RWY 29L, ABEAM TWY Y6 — 283212.5N 0770629.5E ix. NORTH OF RWY 11L AND EAST OF TWY R6 — 283303.7N 0770414.5E x. NORTH OF RWY 11L AND EAST OF TWY R4 — 283258.5N0770437.1E xi. NORTH OF RWY 11L AND EAST OF TWY R1 — 283245.5N0770532.2E xii. NORTH OF RWY 29R AND WEST OF TWY P4 — 283237.1N0770611.4E xiii. NORTH OF RWY 29R AND WEST OF TWY P6 — 283233.2N0770629.8E 3. No entry BAR lights provided on the following TWYs: a.RWY 10/28 : TWY J3, TWY J2, TWY J1, TWY K1, TWY H1, TWY G1, TWY G2 b.RWY11R/29L:TWYY1,TWY Y2,TWY Y4,TWY Z2,TWY Z3 c. RWY 09/27: TWY D5, TWY D6 d. RWY 11L/29R: TWY P1, TWY P3, TWY R2 and RWY R3 4. Runway crossing lights available between TWY K2 and H2, K3 and H3, K6 and H6, J5 and G5, D3 and A2, P8 and S8, P7 and S7, P6 and S6, P5 and S5, P2 and S2, R1 and S1, R4 and T4, R5 and T5, R6 and T6, R7 and T7 and vice versa.

VIDP AD 2.10 AERODROME OBSTACLES

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
27/TKOF 09/APCH	TANK	283417.5N 0770447.8E	773 FT	NO	SINTEX TANK ON BUILDING
27/TKOF 09/APCH	BUILDING	283417.8N 0770449.3E	772 FT	NO	BUILDING TOP
27/TKOF 09/APCH	TANK	283419.6N 0770450.1E	776 FT	NO	SINTEX TANK ON BUILDING
27/TKOF 09/APCH	TANK	283419.7N 0770451.4E	774 FT	NO	SINTEX TANK ON BUILDING
27/TKOF 09/APCH	BUILDING	283417.7N 0770448.4E	777 FT	NO	BUILDING STAIRCASE ROOM TOP, BUILDING TOP - 234.3 M
27/TKOF 09/APCH	BUILDING	283418.8N 0770449.5E	772 FT	NO	BUILDING STAIRCASE ROOM TOP - 235.3 M, BUILDING TOP - 233.8 M
27/TKOF 09/APCH	TANK	283419.1N 0770445.7E	776 FT	NO	SINTEX TANK ON BUILDING - 236.60 M, STAIRCASE TOP - 235.36 M
28/TKOF 10/APCH	BUILDING	283413.8N 0770444.2E	765 FT	NO	BUILDING MUMTI TOP
28/TKOF 10/APCH	TANK	283414.0N 0770445.0E	762 FT	NO	SINTEX TANK ON BUILDING
28/TKOF 10/APCH	TOWER	283419.5N 0770430.2E	792 FT	NO	MOBILE TOWER ON BUILDING
28/TKOF 10/APCH	BUILDING	283416.3N 0770439.4E	774 FT	NO	BUILDING STRAIRCASE ROOM TOP, BUILDING TOP - 234.7 M
28/TKOF 10/APCH	BUILDING	283413.2N 0770444.2E	761 FT	NO	BUILDING TOP
28/TKOF 10/APCH	OTHER	283414.2N 0770441.5E	768 FT	NO	BUILDING, GRILL TOP - 234.15 M
28/TKOF 10/APCH	TANK	283413.1N 0770442.0E	767 FT	NO	SINTEX TANK ON BUILDING - 233.81 M, STAIRCASE TOP - 232.69 M
28/TKOF 10/APCH	TANK	283413.3N 0770441.5E	768 FT	NO	SINTEX TANK ON BUILDING - 234.00 M
28/TKOF 10/APCH	BUILDING	283413.3N 0770441.9E	765 FT	NO	BUILDING TOP
28/TKOF 10/APCH	BUILDING	283413.8N 0770442.4E	769 FT	NO	BUILDING STAIRCASE ROOM TOP - 234.43 M, BUILDING TOP - 232.01 M

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
28/APCH 10/TKOF	TREE	283318.7N 0770756.1E	846 FT	NO	TREE
28/APCH 10/TKOF	TREE	283315.5N 0770752.9E	845 FT	NO	GROUP OF TREES
28/APCH 10/TKOF	TREE	283316.7N 0770754.0E	842 FT	NO	GROUP OF TREES
28/APCH 10/TKOF	TREE	283322.5N 0770743.1E	824 FT	NO	TREE
28/APCH 10/TKOF	OTHER	283326.0N 0770732.5E	804 FT	NO	L/C ROD ON LOCALIZER HUT (HUT TOP 242.1 M)
28/APCH 10/TKOF	OTHER	283333.0N 0770726.7E	784 FT	NO	ELECTRICAL JUNCTION BOX
28/APCH 10/TKOF	OTHER	283332.7N 0770730.3E	790 FT	NO	ELECTRICAL JUNCTION BOX
28/APCH 10/TKOF	ANTENNA	283328.1N 0770732.7E	800 FT	NO	ANTENNA NEAR LLZ
27/APCH 09/TKOF	NAVAID	283411.5N 0770706.4E	762 FT	LGTD	LOCALIZER MAIN ANTENNA
27/APCH 09/TKOF	BUILDING	283419.8N 0770757.9E	857 FT	NO	BUILDING TOP (DEFENCE ESTATES OFFICERS)
27/APCH 09/TKOF	TREE	283409.6N 0770727.7E	803 FT	NO	TREE
27/APCH 09/TKOF	TREE	283410.5N 0770725.3E	800 FT	NO	TREE
27/APCH 09/TKOF	OTHER	283407.9N 0770712.0E	777 FT	NO	WATCH TOWER
27/APCH 09/TKOF	POLE	283409.2N 0770727.1E	799 FT	NO	LIGHT POLE
27/APCH 09/TKOF	OTHER	283413.1N 0770708.5E	764 FT	NO	WATCH TOWER
27/APCH 09/TKOF	OTHER	283413.1N 0770706.9E	772 FT	NO	MOBILE ROAD TRAFFIC (ROAD ELEV. 230.2 M. + 5.0 M. TRAFFIC HT.)
27/APCH 09/TKOF	OTHER	283402.8N 0770756.7E	855 FT	NO	METRO ELECTRIC TRACTION OVERHEAD FRAME
27/APCH 09/TKOF	OTHER	283401.2N 0770756.4E	854 FT	NO	METRO ELECTRIC TRACTION OVERHEAD FRAME
27/APCH 09/TKOF	OTHER	283405.3N 0770757.5E	855 FT	NO	METRO ELECTRIC TRACTION OVERHEAD FRAME

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
27/APCH 09/TKOF	OTHER	283403.7N 0770756.9E	856 FT	NO	METRO ELECTRIC TRACTION OVERHEAD FRAME
27/APCH 09/TKOF	OTHER	283411.6N 0770703.2E	759 FT	NO	APPROACH LIGHT
27/APCH 09/TKOF	OTHER	283411.6N 0770704.3E	760 FT	NO	APPROACH LIGHT
27/APCH 09/TKOF	OTHER	283411.5N 0770705.4E	762 FT	NO	APPROACH LIGHT
27/APCH 09/TKOF	OTHER	283411.5N 0770707.7E	766 FT	NO	APPROACH LIGHT
27/APCH 09/TKOF	OTHER	283412.8N 0770709.0E	773 FT	NO	MOBILE ROAD TRAFFIC (ROAD ELEV.230.5 M. + 5.0 M. TRAFFIC HT.)
27/APCH 09/TKOF	OTHER	283413.0N 0770710.4E	774 FT	NO	MOBILE ROAD TRAFFIC (ROAD ELEV.230.9 M. + 5.0 M. TRAFFIC HT.)
27/APCH 09/TKOF	OTHER	283402.0N 0770756.6E	855 FT	NO	METRO ELECTRIC TRACTION OVERHEAD FRAME
27/APCH 09/TKOF	OTHER	283412.0N 0770706.4E	763 FT	NO	LIGHTNING ARRESTOR ON LOCALIZER
27/APCH 09/TKOF	OTHER	283408.8N 0770706.9E	763 FT	NO	LOCALIZER HUT
27/APCH 09/TKOF	OTHER	283411.5N 0770709.9E	768 FT	NO	APPROACH LIGHT
27/APCH 09/TKOF	OTHER	283411.4N 0770712.1E	771 FT	NO	APPROACH LIGHT
29R/APCH 11L/TKOF	OTHER	283233.2N 0770629.9E	800 FT	LGTD	WDI (LGT)
29L/TKOF 11R/APCH	BUILDING	283320.0N 0770244.2E	857 FT	NO	BUILDING TOP
29L/TKOF 11R/APCH	TREE	283253.7N 0770341.2E	749 FT	NO	TREE
29L/TKOF 11R/APCH	POLE	283247.6N 0770350.7E	731 FT	NO	LIGHT POLE
29L/TKOF 11R/APCH	POLE	283246.8N 0770351.3E	731 FT	NO	LIGHT POLE
29L/TKOF 11R/APCH	POLE	283248.1N 0770349.8E	729 FT	NO	LIGHT POLE
29L/TKOF 11R/APCH	POLE	283257.7N 0770343.8E	743 FT	NO	RAILWAY ELECTRIC POLE
29L/TKOF 11R/APCH	POLE	283258.0N 0770345.3E	740 FT	NO	LIGHT POLE

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
29L/TKOF 11R/APCH	FENCE	283248.1N 0770349.7E	730 FT	NO	AIRPORT BOUNDARY WALL WITH FENCING ON TOP
29L/TKOF 11R/APCH	OTHER	283256.5N 0770349.8E	735 FT	NO	MOBILE ROAD TRAFFIC (ROAD ELEV. 218.9 M. + 5.0 M. TRAFFIC HT.)
29L/TKOF 11R/APCH	OTHER	283247.9N 0770350.6E	735 FT	NO	MOBILE ROAD TRAFFIC (ROAD ELEV. 219.1 M. + 5.0 M.. TRAFFIC HT.)
29L/TKOF 11R/APCH	OTHER	283250.4N 0770342.8E	741 FT	LGTD	L/C ROD ON LOCALIZER HUT (LGT TOP 224.5 M)
29L/TKOF 11R/APCH	OTHER	283256.6N 0770342.3E	745 FT	NO	MOBILE RAIL TRAFFIC (RAILWAY TRACK ELEV. 219.5 M. + 7.6 M. TRAFFIC HT.)
29L/TKOF 11R/APCH	OTHER	283258.6N 0770344.0E	745 FT	NO	MOBILE RAIL TRAFFIC (RAILWAY TRACK ELEV. 219.4 M. + 7.6 M. TRAFFIC HT.)
29L/TKOF 11R/APCH	OTHER	283254.8N 0770349.1E	739 FT	NO	SECURITY HUT
29L/TKOF 11R/APCH	POLE	283253.8N 0770339.8E	747 FT	NO	RAILWAY ELECTRIC POLE (TF2232)
29L/TKOF 11R/APCH	POLE	283256.3N 0770342.6E	746 FT	NO	RAILWAY ELECTRIC POLE NEW (WITHOUT WIRE)
29L/TKOF 11R/APCH	POLE	283255.0N 0770341.6E	747 FT	NO	RAILWAY ELECTRIC POLE NEW (WITHOUT WIRE)
29L/TKOF 11R/APCH	POLE	283254.8N 0770341.8E	746 FT	NO	RAILWAY ELECTRIC POLE NEW (WITHOUT WIRE)
29L/TKOF 11R/APCH	POLE	283256.7N 0770342.8E	747 FT	NO	RAILWAY ELECTRIC POLE (TF2229 POLE)
29L/APCH 11R/TKOF	BUILDING	283213.9N 0770655.7E	816 FT	NO	BUILDING TOP (CHICCO)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
29L/APCH 11R/TKOF	OTHER	283211.5N 0770652.7E	825 FT	NO	BUILDING, PIPE TOP - 251.4 M, SINTEX TANK TOP - 251.1 M, STAIRCASE ROOM TOP - 249.9 M (PETER ENGLAND)
29L/APCH 11R/TKOF	BUILDING	283205.5N 0770651.8E	831 FT	NO	Building Top
29L/APCH 11R/TKOF	BUILDING	283206.1N 0770650.0E	820 FT	NO	Top
29L/APCH 11R/TKOF	TANK	283139.6N 0770806.1E	954 FT	NO	SINTEX TANK ON BUILDING
29L/APCH 11R/TKOF	TANK	283206.6N 0770717.3E	876 FT	NO	SINTEX TANK ON BUILDING
29L/APCH 11R/TKOF	BUILDING	283207.0N 0770653.6E	821 FT	NO	BUILDING TOP
29L/APCH 11R/TKOF	TOWER	283207.5N 0770711.8E	844 FT	NO	MOBILE TOWER ON BUILDING
29L/APCH 11R/TKOF	OTHER	283207.6N 0770703.0E	833 FT	NO	FACTORY SHED
29L/APCH 11R/TKOF	BUILDING	283208.0N 0770707.7E	835 FT	NO	BUILDING TOP
29L/APCH 11R/TKOF	TANK	283210.0N 0770701.2E	836 FT	NO	SINTEX TANK ON BUILDING
29L/APCH 11R/TKOF	BUILDING	283210.9N 0770650.5E	808 FT	NO	BUILDING TOP
29L/APCH 11R/TKOF	OTHER	283211.1N 0770652.4E	826 FT	NO	FLAG ON BUILDING
29L/APCH 11R/TKOF	TANK	283212.1N 0770658.9E	836 FT	NO	SINTEX TANK ON BUILDING
29L/APCH 11R/TKOF	TOWER	283212.1N 0770722.0E	860 FT	NO	MOBILE TOWER ON BUILDING
29L/APCH 11R/TKOF	BUILDING	283212.3N 0770700.1E	822 FT	NO	BUILDING TOP
29L/APCH 11R/TKOF	OTHER	283215.4N 0770658.7E	820 FT	NO	FACTORY SHED
29L/APCH 11R/TKOF	TANK	283216.0N 0770701.6E	828 FT	NO	SINTEX TANK ON BUILDING
29L/APCH 11R/TKOF	TANK	283215.6N 0770700.4E	834 FT	NO	SINTEX TANK ON BUILDING, BUILDING TOP - 251.2 M
29L/APCH 11R/TKOF	BUILDING	283213.3N 0770656.5E	832 FT	NO	BUILDING TOP (339)
29L/APCH 11R/TKOF	BUILDING	283213.4N 0770659.0E	836 FT	NO	BUILDING STAIRCASE ROOM TOP, BUILDING TOP - 253.5 M
29L/APCH 11R/TKOF	BUILDING	283213.5N 0770701.2E	833 FT	NO	BUILDING TOP

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
29L/APCH 11R/TKOF	TANK	283213.0N 0770705.1E	835 FT	NO	SINTEX TANK ON BUILDING, STAIRCASE ROOM TOP - 253.1 M, BUILDING TOP - 250.4 M.
29L/APCH 11R/TKOF	BUILDING	283212.0N 0770706.5E	842 FT	NO	BUILDING STAIRCASE ROOM TOP, BUILDING TOP - 253.2 M
29L/APCH 11R/TKOF	BUILDING	283216.7N 0770700.8E	846 FT	NO	BUILDING STAIRCASE ROOM TOP - 257.1 M, SINTEX TANK TOP - 255.5 M, BUILDING TOP - 254.7 M (FEDERATION OF SEED INDUSTRY OF INDIA)
29L/APCH 11R/TKOF	POLE	283217.3N 0770700.8E	847 FT	NO	BUILDING, POLE TOP - 258.34 M,SINTEX TOP - 255.30 M, BUILDING TOP - 254.29M (A-99)
29L/APCH 11R/TKOF	BUILDING	283216.9N 0770659.5E	848 FT	NO	BUILDING STAIRCASE ROOM TOP - 258.44 M, BUILDING TOP - 256.26 M (SHRAY LOGISTICS)
29L/APCH 11R/TKOF	BUILDING	283216.7N 0770659.0E	831 FT	NO	BUILDING TOP (LEVIS)
29L/APCH 11R/TKOF	BUILDING	283213.7N 0770656.3E	834 FT	NO	BUILDING STAIRCASE ROOM TOP, BUILDING TOP - 252.29 M (338/2)
29L/APCH 11R/TKOF	BUILDING	283211.1N 0770650.9E	815 FT	NO	BUILDING TOP (INDIAN OIL)
29L/APCH 11R/TKOF	BUILDING	283211.4N 0770651.7E	824 FT	NO	BUILDING TOP (EKKA DECOR PVT LTD)
29L/APCH 11R/TKOF	ANTENNA	283209.6N 0770647.9E	824 FT	NO	ANTENNA ON BUILDING - 251.06 M, BUILDING TOP - 250.37 M (DEE MARKS HOTEL OF RESORTS OFFICE)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
29L/APCH 11R/TKOF	BUILDING	283215.1N 0770658.1E	826 FT	NO	BUILDING STAIRCASE ROOM TOP - 251.75 M , BUILDING TOP - 250.18 M (BENTTON)
29L/APCH 11R/TKOF	BUILDING	283214.6N 0770657.2E	824 FT	NO	BUILDING TOP (340)
29L/APCH 11R/TKOF	BUILDING	283213.0N 0770708.6E	847 FT	NO	BUILDING TOP (34)
29L/APCH 11R/TKOF	BUILDING	283215.2N 0770716.7E	861 FT	NO	BUILDING TOP
29L/APCH 11R/TKOF	BUILDING	283214.2N 0770716.3E	861 FT	NO	BUILDING STAIRCASE ROOM TOP - 262.29 M, SINTEX TANK TOP - 259.75 M, BUILDING TOP - 259.50 M (LAL)
29L/APCH 11R/TKOF	BUILDING	283209.6N 0770713.9E	850 FT	NO	BUILDING STAIRCASE ROOM TOP - 258.99 M, BUILDING TOP - 257.53 M
29L/APCH 11R/TKOF	OTHER	283209.4N 0770713.0E	847 FT	NO	BUILDING, SHED TOP - 258.25 M, BUILDING TOP - 257.00 M
29L/APCH 11R/TKOF	BUILDING	283210.2N 0770711.9E	850 FT	NO	BUILDING TOP
29L/APCH 11R/TKOF	OTHER	283210.6N 0770712.0E	857 FT	NO	BUILDING, FLAG TOP - 261.22 M, BUILDING TOP - 259.43 M (OMKARA (A-9))
29L/APCH 11R/TKOF	TANK	283211.1N 0770710.7E	850 FT	NO	SINTEX TANK ON BUILDING - 259.21 M, BUILDING TOP - 258.69 M
29L/APCH 11R/TKOF	OTHER	283212.1N 0770710.7E	849 FT	NO	BUILDING, GRILL TOP - 258.7M, BUILDING TOP - 258.56 M
29L/APCH 11R/TKOF	TANK	283212.5N 0770708.7E	846 FT	NO	BUILDING SINTEX TANK TOP - 257.85 M, BUILDING TOP - 256.26 M (ROTI FACTORY)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
29L/APCH 11R/TKOF	BUILDING	283215.0N 0770700.8E	844 FT	NO	BUILDING STAIRCASE ROOM TOP - 257.29 M, BUILDING TOP - 255.61 M
29L/APCH 11R/TKOF	BUILDING	283213.3N 0770706.0E	845 FT	NO	BUILDING TOP (SS CARGO LOGISTICS PVT. LTD)
29L/APCH 11R/TKOF	TANK	283216.4N 0770718.1E	859 FT	NO	SINTEX TANK ON BUILDING- 261.85 M, STAIRCASE ROOM - 261.36 M, BUILDING TOP - 259.19 M
29L/APCH 11R/TKOF	BUILDING	283214.0N 0770717.7E	854 FT	NO	BUILDING TOP
29L/APCH 11R/TKOF	BUILDING	283214.4N 0770700.3E	838 FT	NO	BUILDING STAIRCASE ROOM TOP - 255.52 M, BUILDING TOP - 253.64 M
29L/APCH 11R/TKOF	BUILDING	283214.6N 0770702.3E	851 FT	NO	BUILDING STAIRCASE ROOM TOP - 259.48 M, BUILDING TOP - 257.86 M
29L/APCH 11R/TKOF	ANTENNA	283214.4N 0770703.1E	851 FT	NO	ANTENNA ON BUILDING - 259.43 M, SINTEX TANK TOP - 254.62 M, BUILDING TOP - 253.35 M.
29L/APCH 11R/TKOF	BUILDING	283216.5N 0770704.0E	835 FT	NO	BUILDING STAIRCASE ROOM TOP - 254.37 M, BUILDING TOP - 252.22 M
29L/APCH 11R/TKOF	ANTENNA	283217.5N 0770700.6E	844 FT	NO	ANTENNA ON BUILDING, GRILL TOP - 256.92 M, STAIRCASE ROOM TOP - 256.92 M, BUILDING TOP - 254.78 M.
29L/APCH 11R/TKOF	ANTENNA	283217.3N 0770702.0E	847 FT	NO	ANTENNA ON BUILDING - 258.08 M, SINTEX TANK TOP - 255.37 M, BUILDING TOP - 254.92 M.

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
29L/APCH 11R/TKOF	OTHER	283217.2N 0770708.3E	855 FT	NO	BUILDING, PIPE TOP - 260.61 M, STAIRCASE ROOM TOP - 259.87 M, BUILDING TOP - 257.26 M.
29L/APCH 11R/TKOF	TANK	283213.8N 0770709.9E	851 FT	NO	SINTEX TANK ON BUILDING - 259.37 M, BUILDING TOP - 258.32 M
29L/APCH 11R/TKOF	TANK	283214.0N 0770710.2E	851 FT	NO	SINTEX TANK ON BUILDING - 259.50 M, STAIRCASE ROOM TOP - 258.55 M, BUILDING TOP - 256.23 M
29L/APCH 11R/TKOF	OTHER	283212.8N 0770710.6E	860 FT	NO	BUILDING, PIPE TOP - 262.13 M, PIPE TOP - 260.70 M, SINTEX TANK TOP - 260.38 M, STAIRCASE ROOM TOP - 259.15 M, BUILDING TOP - 257.35 M (BLUE BIRDS)
29L/APCH 11R/TKOF	BUILDING	283210.9N 0770709.8E	855 FT	NO	BUILDING STAIRCASE ROOM TOP - 260.47 M, PIPE TOP - 259.83 M, BUILDING TOP - 258.85 M
29L/APCH 11R/TKOF	BUILDING	283211.3N 0770712.0E	845 FT	NO	BUILDING STAIRCASE ROOM TOP - 257.71 M, BUILDING - 256.06 M
29L/APCH 11R/TKOF	BUILDING	283210.6N 0770712.4E	851 FT	NO	BUILDING STAIRCASE ROOM TOP - 259.36 M, BUILDING TOP - 257.40 M
29L/APCH 11R/TKOF	BUILDING	283211.5N 0770716.4E	849 FT	NO	BUILDING STAIRCASE ROOM TOP - 258.77 M, BUILDING TOP - 257.64 M

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
29L/APCH 11R/TKOF	OTHER	283213.0N 0770717.5E	854 FT	NO	BUILDING, PIPE TOP - 260.19 M, SINTEX TANK TOP - 259.91 M, BUILDING TOP - 258.66 M.
29L/APCH 11R/TKOF	ANTENNA	283214.2N 0770715.5E	868 FT	NO	BUILDING DISH ANTENNA TOP - 264.48 M, STAIRCASE ROOM TOP - 263.52 M, BUILDING TOP - 262.42 M.
29L/APCH 11R/TKOF	BUILDING	283214.3N 0770715.4E	864 FT	NO	BUILDING STAIRCASE ROOM TOP - 263.45 M, BUILDING TOP - 262.39 M
29L/APCH 11R/TKOF	BUILDING	283215.2N 0770715.1E	845 FT	NO	BUILDING STAIRCASE ROOM TOP - 257.64 M, BUILDING TOP - 257.16 M
29L/APCH 11R/TKOF	TANK	283215.3N 0770716.2E	861 FT	NO	SINTEX TANK ON BUILDING - 262.32 M, STAIRCASE ROOM TOP - 261.18 M, BUILDING TOP - 259.27 M.
29L/APCH 11R/TKOF	TANK	283216.5N 0770718.3E	860 FT	NO	SINTEX TANK ON BUILDING- 262.27 M, DISH ANTENNA TOP - 262.23 M, STAIRCASE ROOM TOP - 261.63 M, BUILDING TOP - 259.47 M.
29L/APCH 11R/TKOF	TANK	283213.5N 0770714.4E	861 FT	NO	SINTEX TANK ON BUILDING - 262.48 M, STAIRCASE ROOM TOP - 261.76 M, BUILDING TOP - 259.53 M
29L/APCH 11R/TKOF	OTHER	283211.4N 0770706.3E	857 FT	NO	BUILDING, PIPE TOP - 261.18 M, SINTEX TANK TOP - 260.90 M, STAIRCASE ROOM TOP - 259.62 M, BUILDING TOP - 258.10 M

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
29L/APCH 11R/TKOF	TANK	283209.7N 0770713.0E	851 FT	NO	SINTEX TANK ON BUILDING - 259.26 M, STAIRCASE ROOM TOP - 258.73 M, BUILDING TOP - 256.98 M
29L/APCH 11R/TKOF	TREE	283212.8N 0770708.5E	849 FT	NO	GROUP OF TREES
29L/APCH 11R/TKOF	TREE	283208.4N 0770645.2E	829 FT	NO	TREE
29L/APCH 11R/TKOF	TREE	283208.5N 0770646.0E	844 FT	NO	TREE
29L/APCH 11R/TKOF	TREE	283211.1N 0770654.4E	833 FT	NO	GROUP OF TREES
29L/APCH 11R/TKOF	TREE	283207.9N 0770650.7E	826 FT	NO	TREE
29L/APCH 11R/TKOF	TREE	283209.7N 0770652.1E	828 FT	NO	TREE
29L/APCH 11R/TKOF	TREE	283209.7N 0770650.3E	829 FT	NO	GROUP OF TREES
29L/APCH 11R/TKOF	TREE	283209.8N 0770652.9E	831 FT	NO	TREE
29L/APCH 11R/TKOF	TREE	283210.4N 0770654.4E	838 FT	NO	GROUP OF TREES
29L/APCH 11R/TKOF	TREE	283210.4N 0770655.8E	837 FT	NO	GROUP OF TREES
29L/APCH 11R/TKOF	TREE	283210.8N 0770653.1E	819 FT	NO	TREE
29L/APCH 11R/TKOF	TREE	283211.2N 0770655.6E	840 FT	NO	GROUP OF TREES
29L/APCH 11R/TKOF	TREE	283212.1N 0770654.3E	822 FT	NO	TREE
29L/APCH 11R/TKOF	TREE	283212.3N 0770650.6E	812 FT	NO	GROUP OF TREES
29L/APCH 11R/TKOF	TREE	283212.9N 0770654.3E	823 FT	NO	TREE
29L/APCH 11R/TKOF	TREE	283213.5N 0770657.2E	837 FT	NO	TREE
29L/APCH 11R/TKOF	TREE	283213.5N 0770652.9E	813 FT	NO	GROUP OF TREES
29L/APCH 11R/TKOF	TREE	283215.6N 0770649.4E	806 FT	NO	GROUP OF TREES
29L/APCH 11R/TKOF	TREE	283216.5N 0770655.1E	816 FT	NO	GROUP OF TREES
29L/APCH 11R/TKOF	TREE	283214.2N 0770659.0E	828 FT	NO	GROUP OF TREES
29L/APCH 11R/TKOF	POLE	283218.2N 0770657.0E	815 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283217.1N 0770656.1E	814 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283216.6N 0770655.6E	814 FT	NO	LIGHT POLE

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
29L/APCH 11R/TKOF	POLE	283216.2N 0770655.1E	814 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283215.7N 0770654.5E	813 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283215.4N 0770653.9E	814 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283215.0N 0770653.3E	814 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283214.6N 0770652.7E	815 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283214.3N 0770652.1E	815 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283214.0N 0770651.5E	815 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283213.6N 0770650.8E	815 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283213.3N 0770650.2E	815 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283213.0N 0770649.6E	815 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283212.6N 0770648.9E	814 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283213.7N 0770652.0E	808 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283213.2N 0770650.9E	808 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283212.6N 0770649.8E	808 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283211.0N 0770646.6E	807 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283211.7N 0770647.0E	814 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283212.0N 0770647.7E	813 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283212.3N 0770648.3E	814 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283208.8N 0770642.1E	811 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283209.1N 0770642.8E	812 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283209.5N 0770643.3E	812 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283209.8N 0770644.0E	813 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283210.1N 0770644.6E	812 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283210.4N 0770645.3E	812 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283210.8N 0770645.9E	812 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283211.1N 0770646.6E	813 FT	NO	LIGHT POLE

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
29L/APCH 11R/TKOF	POLE	283209.3N 0770643.3E	806 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283209.8N 0770644.3E	806 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283210.3N 0770645.3E	806 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283210.8N 0770646.3E	807 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283211.8N 0770648.2E	807 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283211.9N 0770649.7E	809 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283211.9N 0770651.0E	808 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283211.3N 0770649.9E	807 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283210.8N 0770648.9E	807 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283210.5N 0770649.2E	808 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283210.3N 0770648.8E	807 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283210.8N 0770647.6E	808 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283209.7N 0770647.5E	807 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283210.1N 0770646.2E	808 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283209.2N 0770645.9E	806 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283217.6N 0770656.6E	814 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283211.2N 0770647.1E	807 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283217.9N 0770656.7E	813 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283216.8N 0770655.9E	815 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	TOWER	283208.2N 0770755.3E	925 FT	NO	MOBILE TOWER
29L/APCH 11R/TKOF	POLE	283208.1N 0770643.9E	807 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283208.5N 0770643.2E	808 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283208.7N 0770644.9E	806 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283209.3N 0770644.7E	809 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283209.3N 0770646.8E	808 FT	NO	ELECTRIC POLE
29L/APCH 11R/TKOF	POLE	283209.8N 0770645.7E	809 FT	NO	LIGHT POLE

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
29L/APCH 11R/TKOF	POLE	283209.9N 0770647.3E	808 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283211.9N 0770651.8E	812 FT	NO	ELECTRIC POLE
29L/APCH 11R/TKOF	POLE	283212.3N 0770652.8E	813 FT	NO	ELECTRIC POLE
29L/APCH 11R/TKOF	POLE	283212.8N 0770653.7E	812 FT	NO	ELECTRIC POLE
29L/APCH 11R/TKOF	POLE	283212.9N 0770653.2E	812 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	POLE	283213.5N 0770654.7E	815 FT	NO	ELECTRIC POLE
29L/APCH 11R/TKOF	POLE	283214.2N 0770655.3E	815 FT	NO	LIGHT POLE
29L/APCH 11R/TKOF	TOWER	283216.3N 0770706.9E	853 FT	NO	MOBILE TOWER
29L/APCH 11R/TKOF	FENCE	283209.8N 0770643.3E	797 FT	NO	AIRPORT FENCING
29L/APCH 11R/TKOF	FENCE	283210.9N 0770635.7E	785 FT	NO	AIRPORT FENCING
29L/APCH 11R/TKOF	OTHER	283212.5N 0770648.6E	805 FT	NO	MOBILE ROAD TRAFFIC (ROAD ELEV. 240.3 M. + 5.0 M. TRAFFIC HT.)
29L/APCH 11R/TKOF	OTHER	283211.8N 0770647.3E	804 FT	NO	MOBILE ROAD TRAFFIC (ROAD ELEV. 240.2 M. + 5.0 M. TRAFFIC HT.)
29L/APCH 11R/TKOF	OTHER	283215.6N 0770646.3E	800 FT	LGTD	OBSTRUCTION LIGHT ON LOCALIZER HUT (HUT TOP 241.2 M)
29L/APCH 11R/TKOF	OTHER	283216.0N 0770658.0E	833 FT	NO	HOARDING LIGHT TOP - 254.04 M, HOARDING TOP - 253.59 M
29L/APCH 11R/TKOF	ANTENNA	283215.7N 0770703.8E	848 FT	NO	Antenna On Building - 258.60 M, Sintex Tank Top: 253.87 M, Building Top: 252.90 M
29L/APCH 11R/TKOF	ANTENNA	283207.1N 0770704.2E	864 FT	NO	COMMUNICATION ANTENNA
29L/APCH 11R/TKOF	OTHER	283218.8N 0770647.5E	801 FT	NO	L/C ROD TOP
In circling area and at AD	NAVAID	283403.2N 0770518.0E	772 FT	LGTD	L/C ROD ON GP MAIN ANTENNA (LGT TOP 234.5 M)
In circling area and at AD	NAVAID	283403.1N 0770518.2E	738 FT	NO	GP DME

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	NAVAID	283403.9N 0770515.4E	740 FT	LGTD	L/C ROD ON GP MONITOR ANTENNA (LGT) (ANTENNA TOP 225.1 M.)
In circling area and at AD	NAVAID	283417.7N 0770528.8E	770 FT	NO	L/C ROD ON GP MAIN ANTENNA (ANTENNA TOP 234.6 M)
In circling area and at AD	NAVAID	283238.5N 0770429.4E	785 FT	LGTD	L/C ROD ON GP MAIN ANTENNA (LGT) (ANTENNA TOP 238.5 M)
In circling area and at AD	NAVAID	283238.4N 0770429.5E	742 FT	NO	GP DME
In circling area and at AD	NAVAID	283336.9N 0770712.5E	789 FT	LGTD	GP MONITOR ANTENNA LGT TOP (ANTENNA TOP 240.3 M)
In circling area and at AD	NAVAID	283408.5N 0770642.1E	799 FT	LGTD	L/C ROD ON GP MAIN ANTENNA (LGT TOP 243.1 M) (ANTENNA TOP 242.0 M)
In circling area and at AD	NAVAID	283408.7N 0770642.0E	763 FT	NO	GLIDE PATH DME
In circling area and at AD	NAVAID	283337.7N 0770708.9E	827 FT	LGTD	GP MAIN ANTENNA L/C ROD TOP (LGT TOP 251.7 M, ANTENNA TOP 251.3 M)
In circling area and at AD	NAVAID	283337.8N 0770708.7E	783 FT	NO	GP DME
In circling area and at AD	NAVAID	283402.1N 0770530.5E	757 FT	LGTD	DVOR MONITOR ANTENNA (LGT TOP)(ANTENNA TOP 230.6 M.)
In circling area and at AD	NAVAID	283329.5N 0770538.9E	928 FT	NO	SMR RADAR ON OLD ATC BUILDING
In circling area and at AD	NAVAID	283225.6N 0770508.8E	814 FT	NO	ASR 29 (ELDIS RADAR) L/C ROD TOP (RADAR TOP 244.4 M.) BUILDING TOP 237.1 M)
In circling area and at AD	NAVAID	283226.1N 0770527.8E	761 FT	LGTD	GP DME (LGT)
In circling area and at AD	NAVAID	283226.0N 0770527.9E	804 FT	NO	L/C ROD ON GP MAIN ANTENNA (ANTENNA TOP 243.1 M)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	NAVAID	283225.3N 0770531.1E	766 FT	LGTD	L/C ROD ON GP MONITOR ANTENNA (ANTENNA TOP 233.0 M, LGT TOP 233.1 M)
In circling area and at AD	NAVAID	283400.9N 0770536.7E	754 FT	NO	DVOR MAIN ANTENNA
In circling area and at AD	NAVAID	283400.8N 0770537.1E	763 FT	LGTD	L/C ROD ON DVOR DME (LGT) (DME TOP 232.5 M)
In circling area and at AD	NAVAID	283232.0N 0770450.1E	759 FT	LGTD	DVOR DME (LGT)
In circling area and at AD	NAVAID	283408.4N 0770645.8E	769 FT	LGTD	L/C ROD ON GP MONITOR ANTENNA (LGT TOP 234.4 M) (ANTENNA TOP 234.0 M)
In circling area and at AD	NAVAID	283249.0N 0770440.1E	778 FT	NO	L/C ROD ON GP MAIN ANTENNA (ANTENNA TOP 235.5)
In circling area and at AD	NAVAID	283248.9N 0770440.1E	748 FT	NO	L/C ROD ON GP DME (DME TOP 227.8 M)
In circling area and at AD	NAVAID	283249.6N 0770437.4E	742 FT	NO	L/C ROD ON GP MONITOR ANTENNA (ANTENNA TOP 225.5 M)
In circling area and at AD	NAVAID	283239.2N 0770426.5E	744 FT	NO	GP MONITOR ANTENNA
In circling area and at AD	NAVAID	283230.7N 0770605.2E	782 FT	LGTD	GP MONITOR ANTENNA (L/C ROD and LGT TOP - 238.5 M, ANTENNA TOP 238.1 M)
In circling area and at AD	NAVAID	283231.5N 0770600.9E	779 FT	LGTD	GP DME LGT TOP (ANTENNA TOP - 237.3 M)
In circling area and at AD	NAVAID	283231.6N 0770601.0E	815 FT	LGTD	GP MAIN ANTENNA L/C ROD TOP (LGT TOP - 248.1 M, ANTENNA TOP - 247.8 M)
In circling area and at AD	OTHER	283403.2N 0770518.3E	732 FT	NO	GP HUT TOP

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	283415.2N 0770443.0E	770 FT	NO	BUILDING STAIRCASE ROOM TOP - 234.63 M, BUILDING STAIRCASE ROOM TOP - - 233.45 M
In circling area and at AD	TANK	283415.6N 0770442.0E	772 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	BUILDING	283421.0N 0770447.3E	782 FT	NO	BUILDING TOP
In circling area and at AD	TANK	283420.7N 0770449.2E	778 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	TANK	283420.8N 0770450.0E	780 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	TANK	283420.5N 0770450.7E	774 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	BUILDING	283135.3N 0770806.0E	962 FT	NO	BUILDING STAIRCASE ROOM TOP - 293.3 M, BUILDING TOP - 290.1 M
In circling area and at AD	BUILDING	283133.0N 0770800.8E	947 FT	NO	BUILDING STAIRCASE ROOM TOP - 288.7 M, BUILDING TOP - 286.8 M
In circling area and at AD	TANK	283136.2N 0770809.9E	952 FT	NO	SINTEX TANK ON BUILDING - 290.2 M, BUILDING TOP - 286.1 M
In circling area and at AD	TANK	283135.2N 0770802.4E	952 FT	NO	SINTEX TANK ON BUILDING - 290.2 M, STAIRCASE TOP - 288.8 M, BUILDING TOP - 286.5 M
In circling area and at AD	BUILDING	283136.4N 0770803.6E	949 FT	NO	BUILDING STAIRCASE ROOM TOP - 289.2 M, BUILDING TOP - 286.4 M
In circling area and at AD	TANK	283132.8N 0770807.6E	960 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	BUILDING	283126.7N 0770816.3E	961 FT	NO	BUILDING TOP
In circling area and at AD	TANK	283129.4N 0770800.3E	960 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	BUILDING	283128.4N 0770809.8E	953 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283129.0N 0770836.6E	979 FT	NO	BUILDING TOP
In circling area and at AD	TANK	283129.7N 0770758.3E	946 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	TANK	283129.8N 0770826.4E	939 FT	NO	SINTEX TANK ON BUILDING

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	TANK	283130.7N 0770756.9E	945 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	BUILDING	283131.8N 0770825.0E	953 FT	NO	BUILDING TOP
In circling area and at AD	TANK	283131.9N 0770804.4E	956 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	BUILDING	283132.1N 0770819.4E	952 FT	NO	BUILDING TOP
In circling area and at AD	TANK	283132.1N 0770816.5E	957 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	TANK	283132.2N 0770759.5E	948 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	TANK	283132.3N 0770818.1E	954 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	BUILDING	283132.3N 0770822.0E	935 FT	NO	BUILDING (U/C)
In circling area and at AD	BUILDING	283132.7N 0770840.5E	980 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283133.0N 0770836.8E	981 FT	NO	BUILDING TOP
In circling area and at AD	TANK	283133.2N 0770825.5E	947 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	BUILDING	283133.4N 0770828.0E	940 FT	NO	BUILDING TOP
In circling area and at AD	TANK	283133.7N 0770805.9E	952 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	BUILDING	283134.1N 0770816.9E	955 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283134.2N 0770815.1E	949 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283134.4N 0770825.9E	951 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283134.6N 0770828.4E	945 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283134.8N 0770840.5E	974 FT	NO	BUILDING TOP
In circling area and at AD	TANK	283134.9N 0770816.0E	954 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	BUILDING	283134.9N 0770824.8E	942 FT	NO	BUILDING (U/C)
In circling area and at AD	TANK	283134.9N 0770813.5E	953 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	TANK	283135.5N 0770804.6E	953 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	TANK	283135.9N 0770824.0E	938 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	BUILDING	283136.3N 0770814.2E	954 FT	NO	BUILDING TOP
In circling area and at AD	TANK	283136.5N 0770828.7E	949 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	BUILDING	283137.4N 0770822.1E	953 FT	NO	BUILDING TOP

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	283137.6N 0770809.3E	951 FT	NO	BUILDING TOP
In circling area and at AD	TANK	283138.7N 0770806.4E	949 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	BUILDING	283139.5N 0770825.0E	936 FT	NO	TEMPLE TOP
In circling area and at AD	TANK	283139.5N 0770823.1E	946 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	TANK	283140.2N 0770807.2E	951 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	TANK	283140.6N 0770808.5E	952 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	TANK	283142.4N 0770808.5E	948 FT	NO	SINTEX TANK ON BUILDING
In circling area and at AD	BUILDING	283147.3N 0770809.0E	942 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283154.5N 0770808.4E	929 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283218.9N 0770708.6E	856 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283240.1N 0770847.1E	956 FT	NO	BUILDING (U/C)
In circling area and at AD	BUILDING	283242.2N 0770844.8E	955 FT	NO	BUILDING (U/C)
In circling area and at AD	BUILDING	283134.9N 0770826.8E	945 FT	NO	BUILDING GENERATOR ROOM TOP - 288.1 M, BUILDING TOP - 286.0 M (D-60)
In circling area and at AD	BUILDING	283154.6N 0770858.6E	927 FT	NO	BUILDING TOP (GURUDWARA)
In circling area and at AD	BUILDING	282951.7N 0770617.3E	1051 FT	NO	BUILDING STAIRCASE ROOM TOP, BUILDING TOP - 315.4 M
In circling area and at AD	BUILDING	283208.2N 0770915.2E	936 FT	NO	BUILDING STAIRCASE ROOM TOP - 285.31 M, SINTEX TANK TOP - 284.49 M, BUILDING TOP - 283.82 M (ALL INDIA COUNCIL FOR TECHNICAL EDUCATION)
In circling area and at AD	BUILDING	283218.2N 0770701.7E	837 FT	NO	BUILDING TOP (CITY CARK)
In circling area and at AD	BUILDING	283218.8N 0770703.6E	845 FT	NO	BUILDING TOP (JPS FLAT)
In circling area and at AD	TANK	283127.8N 0770818.1E	954 FT	NO	SINTEX TANK ON BUILDING-290.83 M, BUILDING TOP-287.64 M (B-94)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	283126.3N 0770822.4E	940 FT	NO	BUILDING TOP (C-25)
In circling area and at AD	BUILDING	283130.7N 0770819.4E	953 FT	NO	BUILDING TOP (B-18)
In circling area and at AD	BUILDING	283129.4N 0770822.3E	946 FT	NO	BUILDING TOP (C-38)
In circling area and at AD	BUILDING	283135.4N 0770809.6E	956 FT	NO	BUILDING STAIRCASE ROOM TOP - 291.28 M, BUILDING TOP- 289.80 M (B-60/61)
In circling area and at AD	BUILDING	283133.3N 0770810.6E	960 FT	NO	BUILDING TOP (A-109)
In circling area and at AD	BUILDING	283132.0N 0770812.3E	954 FT	NO	BUILDING TOP (JVD HOMES B1/ 2)
In circling area and at AD	BUILDING	283130.0N 0770808.3E	951 FT	NO	BUILDING STAIRCASE ROOM TOP - 289.94 M, BUILDING TOP- 286.70 M (B-92)
In circling area and at AD	BUILDING	283130.1N 0770806.2E	957 FT	NO	BUILDING TOP (B-423)
In circling area and at AD	BUILDING	283130.4N 0770811.9E	958 FT	NO	BUILDING TOP (A-11)
In circling area and at AD	BUILDING	283129.5N 0770812.2E	943 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283131.3N 0770800.8E	949 FT	NO	BUILDING STAIRACASE ROOM TOP - 289.25 M, BUILDING TOP- 285.96 M (B-352A)
In circling area and at AD	TANK	283146.0N 0770856.6E	928 FT	NO	BUILDING SINTEX TANK TOP - 282.9 M, BUILDING TOP - 280.26 M
In circling area and at AD	TANK	283135.5N 0770817.9E	950 FT	NO	BUILDING SINTEX TANK TOP - 289.45 M, BUILDING TOP - 287.66 M (MEER SINGH TATHI)
In circling area and at AD	BUILDING	283136.1N 0770817.3E	949 FT	NO	BUILDING TOP
In circling area and at AD	TANK	283207.4N 0770908.5E	930 FT	NO	SINTEX TANK TOP ON BUILDING, BUILDING TOP - 279.34 M (LOHIA BHAWAN)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	TANK	283135.7N 0770830.2E	935 FT	NO	SINTEX TANK ON BUILDING-284.89 M, STAIRCASE ROOM TOP-283.71 M, BUILDING TOP-281.71 M (JAI JUDICIAL ACADEMY)
In circling area and at AD	BUILDING	283137.4N 0770830.2E	943 FT	NO	BUILDING TOP - 287.30 M (D 82)
In circling area and at AD	BUILDING	283138.0N 0770830.8E	941 FT	NO	BUILDING TOP (D-81)
In circling area and at AD	TANK	283139.4N 0770829.8E	932 FT	NO	SINTEX TANK ON BUILDING-284.10 M, BUILDING TOP-283.43 M (D-74)
In circling area and at AD	BUILDING	283130.5N 0770759.9E	946 FT	NO	BUILDING STAIRCASE ROOM TOP - 288.21 M, BUILDING TOP - 285.98 M (B 338, YADAV MWASH)
In circling area and at AD	BUILDING	283128.9N 0770801.7E	953 FT	NO	BUILDING TOP (B 262)
In circling area and at AD	BUILDING	283128.2N 0770803.3E	946 FT	NO	BUILDING TOP (B 491)
In circling area and at AD	BUILDING	283128.0N 0770806.2E	951 FT	NO	BUILDING STAIRCASE ROOM TOP - 290.0 M, BUILDING TOP - 288.75 M (B 457)
In circling area and at AD	BUILDING	283127.6N 0770807.4E	953 FT	NO	BUILDING TOP (B 442)
In circling area and at AD	BUILDING	283126.8N 0770810.3E	955 FT	NO	BUILDING TOP
In circling area and at AD	TANK	283132.9N 0770800.2E	948 FT	NO	SINTEX TANK ON BUILDING- 288.93 M, STAIRCASE - 287.81 M, BUILDING TOP - 284.87 M. (SF 577)
In circling area and at AD	TANK	283134.4N 0770802.1E	953 FT	NO	SINTEX TANK ON BUILDING- 290.06 M, BUILDING TOP 289.39 M
In circling area and at AD	TANK	283134.4N 0770800.3E	941 FT	NO	SINTEX TANK ON BUILDING- 286.76 M, BUILDING TOP 285.45 M (B 278)
In circling area and at AD	TANK	283135.2N 0770800.6E	941 FT	NO	SINTEX TANK ON BUILDING- 286.85 M, BUILDING TOP 286.23 M (B 275)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	283136.5N 0770759.2E	949 FT	NO	BUILDING STAIRCASE ROOM TOP - 289.19 M, SINTEX TANK - 288.69 M, BUILDING TOP - 288.05 M (B 273)
In circling area and at AD	BUILDING	283126.3N 0770820.6E	957 FT	NO	BUILDING STAIRCASE ROOM TOP - 291.70 M, BUILDING TOP- 288.04 M (B-10)
In circling area and at AD	BUILDING	283137.0N 0770812.1E	947 FT	NO	BUILDING STAIRCASE ROOM TOP - 288.61 M, SINTEX TANK TOP - 288.46 M, BUILDING TOP - 285.53 M (A 115)
In circling area and at AD	TANK	283135.3N 0770812.2E	951 FT	NO	SINTEX TANK ON BUILDING - 289.92 M, BUILDING TOP - 287.02 M (A 110)
In circling area and at AD	BUILDING	283136.9N 0770813.6E	948 FT	NO	BUILDING STAIRCASE ROOM TOP - 288.75 M, SINTEX TANK TOP - 288.39 M, BUILDING TOP - 287.19 M (A 31)
In circling area and at AD	TANK	283136.8N 0770814.7E	949 FT	NO	SINTEX TANK ON BUILDING - 289.36 M, BUILDING TOP - 288.40 M (A 28)
In circling area and at AD	TANK	283125.9N 0770801.3E	961 FT	NO	SINTEX TANK ON BUILDING- 292.77 M, BUILDING TOP - 292.33 M. (B 153)
In circling area and at AD	TANK	283135.5N 0770809.3E	956 FT	NO	SINTEX TANK ON BUILDING- 291.50 M, STAIRCASE ROOM - 290.32 M, BUILDING TOP - 286.18 M (B 61)
In circling area and at AD	BUILDING	283135.5N 0770809.7E	957 FT	NO	BUILDING STAIRCASE ROOM TOP - 291.63 M, BUILDING TOP - 289.68 M (B 60)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	TANK	283218.6N 0770706.9E	858 FT	NO	BUILDING SINTEX TANK TOP - 261.64 M, STAIRCASE ROOM - 260.66 M, BUILDING TOP - 258.49 M (DALAL COMPLEX 1)
In circling area and at AD	BUILDING	283131.3N 0770808.2E	949 FT	NO	BUILDING STAIRCASE ROOM TOP - 289.18 M (B 97)
In circling area and at AD	BUILDING	283130.9N 0770806.9E	942 FT	NO	BUILDING TOP (B 129)
In circling area and at AD	BUILDING	283131.9N 0770810.1E	952 FT	NO	BUILDING STAIRCASE ROOM TOP - 290.21 M, BUILDING TOP - 286.73 M (B 77A)
In circling area and at AD	TANK	283132.1N 0770813.3E	952 FT	NO	SINTEX TANK ON BUILDING- 290.30 M, BUILDING TOP - 289.68 M (B 1/5)
In circling area and at AD	TANK	283132.2N 0770814.2E	950 FT	NO	SINTEX TANK ON BUILDING- 289.64 M, BUILDING TOP - 288.61 M (B 72)
In circling area and at AD	TANK	283134.8N 0770814.5E	950 FT	NO	SINTEX TANK ON BUILDING- 289.63 M, BUILDING TOP - 288.57 M (A 59)
In circling area and at AD	BUILDING	283133.3N 0770817.6E	952 FT	NO	BUILDING TOP (A12)
In circling area and at AD	TANK	283131.7N 0770814.5E	950 FT	NO	SINTEX TANK ON BUILDING- 289.46 M, STAIRCASE ROOM - 288.65 M, BUILDING TOP - 287.21 M (B 65)
In circling area and at AD	TANK	283131.8N 0770814.9E	951 FT	NO	SINTEX TANK ON BUILDING- 290.03 M, BUILDING TOP - 289.21 M (B 59)
In circling area and at AD	BUILDING	283201.6N 0770909.2E	929 FT	NO	BUILDING STAIRCASE ROOM TOP - 283.2 M, BUILDING TOP - 281.75 M
In circling area and at AD	TANK	283131.2N 0770817.3E	953 FT	NO	SINTEX TANK ON BUILDING- 290.36 M, BUILDING TOP - 289.72 M (B 35)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	283150.2N 0770856.3E	933 FT	NO	TEMPLE TOP
In circling area and at AD	BUILDING	283152.3N 0770857.5E	935 FT	NO	BUILDING STAIRCASE ROOM TOP - 284.92 M, BUILDING TOP - 282.20 M (SOS CHILDRENS VILLAGE)
In circling area and at AD	BUILDING	283153.2N 0770858.1E	929 FT	NO	BUILDING STAIRCASE ROOM TOP - 283.28 M, BUILDING TOP - 280.25 M (IRCEN BHAWAN)
In circling area and at AD	BUILDING	283203.3N 0770909.3E	937 FT	NO	BUILDING STAIRCASE ROOM TOP - 285.6 M, BUILDING TOP - 281.68 M (SUKHMANI HOSPITAL)
In circling area and at AD	BUILDING	283218.5N 0770705.6E	846 FT	NO	BUILDING STAIRCASE ROOM TOP - 257.85 M, BUILDING TOP - 255.73 M
In circling area and at AD	TANK	283219.0N 0770706.8E	859 FT	NO	SINTEX TANK ON BUILDING - 261.70 M, STAIRCASE ROOM TOP - 260.83 M, BUILDING TOP - 258.87 M.
In circling area and at AD	BUILDING	283218.8N 0770709.3E	856 FT	NO	BUILDING STAIRCASE ROOM TOP - 260.86 M, BUILDING TOP - 258.26 M
In circling area and at AD	TANK	283218.0N 0770709.9E	851 FT	NO	SINTEX TANK ON BUILDING - 259.36 M, STAIRCASE ROOM TOP - 258.35 M, BUILDING TOP - 256.46 M.
In circling area and at AD	BUILDING	283217.4N 0770709.0E	849 FT	NO	BUILDING STAIRCASE ROOM TOP - 258.69 M, BUILDING TOP - 257.24 M

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	TANK	283127.5N 0770806.2E	947 FT	NO	SINTEX TANK ON BUILDING - 288.72 M, BUILDING TOP - 287.53 M.
In circling area and at AD	BUILDING	283127.9N 0770808.2E	954 FT	NO	BUILDING STAIRCASE ROOM TOP - 290.81 M, BUILDING TOP - 287.12 M
In circling area and at AD	TANK	283140.9N 0770848.1E	930 FT	NO	SINTEX TANK ON BUILDING - 283.59 M, BUILDING TOP - 282.14 M.
In circling area and at AD	BUILDING	283201.1N 0770906.9E	927 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283138.4N 0770835.0E	928 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283224.1N 0770917.1E	928 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283126.9N 0770805.3E	938 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283137.7N 0770805.3E	944 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283134.5N 0770810.7E	954 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283125.6N 0770850.2E	986 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283125.6N 0770853.0E	980 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283129.3N 0770851.9E	984 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283127.3N 0770900.6E	981 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283125.8N 0770855.9E	979 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283124.7N 0770856.9E	977 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283125.6N 0770901.5E	979 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283133.8N 0770840.3E	978 FT	NO	BUILDING TOP (SARASWATI APARTMENT)
In circling area and at AD	BUILDING	283131.5N 0770837.4E	982 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283126.5N 0770840.6E	984 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283129.2N 0770836.0E	976 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283125.2N 0770800.6E	955 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283130.0N 0770837.8E	975 FT	NO	BUILDING TOP

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	283140.4N 0770810.6E	939 FT	NO	BUILDING TOP
In circling area and at AD	BUILDING	283126.3N 0770810.7E	957 FT	NO	BUILDING STAIRCASE ROOM TOP - 291.73 M, BUILDING , TOP - 287.80 M
In circling area and at AD	TANK	283127.5N 0770802.8E	950 FT	NO	SINTEX TANK ON BUILDING - 289.50 M, STAIRCASE ROOM TOP - 288.39 M, BUILDING TOP- 286.75 M.
In circling area and at AD	TANK	283135.9N 0770801.2E	950 FT	NO	SINTEX TANK ON BUILDING - 289.55 M, STAIRCASE TOP - 287.84 M, BUILDING TOP - 286.48 M
In circling area and at AD	TANK	283133.5N 0770801.5E	950 FT	NO	SINTEX TANK ON BUILDING - 289.66 M, STAIRCASE TOP - 288.58 M, BUILDING TOP - 286.54 M
In circling area and at AD	TANK	283131.9N 0770806.7E	950 FT	NO	SINTEX TANK ON BUILDING - 289.69 M, STAIRCASE ROOM TOP - 288.41 M, BUILDING TOP- 286.69 M.
In circling area and at AD	TANK	283134.6N 0770803.6E	953 FT	NO	SINTEX TANK ON BUILDING - 290.35 M, STAIRCASE TOP - 289.20 M, BUILDING TOP - 287.76 M
In circling area and at AD	TANK	283131.2N 0770806.3E	953 FT	NO	SINTEX TANK ON BUILDING - 290.48 M, STAIRCASE ROOM TOP - 288.82 M, BUILDING TOP- 286.47 M.
In circling area and at AD	TANK	283135.1N 0770811.0E	955 FT	NO	SINTEX TANK ON BUILDING - 291.24 M, STAIRCASE ROOM TOP - 290.21 M, BUILDING TOP- 287.96 M.

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	TANK	283130.2N 0770806.9E	957 FT	NO	SINTEX TANK ON BUILDING - 291.82 M, STAIRCASE ROOM TOP - 290.94 M, BUILDING TOP- 289.21 M.
In circling area and at AD	TANK	283137.2N 0770802.2E	950 FT	NO	SINTEX TANK ON BUILDING - 298.46 M, STAIRCASE TOP - 289.02 M, BUILDING TOP - 288.64 M
In circling area and at AD	BUILDING	283131.6N 0770801.5E	950 FT	NO	BUILDING STAIRCASE ROOM TOP - 289.46 M, BUILDING TOP - 287.23 M
In circling area and at AD	BUILDING	283136.3N 0770805.2E	951 FT	NO	BUILDING STAIRCASE ROOM TOP - 289.76 M, BUILDING TOP - 286.69 M
In circling area and at AD	BUILDING	283128.6N 0770802.4E	958 FT	NO	BUILDING STAIRCASE ROOM TOP - 291.97 M, BUILDING TOP - 290.18 M
In circling area and at AD	BUILDING	283127.2N 0770805.1E	945 FT	NO	BUILDING STAIRCASE ROOM TOP - 287.93 M, SINTEX TANK TOP-287.02 M, BUILDING TOP - 286.53 M
In circling area and at AD	BUILDING	283130.9N 0770759.9E	948 FT	NO	BUILDING STAIRCASE ROOM TOP - 288.9 M, SINTEX TANK TOP-288.16 M, BUILDING TOP - 285.20 M
In circling area and at AD	BUILDING	283128.6N 0770803.7E	952 FT	NO	BUILDING STAIRCASE ROOM TOP - 290.26 M, SINTEX TANK TOP-288.64 M, BUILDING TOP - 286.75 M

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	283128.3N 0770759.8E	952 FT	NO	BUILDING STAIRCASE ROOM TOP - 290.34 M, SINTEX TANK TOP-289.86 M, BUILDING TOP - 288.55 M
In circling area and at AD	BUILDING	283130.7N 0770809.7E	951 FT	NO	BUILDING STAIRCASE ROOM TOP - 289.9 M, SINTEX TANK TOP-287.80 M, BUILDING TOP - 284.22 M
In circling area and at AD	BUILDING	283129.2N 0770807.4E	954 FT	NO	BUILDING STAIRCASE ROOM TOP - 290.66 M, SINTEX TANK TOP-289.38 M, BUILDING TOP - 288.55 M
In circling area and at AD	BUILDING	283127.9N 0770814.4E	956 FT	NO	BUILDING STAIRCASE ROOM TOP - 291.42 M, BUILDING TOP - 290.11 M
In circling area and at AD	BUILDING	283128.3N 0770800.2E	957 FT	NO	BUILDING STAIRCASE ROOM TOP - 291.59 M, SINTEX TANK TOP-291.18 M, BUILDING TOP - 290.09 M
In circling area and at AD	BUILDING	283127.7N 0770801.6E	958 FT	NO	BUILDING STAIRCASE ROOM TOP - 292.05 M, SINTEX TANK TOP-291.99 M, BUILDING TOP - 291.18 M
In circling area and at AD	BUILDING	283127.3N 0770809.2E	959 FT	NO	BUILDING STAIRCASE ROOM TOP - 292.17 M, SINTEX TANK TOP-290.34 M, BUILDING TOP - 289.52 M
In circling area and at AD	BUILDING	283128.5N 0770806.2E	959 FT	NO	BUILDING STAIRCASE ROOM TOP - 292.32 M, SINTEX TANK TOP-289.9 M, BUILDING TOP - 289.36 M

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	283128.6N 0770810.7E	962 FT	NO	BUILDING STAIRCASE ROOM TOP - 293.12 M, BUILDING TOP - 290.26 M
In circling area and at AD	BUILDING	283134.8N 0770800.1E	946 FT	NO	BUILDING STAIRCASE ROOM TOP - 288.33 M, BUILDING TOP - 286.47 M
In circling area and at AD	BUILDING	283132.6N 0770804.3E	947 FT	NO	BUILDING STAIRCASE ROOM TOP - 288.49 M, BUILDING TOP - 286.72 M
In circling area and at AD	BUILDING	283126.2N 0770836.9E	978 FT	NO	BUILDING TOP
In circling area and at AD	TANK	283133.0N 0770812.3E	950 FT	NO	SINTEX TANK ON BUILDING- 289.72 M, BUILDING TOP - 288.73 M (A 85)
In circling area and at AD	TANK	283128.7N 0770804.6E	949 FT	NO	SINTEX TANK ON BUILDING- 289.29 M, STAIRCASE - 287.92 M, BUILDING TOP - 286.25 M. (B 475)
In circling area and at AD	TANK	283129.3N 0770828.0E	954 FT	NO	SINTEX TANK ON BUILDING- 290.67 M, STAIRCASE ROOM - 289.31 M, BUILDING TOP - 287.38 M (C 14)
In circling area and at AD	TANK	283139.8N 0770809.9E	943 FT	NO	SINTEX TANK ON BUILDING - 287.33 M, BUILDING TOP - 286.43 M (B3)
In circling area and at AD	TANK	283131.0N 0770818.0E	953 FT	NO	SINTEX TANK ON BUILDING- 290.35M, BUILDING TOP - 290.30 M (B 34)
In circling area and at AD	TANK	283136.7N 0770815.8E	950 FT	NO	SINTEX TANK ON BUILDING - 289.72 M, BUILDING TOP - 289.14 M (A 26)
In circling area and at AD	TANK	283124.9N 0770801.0E	946 FT	NO	SINTEX TANK ON BUILDING- 288.43 M, STAIRCASE ROOM - 287.82 M, BUILDING TOP - 285.92 M. (B 156)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	283133.2N 0770803.2E	944 FT	NO	BUILDING STAIRCASE ROOM TOP - 287.84 M, BUILDING TOP - 285.83 M (SF 316)
In circling area and at AD	TANK	283135.7N 0770801.4E	949 FT	NO	SINTEX TANK ON BUILDING- 289.43 M, BUILDING TOP 288.44 M (B 270)
In circling area and at AD	BUILDING	283133.1N 0770759.0E	948 FT	NO	BUILDING TOP (B 329)
In circling area and at AD	OTHER	283238.5N 0770429.7E	734 FT	NO	GP HUT TOP
In circling area and at AD	OTHER	283337.7N 0770708.6E	778 FT	NO	GP HUT TOP
In circling area and at AD	OTHER	283226.1N 0770527.5E	753 FT	NO	GP HUT TOP
In circling area and at AD	OTHER	283232.1N 0770457.1E	738 FT	NO	HUT
In circling area and at AD	OTHER	283248.9N 0770440.5E	736 FT	NO	GP HUT
In circling area and at AD	BUILDING	283407.0N 0770714.9E	783 FT	NO	BUILDING STAIRCASE ROOM TOP - 238.75 M, BUILDING TOP - 236.03 M (AIR INDIA)
In circling area and at AD	OTHER	283231.6N 0770600.6E	771 FT	NO	GP HUT TOP
In circling area and at AD	TREE	283152.9N 0770810.5E	929 FT	NO	GROUP OF TREES
In circling area and at AD	TREE	283136.5N 0770848.3E	944 FT	NO	TREE
In circling area and at AD	TREE	283138.2N 0770842.2E	943 FT	NO	TREE
In circling area and at AD	TREE	283141.0N 0770844.9E	935 FT	NO	GROUP OF TREES
In circling area and at AD	TREE	283139.4N 0770847.2E	933 FT	NO	GROUP OF TREES
In circling area and at AD	TREE	283203.8N 0770646.2E	861 FT	NO	GROUP OF TREES
In circling area and at AD	TREE	283204.0N 0770644.7E	868 FT	NO	GROUP OF TREES
In circling area and at AD	TREE	283205.7N 0770643.6E	851 FT	NO	TREE
In circling area and at AD	TREE	283206.3N 0770643.5E	833 FT	NO	TREE
In circling area and at AD	TREE	283247.3N 0770324.8E	794 FT	NO	TREE
In circling area and at AD	TREE	283406.4N 0770715.2E	811 FT	NO	GROUP OF TREES
In circling area and at AD	TREE	283415.7N 0770717.2E	805 FT	NO	TREE

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	283403.3N 0770519.0E	740 FT	NO	WDI
In circling area and at AD	OTHER	283349.1N 0770509.5E	827 FT	NO	FLOOD LIGHT MAST
In circling area and at AD	OTHER	283349.8N 0770506.9E	828 FT	NO	FLOOD LIGHT MAST
In circling area and at AD	OTHER	283350.4N 0770504.3E	827 FT	NO	FLOOD LIGHT MAST
In circling area and at AD	OTHER	283351.0N 0770501.7E	827 FT	NO	FLOOD LIGHT MAST
In circling area and at AD	OTHER	283351.6N 0770459.2E	826 FT	NO	FLOOD LIGHT MAST
In circling area and at AD	POLE	283401.6N 0770520.0E	728 FT	NO	POLE TOP NEAR PAPI LIGHTS
In circling area and at AD	TOWER	283144.9N 0770843.8E	936 FT	NO	MOBILE TOWER ON BUILDING
In circling area and at AD	TOWER	283226.8N 0770847.2E	945 FT	NO	MOBILE TOWER ON BUILDING
In circling area and at AD	OTHER	283138.6N 0770810.2E	947 FT	NO	BUILDING, CELLPHONE MAST TOP - 288.55 M, BUILDING TOP - 283.71 M (B 28)
In circling area and at AD	OTHER	283124.4N 0770832.9E	986 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283134.7N 0770846.5E	991 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283157.8N 0770858.6E	940 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283044.2N 0770538.0E	983 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283253.0N 0770400.5E	744 FT	NO	WIND SOCK
In circling area and at AD	POLE	283208.5N 0770641.8E	804 FT	NO	LIGHT POLE
In circling area and at AD	POLE	283208.7N 0770641.3E	812 FT	NO	LIGHT POLE
In circling area and at AD	POLE	283208.3N 0770640.6E	812 FT	NO	LIGHT POLE
In circling area and at AD	POLE	283208.0N 0770639.9E	812 FT	NO	LIGHT POLE
In circling area and at AD	POLE	283206.5N 0770640.5E	826 FT	NO	LIGHT POLE
In circling area and at AD	OTHER	283210.0N 0770909.2E	934 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283210.0N 0770911.6E	937 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283209.5N 0770908.9E	934 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283124.4N 0770823.6E	953 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283124.9N 0770830.4E	952 FT	NO	PYLON MAST

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	283136.2N 0770839.4E	982 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283136.6N 0770836.7E	983 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283205.8N 0770908.6E	936 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283202.6N 0770909.7E	963 FT	NO	CELLPHONE MAST TOP
In circling area and at AD	OTHER	283148.6N 0770855.5E	930 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283147.1N 0770854.4E	932 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283144.9N 0770852.0E	936 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283142.7N 0770856.4E	942 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283212.7N 0770629.4E	799 FT	NO	WIND SOCK
In circling area and at AD	OTHER	283132.9N 0770755.0E	937 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283129.4N 0770755.5E	938 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283126.2N 0770801.4E	956 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283127.0N 0770801.6E	952 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283133.6N 0770807.6E	983 FT	NO	CELL PHONE MAST ON BUILDING -299.48 M, SINTEX TANK TOP-289.65 M, BUILDING TOP-286.62 M (B-117)
In circling area and at AD	POLE	283207.1N 0770641.9E	827 FT	NO	LIGHT POLE
In circling area and at AD	POLE	283207.6N 0770642.9E	806 FT	NO	LIGHT POLE
In circling area and at AD	OTHER	283010.1N 0770531.6E	958 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283005.6N 0770527.8E	995 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283012.7N 0770533.8E	993 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283044.2N 0770530.9E	978 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283041.1N 0770539.3E	947 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283037.5N 0770541.7E	930 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283032.6N 0770543.6E	934 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283026.4N 0770544.6E	954 FT	NO	PYLON MAST

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	283013.7N 0770540.1E	988 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283018.4N 0770543.0E	968 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283041.0N 0770530.5E	950 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283041.5N 0770529.4E	941 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283044.2N 0770526.6E	930 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283047.5N 0770521.1E	935 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283123.5N 0770822.8E	951 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283124.9N 0770817.4E	954 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283124.5N 0770815.4E	962 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283126.5N 0770804.9E	953 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283125.2N 0770809.2E	957 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283125.6N 0770811.5E	961 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283130.6N 0770834.1E	988 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283415.8N 0770628.3E	771 FT	NO	WDI
In circling area and at AD	POLE	283237.8N 0770848.1E	983 FT	NO	FLAG POLE TOP
In circling area and at AD	POLE	283235.8N 0770848.5E	966 FT	NO	FLAG POLE TOP
In circling area and at AD	OTHER	283202.0N 0770655.9E	873 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283124.3N 0770830.1E	948 FT	NO	PYLON MAST
In circling area and at AD	TOWER	283217.3N 0770858.1E	975 FT	LGTD	MOBILE TOWER (LGT)
In circling area and at AD	TOWER	283122.7N 0770826.3E	955 FT	NO	MOBILE TOWER
In circling area and at AD	TOWER	283135.0N 0770819.7E	954 FT	NO	MOBILE TOWER
In circling area and at AD	OTHER	283139.1N 0770756.2E	930 FT	NO	PYLON MAST
In circling area and at AD	TOWER	283141.4N 0770732.0E	934 FT	LGTD	MOBILE TOWER (LGT)
In circling area and at AD	TOWER	283145.2N 0770821.9E	942 FT	NO	MOBILE TOWER
In circling area and at AD	TOWER	283149.6N 0770911.8E	982 FT	NO	MOBILE TOWER
In circling area and at AD	OTHER	283204.8N 0770902.7E	944 FT	NO	PYLON MAST

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	POLE	283208.2N 0770641.2E	804 FT	NO	LIGHT POLE
In circling area and at AD	OTHER	283211.4N 0770906.5E	945 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283214.1N 0770907.4E	951 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283215.3N 0770904.6E	934 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283217.4N 0770901.7E	938 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283219.4N 0770857.5E	940 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283228.9N 0770841.9E	933 FT	NO	PYLON MAST
In circling area and at AD	TOWER	283230.9N 0770854.0E	928 FT	NO	MOBILE TOWER
In circling area and at AD	OTHER	283258.5N 0770437.2E	752 FT	NO	WINDSOCK
In circling area and at AD	TOWER	283334.7N 0770926.8E	951 FT	LGTD	MOBILE TOWER (LGT)
In circling area and at AD	TOWER	283346.9N 0770902.0E	929 FT	NO	MOBILE TOWER
In circling area and at AD	TOWER	283215.0N 0770914.1E	957 FT	LGTD	MOBILE TOWER (LGT)
In circling area and at AD	OTHER	283133.7N 0770753.3E	934 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283128.9N 0770754.7E	937 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283401.9N 0770529.8E	749 FT	LGTD	WDI (LGT)
In circling area and at AD	OTHER	283417.1N 0770533.0E	753 FT	LGTD	WDI (LGT)
In circling area and at AD	OTHER	283116.2N 0770726.7E	979 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283329.4N 0770706.3E	789 FT	LGTD	WDI (LGT)
In circling area and at AD	OTHER	283114.2N 0770738.0E	988 FT	NO	PYLON MAST
In circling area and at AD	OTHER	283139.1N 0770842.5E	938 FT	NO	FLOOD LIGHT MAST
In circling area and at AD	TOWER	283203.6N 0770909.9E	947 FT	NO	H.T. TOWER
In circling area and at AD	TOWER	283150.6N 0770855.6E	928 FT	NO	H.T. TOWER
In circling area and at AD	TOWER	283157.4N 0770805.4E	928 FT	NO	H.T. TOWER
In circling area and at AD	TOWER	283157.5N 0770804.4E	927 FT	NO	H.T. TOWER
In circling area and at AD	OTHER	283454.7N 0770858.8E	964 FT	LGTD	CELLPHONE MAST (LGT)
In circling area and at AD	TOWER	284150.1N 0770901.3E	1507 FT	LGTD	TV TOWER LGT TOP (TV TOWER TOP 457.4M.)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	283226.3N 0770524.6E	758 FT	NO	WDI
In circling area and at AD	OTHER	283237.2N 0770434.1E	741 FT	NO	WDI
In circling area and at AD	OTHER	283133.2N 0770818.9E	931 FT	NO	FLOOD LIGHT MAST
In circling area and at AD	OTHER	283134.3N 0770820.7E	931 FT	NO	FLOOD LIGHT MAST
In circling area and at AD	OTHER	283135.9N 0770819.5E	929 FT	NO	FLOOD LIGHT MAST
In circling area and at AD	OTHER	283232.1N 0770537.7E	772 FT	LGTD	WINDSOCK (LGT TOP)
In circling area and at AD	FENCE	283245.6N 0770351.6E	730 FT	NO	AIRPORT FENCING
In circling area and at AD	OTHER	283245.9N 0770352.0E	733 FT	NO	MOBILE ROAD TRAFFIC (ROAD ELEV. 218.5 M. + 5.0 M. TRAFFIC HT.)
In circling area and at AD	FENCE	283245.9N 0770351.5E	730 FT	NO	AIRPORT FENCING
In circling area and at AD	OTHER	283254.3N 0770348.8E	735 FT	NO	MOBILE ROAD TRAFFIC (ROAD ELEV. 219.0 M. + 5.0 M. TRAFFIC HT.)
In circling area and at AD	OTHER	283402.5N 0770517.7E	735 FT	NO	L/C ROD NEAR RVR ANTENNA
In circling area and at AD	ANTENNA	283402.0N 0770518.3E	731 FT	NO	RVR ANTENNA
In circling area and at AD	OTHER	283400.1N 0770519.4E	727 FT	NO	PAPI LIGHT TOP
In circling area and at AD	ANTENNA	283402.1N 0770518.4E	730 FT	NO	RVR ANTENNA
In circling area and at AD	ANTENNA	283402.4N 0770517.3E	730 FT	NO	RVR ANTENNA
In circling area and at AD	ANTENNA	283402.2N 0770517.3E	731 FT	NO	RVR ANTENNA
In circling area and at AD	ANTENNA	283135.1N 0770838.9E	995 FT	NO	COMMUNICATION ANTENNA ON BUILDING, BUILDING TOP - 297.7 M (NARMADA BUILDING)
In circling area and at AD	OTHER	283128.9N 0770849.5E	978 FT	LGTD	BUILDING OBSTRUCTION LIGHT TOP (GANGA APARTMENT)
In circling area and at AD	ANTENNA	283126.3N 0770819.3E	961 FT	NO	DISH ANTENNA ON BUILDING
In circling area and at AD	OTHER	283127.5N 0770827.0E	964 FT	NO	ROD ON BUILDING (BUILDING TOP 291.1 M.)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	ANTENNA	283130.7N 0770837.4E	975 FT	NO	DISH ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283132.0N 0770840.0E	982 FT	NO	DISH ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283135.9N 0770827.2E	952 FT	NO	DISH ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283145.7N 0770824.2E	938 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283145.8N 0770820.1E	934 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283145.9N 0770817.7E	935 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283145.9N 0770815.6E	934 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283146.0N 0770813.3E	935 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283147.1N 0770820.1E	943 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283147.5N 0770813.3E	935 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283148.1N 0770821.9E	948 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283149.5N 0770820.3E	934 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283149.6N 0770817.9E	935 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283149.8N 0770813.4E	935 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283150.1N 0770824.7E	948 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283151.0N 0770820.2E	935 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283151.2N 0770815.7E	934 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283151.2N 0770813.4E	935 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283208.0N 0770917.9E	948 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283219.9N 0770906.5E	932 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283226.8N 0770915.0E	948 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283245.7N 0770849.6E	950 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283245.7N 0770846.7E	937 FT	NO	ANTENNA ON BUILDING
In circling area and at AD	ANTENNA	283032.1N 0770414.8E	938 FT	NO	ANTENNA ON BUILDING - 285.97 M, STAIRCASE ROOM TOP-284.54 M,BUILDING TOP-281.69 M (RBS)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	TOWER	283322.1N 0770532.5E	1096 FT	LGTD	ATC TOWER, LIGHTINING ARRESTOR - 334.2 M, BLINKING LIGHT TOP - 333.2 M, BUILDING TOP - 327.5 M.
In circling area and at AD	OTHER	283230.0N 0770931.6E	938 FT	LGTD	BUILDING OBSTRUCTION LIGHT TOP - 285.98 M, SINTEX TANK TOP - 284.27 M, BUILDING TOP - 279.92 M (JNU HOSTEL)
In circling area and at AD	OTHER	283205.9N 0770646.4E	865 FT	LGTD	STATUE OBSTRUCTION LIGHT TOP-263.77 M, HEAD TOP - 262.28 M, BOTTOM - 243.58 M (SHIV MURTI STATUE)
In circling area and at AD	WALL	283130.2N 0770817.9E	956 FT	NO	BUILDING, PARAPET WALL TOP-291.27 M, BUILDING TOP-290.03 M (B-97)
In circling area and at AD	OTHER	283127.9N 0770820.8E	955 FT	NO	BUILDING, GENERATOR TOP-291.07 M, BUILDING TOP-287.50 M (B-7)
In circling area and at AD	OTHER	283129.7N 0770823.3E	954 FT	NO	BUILDING, PIPE TOP- 290.84 M, BUILDING TOP-286.17 M (C-36)
In circling area and at AD	ANTENNA	283131.7N 0770809.3E	962 FT	NO	ANTENNA ON BUILDING-293.06 M, SINTEX TANK TOP-290.99 M, STAIRCASE ROOM TOP-289.41 M, BUILDING TOP-287.74 M (B-53)
In circling area and at AD	OTHER	283131.4N 0770809.1E	958 FT	NO	BUILDING, PIPE TOP- 292.04 M,STAIRCASE ROOM TOP-290.54 M, BUILDING TOP-287.35 M (B-56)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	283131.1N 0770808.3E	952 FT	NO	BUILDING, PIPE TOP-290.1 M, STAIRCASE ROOM TOP-288.77 M, BUILDING TOP-287.12 M (B-96)
In circling area and at AD	OTHER	283132.2N 0770808.2E	953 FT	NO	BUILDING, PIPE TOP-290.60 M, STAIRCASE ROOM TOP-290.43 M, BUILDING TOP-287.24 M (B-99)
In circling area and at AD	OTHER	283134.8N 0770807.5E	954 FT	NO	BUILDING, PIPE TOP-290.87 M, STAIRCASE ROOM TOP-289.50 M, BUILDING TOP-286.96 M (B-215)
In circling area and at AD	OTHER	283136.0N 0770810.8E	951 FT	NO	BUILDING, PIPE TOP-289.91 M, STAIRACSE ROOM TOP-289.02 M, BUILDING TOP-286.80 M (A-156)
In circling area and at AD	OTHER	283134.5N 0770811.8E	951 FT	NO	BUILDING, PIPE TOP-289.93 M, STAIRACSE ROOM TOP-289.08 M, BUILDING TOP-286.75 M (A-186)
In circling area and at AD	OTHER	283133.0N 0770812.0E	953 FT	NO	BUILDING, PIPE TOP-290.59 M, STAIRCASE ROOM TOP-289.36 M, BUILDING TOP-287.40 M (A-85)
In circling area and at AD	ANTENNA	283131.4N 0770814.4E	956 FT	NO	BUILDING COMMUNICATION ANTENNA TOP-291.3 M, BUILDING TOP-287.3 M (B-65)
In circling area and at AD	OTHER	283131.2N 0770805.5E	948 FT	NO	BUILDING, PIPE TOP-288.81 M, STAIRCASE ROOM TOP-287.24 M, BUILDING TOP-284.96 M (B-92)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	283127.8N 0770810.6E	946 FT	NO	BUILDING, PIPE TOP- 288.18 M, BUILDING TOP- 287.04 M (B-413)
In circling area and at AD	WALL	283136.4N 0770822.1E	951 FT	NO	BUILDING PARAPET WALL TOP - 289.81 M, BUILDING TOP - 287.18 M (D9)
In circling area and at AD	ANTENNA	283151.4N 0770857.1E	935 FT	NO	BUILDING ANTENNA - 284.94 M, CENTER FOR SOCIAL RESEARCH BUILDING TOP - 278.38 M (AXIS BANK)
In circling area and at AD	ANTENNA	283127.6N 0770801.0E	947 FT	NO	BUILDING DISH ANTENNA TOP - 288.72 M, SINTEX TANK - 288.39 M, BUILDING TOP - 287.77 M (CH. PRATHIP SINGH)
In circling area and at AD	ANTENNA	283132.4N 0770802.2E	943 FT	NO	BUILDING DISH ANTENNA TOP - 287.52 M, SINTEX TANK - 286.53 M, BUILDING TOP - 284.03 M (B 368)
In circling area and at AD	OTHER	283127.7N 0770809.8E	957 FT	NO	BUILDING, PIPE TOP- 291.69 M, STAIRACASE ROOM TOP-289.08 M, BUILDING TOP- 287.25 M (B-616)
In circling area and at AD	OTHER	283139.0N 0770811.1E	944 FT	LGTD	BUILDING, LIGHTNING ARRESTOR TOP - 287.62 M, SOLAR PANEL TOP - 286.62 M, BUILDING TOP - 283.85 M (B 30)
In circling area and at AD	OTHER	283134.8N 0770812.7E	948 FT	NO	BUILDING, PIPE TOP - 289.10 M, BUILDING TOP - 288.33 M (A 95)
In circling area and at AD	ANTENNA	283133.4N 0770815.9E	961 FT	NO	BUILDING DISH ANTENNA TOP - 292.91 M, STAIRCASE ROOM - 291.86 M, BUILDING TOP - 290.29 M (A 76)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	ANTENNA	283128.4N 0770814.7E	957 FT	NO	BUILDING COMMUNICATION ANTENNA - 291.58 M, DISH ANTENNA - 290.77 M, BUILDING TOP - 290.02 M (B 82)
In circling area and at AD	OTHER	283228.1N 0770930.9E	939 FT	LGTD	BUILDING OBSTRUCTION LIGHT TOP - 286.21 M, SINTEX TANK TOP - 285.06 M, BUILDING TOP - 284.47 M (JNU HOSTEL)
In circling area and at AD	ANTENNA	283035.3N 0770532.2E	932 FT	NO	ANTENNA ON BUILDING - 284.19 M, BUILDING TOP- 280.79 M (COUNTRY INN and SUITES)
In circling area and at AD	OTHER	283019.5N 0770545.3E	953 FT	NO	BUILDING HOARDING TOP-290.45 M, STAIRCASE ROOM TOP-281.60 M, BUILDING TOP-278.47 M (AMBIENCE MALL BUILDING, THE LEELA)
In circling area and at AD	OTHER	283218.1N 0770659.8E	825 FT	NO	BUILDING IRON ROD TOP - 251.54 M, BUILDING TOP - 249.75 M. (NIKE FACTORY STORE)
In circling area and at AD	OTHER	283219.6N 0770702.8E	848 FT	NO	BUILDING, HOARDING TOP - 258.58 M, STAIRCASE ROOM TOP - 255.78 M, BUILDING TOP - 253.93 M.
In circling area and at AD	ANTENNA	283218.7N 0770704.4E	854 FT	NO	ANTENNA ON BUILDING - 260.27 M, ANTENNA TOP - 259.70 M, BUILDING TOP - 257.94 M.

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	283136.4N 0770802.3E	964 FT	NO	BUILDING, PIPE TOP - 293.70 M. SINTEX TANK TOP - 292.95 M, STAIRCASE TOP - 292.42 M, BUILDING TOP - 287.12 M
In circling area and at AD	ANTENNA	283007.7N 0770555.5E	990 FT	LGTD	ANTENNA ON BUILDING - 301.56 M, LIGHTNING ARRESTER TOP-296.81 M, SINTEX TANK TOP-296.36 M, STAIRCASE ROOM TOP-296.26 M, BUILDING TOP-288.01 M (AMBIENCE TOWER)
In circling area and at AD	STACK	282952.6N 0770524.9E	1075 FT	NO	BUILDING, CHIMNEY TOP-327.6 M. BUILDING TOP-311.60 M (DLF)
In circling area and at AD	OTHER	283402.2N 0770530.6E	765 FT	NO	L/C ROD NEAR DVOR MONITOR ANTENNA
In circling area and at AD	ANTENNA	283400.6N 0770522.6E	732 FT	NO	ANTENNA TOP
In circling area and at AD	OTHER	283245.9N 0770355.8E	728 FT	NO	LIGHTNING ARRESTER
In circling area and at AD	ANTENNA	283329.7N 0770709.3E	788 FT	NO	WIND SPEED INDICATOR ANTENNA, AWS ANTENNA
In circling area and at AD	OTHER	283330.1N 0770709.1E	782 FT	NO	L/C ROD NEAR RVR ANTENNA
In circling area and at AD	ANTENNA	283330.3N 0770709.5E	777 FT	NO	RVR ANTENNA
In circling area and at AD	ANTENNA	283330.6N 0770708.4E	777 FT	NO	RVR ANTENNA
In circling area and at AD	ANTENNA	283330.9N 0770706.8E	775 FT	NO	RVR ANTENNA
In circling area and at AD	OTHER	283334.2N 0770700.9E	766 FT	NO	PAPI LIGHT TOP
In circling area and at AD	ANTENNA	283408.5N 0770706.8E	768 FT	NO	ANTENNA NEAR LOCALIZER HUT
In circling area and at AD	ANTENNA	283414.8N 0770647.0E	756 FT	NO	RVR ANTENNA
In circling area and at AD	OTHER	283333.6N 0770724.3E	782 FT	NO	ELECTRICAL JUNCTION BOX
In circling area and at AD	ANTENNA	283330.4N 0770708.3E	777 FT	NO	RVR ANTENNA
In circling area and at AD	ANTENNA	283208.1N 0770634.3E	821 FT	NO	ANTENNA TOP

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	ANTENNA	283209.5N 0770631.5E	809 FT	NO	ANTENNA TOP
In circling area and at AD	OTHER	283214.9N 0770620.8E	783 FT	NO	LIGHTNING ARRESTOR
In circling area and at AD	ANTENNA	283240.1N 0770525.2E	772 FT	LGTD	MET PAR ANTENNA (LGT)
In circling area and at AD	ANTENNA	283227.1N 0770525.2E	749 FT	NO	RVR ANTENNA
In circling area and at AD	ANTENNA	283227.6N 0770523.1E	748 FT	NO	RVR ANTENNA
In circling area and at AD	ANTENNA	283227.7N 0770522.5E	748 FT	NO	RVR ANTENNA
In circling area and at AD	ANTENNA	283232.5N 0770500.0E	739 FT	NO	ANTENNA TOP
In circling area and at AD	ANTENNA	283233.0N 0770457.8E	738 FT	NO	ANTENNA TOP
In circling area and at AD	ANTENNA	283233.1N 0770457.3E	738 FT	NO	ANTENNA TOP
In circling area and at AD	ANTENNA	283226.9N 0770525.1E	748 FT	NO	RVR ANTENNA
In circling area and at AD	ANTENNA	283227.1N 0770524.1E	748 FT	NO	RVR ANTENNA
In circling area and at AD	OTHER	283232.6N 0770458.5E	737 FT	NO	TRANSMISSOMETER
In circling area and at AD	ANTENNA	283246.0N 0770356.0E	723 FT	NO	RVR ANTENNA
In circling area and at AD	ANTENNA	283246.2N 0770355.0E	723 FT	NO	RVR ANTENNA
In circling area and at AD	ANTENNA	283238.5N 0770432.1E	732 FT	NO	RVR ANTENNA
In circling area and at AD	ANTENNA	283238.4N 0770432.6E	732 FT	NO	RVR ANTENNA
In circling area and at AD	ANTENNA	283237.9N 0770433.6E	730 FT	NO	RVR ANTENNA
In circling area and at AD	ANTENNA	283237.9N 0770434.8E	732 FT	NO	RVR ANTENNA
In circling area and at AD	OTHER	283237.7N 0770434.7E	731 FT	NO	TRANSMISSOMETER
In circling area and at AD	ANTENNA	283226.1N 0770524.9E	765 FT	NO	AWS ANTENNA
In circling area and at AD	ANTENNA	283231.6N 0770459.7E	752 FT	NO	AWS ANTENNA
In circling area and at AD	ANTENNA	283237.0N 0770434.5E	748 FT	NO	AWS ANTENNA
In circling area and at AD	ANTENNA	283329.4N 0770709.2E	775 FT	NO	MET ANTENNA
In circling area and at AD	ANTENNA	283330.2N 0770709.4E	777 FT	NO	RVR ANTENNA
In circling area and at AD	OTHER	283346.4N 0770629.5E	764 FT	NO	LIGHTENING ARRESTER
In circling area and at AD	OTHER	283346.6N 0770629.4E	758 FT	NO	TRANSMISSOMETER

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	283346.4N 0770629.4E	758 FT	NO	TRANSMISSOME TER
In circling area and at AD	ANTENNA	283346.8N 0770628.3E	758 FT	NO	RVR ANTENNA
In circling area and at AD	OTHER	283346.7N 0770628.3E	758 FT	NO	TRANSMISSOME TER
In circling area and at AD	OTHER	283215.0N 0770621.7E	778 FT	NO	TRANSMISSOME TER
In circling area and at AD	OTHER	283215.2N 0770620.7E	778 FT	NO	TRANSMISSOME TER
In circling area and at AD	OTHER	283226.7N 0770524.6E	755 FT	NO	LIGHTNING ARRESTOR
In circling area and at AD	OTHER	283232.2N 0770459.1E	744 FT	NO	LIGHTNING ARRESTOR
In circling area and at AD	OTHER	283237.6N 0770434.0E	739 FT	NO	LIGHTNING ARRESTER
In circling area and at AD	ANTENNA	283346.6N 0770629.4E	756 FT	NO	ANTENNA TOP
In circling area and at AD	OTHER	283213.6N 0770857.3E	930 FT	NO	CHIMNEY
In circling area and at AD	OTHER	283235.6N 0770435.8E	748 FT	NO	L/C ROD TOP
In circling area and at AD	OTHER	283330.5N 0770722.4E	780 FT	NO	APPROACH LIGHT
In circling area and at AD	OTHER	283225.8N 0770508.6E	814 FT	NO	L/C ROD ON ASR RADAR PLATFORM
In circling area and at AD	OTHER	283303.8N 0771922.5E	1582 FT	LGTD	BUILDING CRAIN TOP (BUILDING LGT TOP 478.5 M.) (SUPERTECH SUPERNOVA)
In circling area and at AD	OTHER	283248.7N 0770420.7E	747 FT	NO	WDI
In circling area and at AD	TANK	283151.4N 0770903.8E	945 FT	NO	BUILDING, SINTEX TOP - 287.9 M, BUILDING TOP - 286.8 M

Remarks: Pitampura TV Tower Obstruction LGT Available.

VIDP AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

1	Name of the associated meteorological office	Meteorological Watch Office (MWO), IGI Airport, New Delhi
2	Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours	H24
3	Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts	MWO Delhi 9 and 30 HR (3 & 6 hours interval)
4	Availability of the trend forecast for the aerodrome and interval of issuance	Trend 30 Min (valid for next 2 hours)

5	Information on how briefing and/or consultation is provided	Physical and/or through on-line mode.
6	Types of flight documentation supplied and language(s) used in flight documentation	Chart, Image, Text form and Tabular form. English
7	Charts and other information displayed or available for briefing or consultation	S,U85,U70,U50,U30,U25,U15,U10, P30, P25, P20 SW [UPTO FL460]
8	Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images;	Radar and Satellite display work Station
9	The air traffic services unit(s) provided with meteorological information	VIDP Delhi ATC AND ACS
10	Additional information, e.g. concerning any limitation of service.	**1.** Integrated Aviation METROLOGICAL System (DCWIS)*at RWY 27/09, 28/10, 29L/11R and 29R/11L AVBL. The System will provide following information on Digital Display: **PAGE 1:** I Wind Direction- Instantaneous, 2 Minute Average and 10 Minute Average II Wind Speed - Instantaneous, 2 Minute Average and 10 Minute Average **PAGE 2:** I Air Temperature: Actual Dry Bulb Temperature II Humidity: Actual Humidity in Percent III DEW Point Temperature: In Degree Celsius **PAGE 3:** I Air Pressure: in HPA II QNH: in HPA. *DCWIS is available at both side of touchdown zone. **2.** The RVR system at RWY 27/09, 28/10, 29L/11R and 29R/11L will provide RVR and MOR data displayed on PC monitor as follows: I.) RVR: Actual reading displayed from 50m to 2000m in following steps: • below 400M in step of 25M • 400M to 800M in step of 50M • above 800M in step of 100M A value of 2000M is displayed when RVR is equal to or above 2000M. A value of 25M is displayed when RVR is less than 50M. II.) RVR TREND: Displayed upward arrow for uptrend and downward arrow for down trend. ---U for up, D for down, N for no change. **3**. Ceilometers installed at RWY 10 TDZ with Data Updation Frequency 30 Sec. The Display Location is the Met Officer Position at ATC Tower.

VIDP AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

Designations	TRUE Bearings	Dimensions of RWY (M)	Strength of pavement (PCR) and associated data) and surface of runway and associated stopways	Geographical coordinates for threshold and runway end
1	2	3	4	5
10	104.25 DEG	3813 x 45 M	862/F/A/W/T Asphalt	THR: 283402.03N 0770505.54E RWY END: 283330.73N 0770721.27E
28	284.25 DEG	3813 x 45 M	862/F/A/W/T Asphalt	THR: 283330.73N 0770721.27E RWY END: 283402.03N 0770505.54E
11L	103.65 DEG	x M	835/F/B/W/T Asphalt	THR: 283255.30N 0770429.38E RWY END: 283232.30N 0770616.62E
29R	283.65 DEG	x M	835/F/B/W/T Asphalt	THR: 283232.30N 0770616.62E RWY END: 283258.67N 0770413.65E
11R	103.25 DEG		862/F/A/W/T	THR: 283244.86N 0770418.85E RWY END: 283218.30N 0770622.74E
29L	283.25 DEG		862/F/A/W/T	THR: 283227.04N 0770541.97E RWY END: 283249.81N 0770355.80E
09	91.25 DEG	2816 x 45 M	943/F/B/W/T Asphalt	THR: 283413.98N 0770517.28E RWY END: 283411.62N 0770700.84E
27	271.25 DEG	2816 x 45 M	943/F/B/W/T Asphalt	THR: 283411.74N 0770655.30E RWY END: 283413.98N 0770517.28E

THR elevation and highest elevation of TDZ of precision APP RWY	Slope of runway and associated stopway	Dimensions of stopway (M)	Dimensions of clearway (M)	Dimensions of strips (M)
6	7	8	9	10
THR: 720.5FT TDZ: 725.0FT	0.46%	NIL	NIL	3933 x 280 M
THR: 778.3FT TDZ: 775.0FT	-0.46%	NIL	NIL	3933 x 280 M
THR: 726.0FT TDZ: 736.0FT	0.40%	730 x 45 M	NIL	4520 x 280 M
THR: 772.0FT TDZ: 772.0FT	-0.40%	230 x 45 M	NIL	4520 x 280 M
THR: 725.0FT TDZ: 727.0FT	0.39%	320 x 60 M	NIL	4550 x 280 M
THR: 752.2FT TDZ: 750.0FT	-0.38%	NIL	NIL	4550 x 280 M
THR: 718.8FT TDZ: 731.0FT	0.39%	NIL	NIL	2936 x 150 M
THR: 752.4FT TDZ: 752.0FT	-0.39%	NIL	NIL	2936 x 150 M

Dimensions of runway end safety areas	Location and description of arresting system (if any)	Existence of an obstacle-free zone	Remarks.
11	12	13	14
240M x 90M		AVBL	1. PCR: (0 to 98M) 1175/R/A/W/T CONCRETE (98M TO 3661M) 862/F/A/W/T ASPHALT (3661M TO 3813M) 1175/R/A/W/T CONCRETE 2. 1% (Transverse slope)
240M x 90M		AVBL	1. PCR: (0 TO 152M) 1175/R/A/W/T CONCRETE (152M TO 3715M) 862/F/A/W/T ASPHALT (3715M TO 3813M) 1175/R/A/W/T CONCRETE 2. 1% (Transverse slope)
240M x 150M		NIL	1. Dimensions of RWY: 3670 M X 45 M 2. PCR: 0 to 300M - 1506/R/B/W/T (Concrete) 300M to 4100M - 835/F/B/W/T (Asphalt) 4100M to 4400M - 1506/R/B/W/T (Concrete)
240M x 150M		NIL	1. Dimensions of RWY: 4170 M X 45 M 2. PCR: 0 to 300M - 1506/R/B/W/T (Concrete) 300M to 4100M -835/F/B/W/T (Asphalt) 4100M to 4400M - 1506/R/B/W/T (Concrete)
240M x 120M		AVBL	RWY Dimension: 4110 M X 60 M PCR: 0 to 438M: 1175/R/A/W/T (Concrete) 438M to 4166M: 862/F/A/W/T (Flexible) 4166M to 4430M:1175/R/A/W/T (Concrete)

Dimensions of runway end safety areas	Location and description of arresting system (if any)	Existence of an obstacle-free zone	Remarks.
11	12	13	14
240M x 120M		AVBL	RWY Dimension:4430 M X 60 M PCR: 0 to 264M: 1175/R/A/W/T (Concrete) 264M to 3992M:862/F/A/W/T (Flexible) 3992M to 4430M: 1175/R/A/W/T (Concrete)
240M x 90M		AVBL	PCR: 0 to 520M: 1506/R/B/W/T (Concrete) 520M to 2816M: 943/F/B/W/T (Asphalt)
240M x 90M		AVBL	PCR: 0 to 2296M: 943/F/B/W/T (Asphalt) 2296M to 2816M: 1506/R/B/W/T (Concrete)

VIDP AD 2.13 DECLARED DISTANCES

RWY Designator	Take-off run available TORA (M)	Take-off distance available TODA (M)	Accelerate distance available ASDA (M)	Landing distance available LDA (M)	Remarks (including runway entry or start point where alternative reduced declared distances have been declared)
1	2	3	4	5	6
10	3813	3813	3813	3813	
28	3813	3813	3813	3813	
11L	3670	3670	4400	3000	THR displaced by 670M.
29R	4170	4170	4400	3440	THR displaced by 730M.
11R	4110	4110	4430	3465	THR displaced by 645M
29L	4430	4430	4430	2970	THR displaced by 1460M
09	2816	2816	2816	2816	
27	2816	2816	2816	2665	THR displaced by 151M

VIDP AD 2.14 APPROACH AND RUNWAY LIGHTING

Runway Designator	Type, length and intensity of approach lighting system	Runway threshold lights, colour and wing bars	Type of visual slope indicator system	Length of runway touchdown zone lights
1	2	3	4	5
10	CAT III 900 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (68.18FT)	900 M
28	900 M CATIII LIH	Green	PAPI LEFT/3.00 DEG MEHT (76.48FT)	900 M
11L	CAT III 900 M LIH	Green	PAPI BOTH/3.00 DEG MEHT (64.19FT)	900 M

Runway Designator	Type, length and intensity of approach lighting system	Runway threshold lights, colour and wing bars	Type of visual slope indicator system	Length of runway touchdown zone lights
1	**2**	**3**	**4**	**5**
29R	CAT III 900 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (64.19FT)	900 M
11R	900 M CATIII LIH	Green	PAPI BOTH/3.00 DEG MEHT (69.95FT)	900 M
29L	900 M CATIII LIH	Green	PAPI LEFT/3.00 DEG MEHT (69.19FT)	900 M
09	SALS 420 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (73.93FT)	
27	CAT I 570 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (73.93FT)	

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
3813 M 15 M LIH	3813 M 60 M LIH	Red		1. RWY Centreline lights from physical beginning upto 900M from RWY End - variable White; between 900M (from RWY END) and 300M from RWY END – alternate variable White and Red between 300M (from RWY END) and RWY END - Red 2. RWY Edge Lights from physical beginning upto 600M (from RWY END) – variable White; between 600M (from RWY END) and RWY END – Yellow. 3. The portion of RWY 10 CAT III approach lights between450m to 785.5m from RWY 10 THR are spaced at 27.5m except the 22nd light which is placed at 58m beyond railway line . However, approach lights upto 600m from RWY 10 THR having crossbars at 150, 300, 450 and 600m are serviceable. 4. CAT III approach lights from RWY 10 threshold up to 450m and between 785.5m to 900m from RWY 10 threshold are evenly spaced i.e. at30m. 5.Rapid ExitTaxiway Indicator Lights (RETILS) installed at RWY 10 for TWY G1,TWY G2 . TWY J1, TWY J2 and TWY J3.

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
3813 M 15 M LIH	3813 M 60 M LIH	Red		1. RWY Centreline lights from physical beginning upto 900M from RWY End - variable White; between 900M (from RWY END) and 300M from RWY END – alternate variable White and Red between 300M (from RWY END) and RWY END – Red. 2. The centreline lights are uniformly spaced at 30M in CAT I conditions and 15 M in CAT II/III conditions. 3. RWY Edge Lights from physical beginning upto 600M (from RWY END) – variable White; between 600M (from RWY END) and RWY END – Yellow. 4.Rapid Exit Taxiway Indicator Lights (RETILS) installed at RWY 28 for TWY H1, K1and K2.

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
3670 M 15 M LIH	3670 M 60 M LIH	Red	Red 730 M	1. Due to displaced THR, the aeronautical ground LGT arrangement under mixed mode OPS on RWY11L in PRE THR area is a combination of single sourced RWY centreline LGT and Calvert Type APCH LGT. 2. Displaced threshold of runway 11L provided with flashing white RWY threshold identification lights at a distance of 670 M from physical extremity of RWY 11L. 3. RWY Centreline lights from physical beginning up to 900M from RWY End – variable White; between 900M (from RWY END) and 300M (from RWY END) – alternate variable White and Red between 300M (from RWY END) and RWY END – Red. 4. RWY Edge Lights from physical beginning up to 670M (Displaced threshold) – Red; between 670M (from physical beginning) and 600M (from RWY END) – variable White; between 600M (from RWY END) and RWY END – Yellow; beyond RWY END for 730M long Stopway – Red. 5. Rapid Exit Taxiways Indicator Lights (RETILS) installed at RWY 11L for TWY P1, P3.

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
4170 M 15 M LIH	4170 M 60 M LIH	Red	Red 230 M	1. Due to displaced THR, the aeronautical ground LGT arrangement under mixed mode OPS on RWY29R in PRE THR area is a combination of single sourced RWY centreline LGT and Calvert Type APCH LGT. 2. Displaced threshold of runway 29R provided with flashing white RWY threshold identification lights at a distance of 730 M from physical extremity of RWY 29R. 3. RWY Centreline lights from physical beginning up to 900M from RWY End – variable White; between 900M (from RWY END) and 300M (from RWY END) – alternate variable White and Red between 300M (from RWY END) and RWY END – Red. 4. RWY Edge Lights from physical beginning up to 730M (Displaced threshold) – Red; between 730M (from physical beginning) and 600M (from RWY END) – variable White; between 600M (from RWY END) and RWY END – Yellow;beyond RWY END for 230M long Stopway – Red 5. Rapid Exit Taxiways Indicator Lights (RETILS) installed at RWY 29R for TWY R2, R3.

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
4110 M 15 M LIH	4110 M 60 M LIH	Red	Red 320 M	1. RWY Centerline lights from physical beginning upto 900M from RWY End - variable White; between 900M (from RWY END) and 300M from RWY END – alternate variable White and Red between 300M (from RWY END) and RWY END – Red. 2. RWY Edge Lights from physical beginning upto 645M (Displaced threshold) – Red; between 645M (from physical beginning) and 600M (from RWY END) – variable White; between 600M (from RWY END) and RWY END – Yellow;beyond RWY END for 320M long Stopway – Red. 3. Rapid Exit Taxiways Indicator Lights (RETILS) installed at RWY 11R for TWY Y1, Y2 and Y4. 4. 'Displaced threshold of RWY 11R provided with flashing white RWY threshold identification lights at a distance of 645 M from physical extremity of RWY 11R.

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
4430 M 15 M LIH	4430 M 60 M LIH	Red		1. Due to displaced THR, the aeronautical ground LGT arrangement under mixed mode OPS on RWY29L in PRE THR area is as follows: i. Physical beginning of RWY29L to 560M - single sourced RWY center line LGT. ii. FM 560M till THR RWY29L - Combination of single sourced RWY center line LGT and Calvert Type APCH LGT. 2. 'Displaced threshold of runway 29L provided with flashing white RWY threshold identification lights at a distance of 1460 M from physical extremity of RWY 29L. 3. RWY Centerline lights from physical beginning upto 900M from RWY End - variable White; between 900M (from RWY END) and 300M from RWY END – alternate variable White and Red between 300M (from RWY END) and RWY END – Red. 4. RWY Edge Lights from physical beginning upto 1460M (Displaced threshold) – Red; between 1460M (from physical beginning) and 600M (from RWY END) – variable White; between 600M (from RWY END) and RWY END – Yellow. 5. Rapid Exit Taxiways Indicator Lights (RETILS) installed at RWY 29L for TWY Z2, Z3.

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
2816 M 15 M LIH	2816 M 60 M LIH	Red		1. RWY Centrelinelights from physical beginning upto 900M from RWY End - variable White; between 900M (from RWY END) and 300M from RWY END – alternate variable White and Red between 300M (from RWY END) and RWY END - Red 2. RWY Edge Lights from physical beginning upto 600M (from RWY END) – variable White; between 600M (from RWY END) and RWY END – Yellow 3. RWY centre line LGT are uniformly offset to the North side of RWY centerline by not more than 60 cm.

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
2816 M 15 M LIH	2816 M 60 M LIH	Red		1. RWY Centerline lights from physical beginning upto 151M (Displaced Threshold) – variable white; between 151M (from physical beginning) and 900M from RWY END – variable White between 900M (from RWY END) and 300M from RWY END – alternate variable White and Red between 300M (from RWY End) and RWY END – Red. 2. RWY Edge Lights from physical beginning upto 151M (Displaced threshold) – Red; between 151M (from physical beginning) and 600M (from RWY END) – variable White; between 600M (from RWY END) and RWY END – Yellow. 3. RWY centre line LGT are uniformly offset to the North side of RWY centerline by not more than 60 cm. 4. Displaced threshold of RWY 27 provided with flashing white RWY Threshold Identification Lights at a distance of 151M from physical extremity of RWY27. 5. Rapid Exit Taxiways Indicator Lights (RETILS) installed at RWY 27 for TWY D5, D6.

VIDP AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

1	Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any)	ABN	283337N 0770536E, Rotating green /white beacon on top of Apron Management service tower (Apron- II). FLG W AND G, 24FPM, 400,000 Cd, SUNSET TO SUNRISE
		IBN	NIL

2	Location and lighting (if any) of anemometer/ landing direction indicator;	LDI	NIL
		Anemometer	Sl. No. Location Coordinates i North of RWY 09, near GP 09 283417.5N 0770529.6E ii North of RWY 27 and East of TWY A2 283416.4N 0770647.3E iii North of RWY 10, near GP 10 283403.2N 0770518.9E iv South of RWY 28 between TWY J5& J7 283329.7N 0770709.3E v South of RWY 11R near GP 11R 283237.1N 0770434.0E vi South of RWY 29L near GP 29L 283226.3N 0770524.5E vii South of RWY 29R, east of TWY S2 283232.9N 0770558.8E viii South of RWY 29R, between TWY S1 and TWY S2 283240.1N 0770525.2E ix South of RWY 11L, near GP RWY 11L 283249.2N 0770443.1E
3	Taxiway edge and taxiway centre line lights;	Edge	All TWY curves except TWY LINK 16 (Between TWY C up to GA Apron), TWY LINK 13 which have reflective edge markers.
		Centre Line	All Taxiways except portion of TWY LINK 32 south of TWY C, Portion of TWY LINK 30 south of TWY M, LINK 16 (Between LINK 14 and TWY C), LINK 14, LINK 15 and LINK 13.
4	Secondary power supply including switch-over time;	SECONDARY POWER SUPPLY TO ALL LIGHTING. SWITCH OVER TIME : CAT- III : 1 SEC CAT- II : 1 SEC CAT- I : 15 SEC ONLINE UPS AVBL FOR CRITICAL CIRCUITS.	
5	Remarks	CAT-III BSES supply through online UPS as primary source and DG supply as secondary source.	

VIDP AD 2.16 HELICOPTER LANDING AREA

1	Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area	Not Established
2	TLOF and/or FATO area elevation:	Not Established
3	TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking;	Not Established
4	True bearings of FATO;	Not Established
5	Declared distances available	Not Established
6	Approach and FATO lighting;	Not Established
7	Remarks	Not Established

VIDP AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

1	Airspace designation, geographical coordinates and lateral limits	CTR: Area bounded by 290200.0N 0771455.4E, then along counterclockwise arc of circle of radius 30NM centred at 283400.89N 0770536.66E to 283600.2N 0773855.2E to 283700.2N 0772955.3E to 283700.2N 0771555.4E, and then along river Yamuna to 285600.0N 0771155.4E to 290200.0N 0771455.4E excluding airspace of Jewar CTR.
2	Vertical limits	FL 75
3	Airspace classification	D
4	Call sign and language(s) of the air traffic services unit providing service;	Delhi Tower, English
5	Transition altitude	5000 FT
6	Hours of applicability	H24
7	Remarks	1. Airspace above 4000 FT AMSL is Classified as Class ‘C’. 2. ATC Frequencies: Delhi Approach/ Delhi RADAR: 126.350 MHz 124.200 MHz 118.825 MHz

VIDP AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

Service Designation	Call sign	Channel(s)	SATVOICE Number(s), if available
1	2	3	4
OTHER	CLEARANCE DELIVERY	121.800 MHZ	
OTHER	CLEARANCE DELIVERY	121.850 MHZ	
OTHER	CLEARANCE DELIVERY	121.950 MHZ	
OTHER	DELHI APPROACH	121.350 MHZ	
OTHER	DELHI APPROACH	125.675 MHZ	
OTHER	DELHI APPROACH	126.350 MHZ	
OTHER	Delhi Arrival	124.200 MHZ	
OTHER	Delhi Arrival	124.250 MHZ	
OTHER	Delhi Departure	118.825 MHZ	
OTHER	Delhi Departure	124.600 MHZ	
ATFM	Delhi flow control	132.500 MHZ	
SAR	-	123.100 MHZ	
TWR	Delhi Tower	118.100 MHZ	
TWR	Delhi Tower	118.250 MHZ	
TWR	Delhi Tower	118.750 MHZ	
TWR	Delhi Tower	123.825 MHZ	
TWR	Delhi Tower	124.375 MHZ	
TWR	Delhi Tower	125.850 MHZ	
ATIS	Delhi Information	126.400 MHZ	
ACC	ACC CENTRAL	132.775 MHZ	
ACC	DELHI CONTROL	119.500 MHZ	
ACC	DELHI CONTROL	120.900 MHZ	
ACC	DELHI CONTROL	124.550 MHZ	
ACC	DELHI CONTROL	125.700 MHZ	
ACC	DELHI CONTROL	125.950 MHZ	
ACC	DELHI CONTROL	132.150 MHZ	
ACC	DELHI CONTROL	132.850 MHZ	
ACC	DELHI CONTROL	132.975 MHZ	
ACC	DELHI CONTROL	133.900 MHZ	
ACC	DELHI CONTROL	134.075 MHZ	
ACC	DELHI CONTROL	134.500 MHZ	
ALRS	Emergency Frequency	121.500 MHZ	
SMC	Delhi Ground	118.550 MHZ	
SMC	Delhi Ground	119.075 MHZ	
SMC	Delhi Ground	119.425 MHZ	
SMC	Delhi Ground	119.575 MHZ	
SMC	Delhi Ground	119.675 MHZ	
SMC	Delhi Ground	121.625 MHZ	
SMC	Delhi Ground	121.750 MHZ	
SMC	Delhi Ground	121.900 MHZ	

Logon address, as appropriate	Hours of operation	Remarks
5	6	7
	H24	Clearance Delivery SDBY
	H24	Clearance Delivery 2
	H24	Clearance Delivery 1
	H24	Delhi Approach SDBY
	H24	Delhi Approach
	H24	DELHI APPROACH
	H24	Delhi Arrival
	H24	Delhi Arrival SDBY
	H24	Delhi Departure
	H24	Delhi Departure
	H24	Delhi Flow Control
	H24	SEARCH AND RESCUE
	H24	TWR MIDDLE
	H24	TWR NORTH SDBY
	H24	TWR NORTH
	H24	TWR SOUTH-2
	H24	TWR MIDDLE AND TWR SDBY
	H24	TWR SOUTH
	H24	ATIS
	H24	ACC CENTRAL
	H24	ACC NORTH EAST
	H24	ACC NORTH EAST
	H24	ACC NORTH WEST
	H24	ACC SOUTH EAST
	H24	ACC SOUTH EAST
	H24	ACC SOUTH WEST
	H24	ACC SOUTH WEST SDBY
	H24	ACC NORTH WEST
	H24	ACC SOUTH EAST SDBY
	H24	ACC NORTH WEST SDBY
	H24	NORTH EAST SDBY
	H24	Emergency
	H24	1. SMC South-2 2. Main Frequency
	H24	1. SMC North-2 2. Main Frequency.
	H24	1. SMC Southeast 2. Main Frequency
	H24	1. SMC West 2. Main Frequency
	H24	1. Combined Backup frequency for SMC Control.
	H24	1. SMC South-1 2. Main Frequency.

Logon address, as appropriate	Hours of operation	Remarks
	H24	1. SMC North 2. Main Frequency
	H24	1. SMC Middle 2. Main Frequency

VIDP AD 2.19 RADIO NAVIGATION AND LANDING AIDS

Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid	Identification	Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate	Hours of operation, as appropriate;
1	2	3	4
LOC 10 CAT III	IDEL	109.500 MHz	H24
LOC 11R CAT III	IDMR	111.300 MHz	H24
LOC 29L CAT III	IDGM	110.900 MHz	H24
LOC 28 CAT III	IPLM	110.300 MHz	H24
LOC 27 CAT I	IDLH	110.500 MHz	H24
LOC 09 CAT I	IDIA	108.500 MHz	H24
LOC 11L CAT III	IDLP	108.700 MHz	H24
LOC 29R CAT III	IDPN	111.100 MHz	H24
GP 27		329.600 MHz	H24
GP 10	IDEL	332.600 MHz	H24
GP 29L	IDGM	330.800 MHz	
GP 11R	IDMR	332.300 MHz	H24
GP 28	IPLM	335.000 MHz	H24
GP 09	IDIA	329.900 MHz	H24
GP 11L	IDLP	330.500 MHz	H24
GP 29R		331.700 MHz	H24
DME ILS 10	IDEL	CH32X	H24
DME ILS 28	IPLM	CH40X	H24
DME ILS 27	IDLH	CH42X	H24
DME ILS 09	IDIA	CH22X	H24
DME ILS 11R	IDMR	CH58X	H24
DME ILS 29L	IDGM	CH46X	H24
DME ILS 11L	IDLP	CH24X	H24
DME ILS 29R	IDPN	CH48X	H24
DVOR/DME	DPN	116.100 MHz CH108X	H24

Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid	Identification	Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate	Hours of operation, as appropriate;
DVOR/DME	DIG	114.600 MHz CH93X	H24

Geographical coordinates of the position of the transmitting antenna	Elevation of transmitting antenna of DME/ elevation of GBAS reference point	Service volume radius from the GBAS reference point	Remarks
5	6	7	8
283328.0N 0770733.2E			CAT-I
283213.4N 0770645.4E			
283252.4N 0770343.7E			i. Localiser restricted to 26 DEG on 150 Hz side due off-course clearance below limits. ii.Off-course clearance on 90 Hz side is within limit.
283405.4N 0770450.8E			LOC Coverage restricted up to 22NM within plus minus 10 DEG sector and 17NM within plus minus 35 DEG at and above 2000 FT height above Threshold (HAT).
283414.5N 0770456.2E			
283411.5N 0770706.3E			
283224.3N 0770654.2E			
283302.9N 0770354.0E			
283408.5N 0770642.1E			
283403.2N 0770517.9E			3 DEG
283226.0N 0770527.9E			3 DEG
283238.5N 0770429.4E			3 DEG
283337.8N 0770708.5E			Glide Angle: 3 DEG
283417.7N 0770528.8E			
283249.0N 0770440.2E			
283231.6N 0770600.9E			GLIDE ANGLE 3 DEG
283403.4N 0770518.0E	746 FT		
283337.9N 0770708.5E	791 FT		
283408.5N 0770642.1E	760 FT		Collocated with GP27
283417.9N 0770528.9E	738 FT		Collocated with GP09
283238.4N 0770429.5E	742 FT		Collocated with GP11R

283226.1N 0770527.8E	760 FT		Collocated with GP29L
283248.9N 0770440.1E	748 FT		1. COLOCATED WITH GP. 2. GRAVITY MODEL: EGM-08.
283231.5N 0770600.9E	756 FT		colocated with GP 29R
283400.9N 0770536.7E	760 FT		
283231.9N 0770449.7E	760 FT		

VIDP AD 2.20 LOCAL AERODROME REGULATIONS

1. AIRPORT REGULATIONS

Note: The procedures mentioned in this para are applicable for all Aprons (Except GA Apron) subject to conditions/ restrictions given in subsequent para 2 of AD 2.20 TAXING PROCEDURES FOR DIFFERENT APRONS.

1.1 To reduce RT congestion, flight crew is encouraged not to request direct routings in Delhi TMA ATC will provide direct routings as and when feasible subject to traffic conditions

1.2 Startup of engines at idle power is permitted during pushback and on stands. However, startup of engines at idle power at aerobridge stands is permitted only after removal of aerobridge. Airlines/GHAs are responsible to adhere to all ground precautions e.g. Wing Walkers/Traffic Marshallers, Ground Equipment, Vehicular Traffic on tail of the stand road etc. against jet blast/ingestion to ensure safety. Indian registered Aircraft should not seek any permission from ATC, ASO & AOCC to carry startup of engine/idle power engine run up on stand. However, ATC may be informed through RT in case of any emergency situation. During idle power engine run up, AME will take all ground precautions including switching on aircraft anti-collision lights, positioning of wing walkers and RT trained engineer to be available in in cockpit. In case of foreign registered aircraft, if pilot conducts idle power engine start/ run up, he will seek permission from ATC.

1.3 In case of APU failure, the aircraft may be permitted to start one or both starboard side engines at idle power on the stand with the aerobridge attached. Further, in case of APU failure, a cross-bleed start is permitted only after completion of pushback, pull forward (if applicable) and disconnection of Tug. No cross bleed is permitted at stand or during push back. Pilots to inform ATC and take all ground precautions against jet blast before starting cross-bleed. Airlines/GHAs are responsible to adhere to all ground precautions to ensure safety.

1.4 In all Aprons except Apron 32 and 34, pushback of aircraft to be separated by one stand, however in Apron 32, Apron 34, Apron II (bays 201-215, 220-231) and Cargo Apron which still have the concept of TDP/ Block system pushback to be given as per defined blocks

1.5 For datalink, capable aircraft while operating in DELHI FIR the AFN logon address for Delhi FIR is VIDF.

1.6 Left turn towards stand 800 from TWY D4 not available while aircraft taxing on TWY D4 facing north. Similarly Right turn on TWY D4 from stand 800 not available while aircraft taxiing from stand 800 facing east.

1.7 The AFN logon address for Departure Clearance (DCL) connection for IGI Airport,New Delhi is VIDP.

1.8 TWY D1 is available for taxiing up to Code Letter C aircraft with OMGWS up to but not including 9.6m.

1.9 The usage of Taxiway T4 & T5 and S5 will be regulated during operations of GP 11L and GP 29R, respectively.

1.10 During Segregated Dependent Easterly Mode of operations, RWY 11R for Arrival and RWY 11L for Departure, for the application of Wake Turbulence Separation Minima with respect to heavier arrival on RWY 11R, RWY 11L intend departure from:

a. TWY R5 requiring a Take-Off Run less than 600m or

b. TWY R6 or R7 requiring a Take-Off Run less than 900m,

shall notify ATC at or before the issuance of taxi clearance.

1.11 During Segregated Dependent Westerly Mode of operations, RWY 29L for Arrival and RWY 29R for Departure:

a. TWY P2 will be preferred entry point for departure from RWY 29R with Wake Turbulence Category Light, ATC may use TWY P2 as preferred entry point for other departure also as per the aircraft type and it's characteristics.

TWY P4 or TWY P5 will be the preferred entry point for departure from RWY 29R with Wake Turbulence Category Medium or Heavy, such departure requiring line up RWY 29R from any other entry point shall notify ATC at or before the issuance of taxi clearance.

b) For the application of Wake Turbulence Separation Minima with respect to heavier arrival on RWY 29L, RWY 29R intend departure from:

i) TWY P2 requiring Take-Off Run less than 600m OR

ii) TWY P4 or TWY P5 requiring Take-Off Run less than 1400m OR

iii) TWY P6 or TWY P7 or TWY P8 requiring Take-Off Run less than 2000m shall notify ATC at or before issuance of taxi clearance

1.12Departure from RWY 11R will not be permitted when RWY 11L available. Pilots to plan aircraft performance considering maximum TORA for RWY 11L 3670M.

1.13 When instructed by ATC to hold short of a TWY/TWY intersection on a TWY not provided with intermediate holding position marking/lights, Pilot must stop ACFT at a point which provides adequate clearance from all ACFT types on intersecting TWY

1.14 Arrival on RWY 29R not permitted when RWY 29L available except for ACFT in a state of emergency. Pilots to plan ACFT performance considering maximum LDA 2970M for RWY 29L.

1.15 Aircraft exercise caution while taxiing on parallel taxi lanes against another aircraft pushing back on the adjacent taxilane.

1.16 All code D/E aircraft intending to hold on taxiway D2 will hold on taxiway D facing east at intermediate holding position (IHP) short of taxiway D2.

1.17 Right turn from TWY N7 on TWY N6 not available.

1.18 Left turn from TWY N6 on TWY N7 not available.

1.19 Engine power more than idle thrust is not permitted during pushback and taxiing in the Apron.

2. TAXING PROCEDURES FOR DIFFERENT APRONS

2.1-A PROCEDURES FOR APRON I

Apron I (STAND 101 - 178)	
Aircraft Stands	**Arrival Procedure**
170-178	The aircraft shall taxi in via taxilane F6 to respective stand.
150-159	The aircraft shall taxi in via taxilane F3 to respective stand.
160R-165	The aircraft shall taxi in via taxilane F5 to respective stand.
140-149B	The aircraft shall taxi in via taxilane E9 to respective stand.
123-132	The aircraft shall taxi in via taxilane E3 to respective stand.
Contact Stands of odd No. (101-121)	The aircraft shall taxi in via taxilane E2 to respective stand.
Contact Stands of even No. (102-122)	The aircraft shall taxi in via taxilane E4 to respective stand.
133L-137	The aircraft shall taxi in via taxilane E6 to respective stand.
Aircraft Stands	**Departure Procedure**
160R-165	The aircraft shall pushback onto taxilane 'F5' facing North or South and to be towed abeam respective stand and follow further instructions from ATC.
170-178	The aircraft shall pushback onto taxilane 'F6' facing North or South and to be towed abeam respective stand and follow further instructions from ATC.
150-159	The aircraft shall pushback onto taxilane 'F3' facing North or South and to be towed abeam respective stand and follow further instructions from ATC.

140-149B	The aircraft shall pushback onto taxilane 'E9' facingNorth or South and to be towed abeam respective stand and follow further instructions from ATC.
133L-137	The aircraft shall pushback onto taxilane E6 facing East or West and to be towed abeam respective stand and follow further instructions from ATC.
123-125	The aircraft shall pushback onto taxilane E3 facing north or south and to be towed abeam respective stand and follow further instructions from ATC.
126-132	The aircraft shall pushback onto taxilane E3 facing east or west and to be towed abeam respective stand and follow further instructions from ATC.
Contact Stands 101 and 103	The aircraft shall pushback onto taxilane 'E2' facing east or west and to be pushed back/towed abeam stand 105 and follow further instructions from ATC.
Contact Stands of odd No. (105-121)	The aircraft shall pushback onto taxilane 'E2' facing east or west and to be pushed back/towed abeam respective stand and follow further instructions from ATC.
Contact Stands 102 and 104	The aircraft shall pushback onto taxilane 'E4' facing east or west and to be pushed back/towed abeam stand 106 and follow further instructions from ATC
Contact Stands of even No. (106-122)	The aircraft shall pushback onto taxilane 'E4' facing east or west and to be pushed back/towed abeam respective stand and follow further instructions from ATC.

Note:

i. Two Code letter C aircraft can taxi simultaneously on Taxilane F3 and F5 between IHPs located short of TWY D and TWY G, respectively.

ii. Simultaneous pushback of two Code letter C aircraft on Taxilane F3 and F5 permitted till visibility up to or more than 100m.

iii. During pushback from stand 165 facing North, aircraft taxiing on TWY G towards west, shall hold short of taxilane F3 and aircraft taxiing on TWY G towards east, shall hold short of taxilane F5.

iv. During pushback from stand 178 facing North, aircraft taxiing on TWY G shall hold short of taxilane F6.

v. During pushback from stand 165 and 178 facing North onto respective taxilane, aircraft shall pull forward abeam stand 164L and 177 respectively.

vi. During pushback from stand 170 facing South, aircraft taxiing on TWY D shall hold short of taxilane F6.

vii. During pushback from stand 170 facing South onto taxilane F6, aircraft shall pull forward abeam stand 171.

viii. During pushback from stand 140 facing south, aircraft taxiing on TWY D towards east, shall hold short of taxilane E9 and aircraft taxiing on TWY D towards west, shall hold short of taxilane E7.Similarly, during pushback from stand 149B facing north, aircraft taxiing on TWY G towards east, shall hold short of taxilane E9 and aircraft taxiing on TWY G towards west, shall hold short of taxilane E7.

ix. During pushback from stand 140 facing south and from stand 149B facing north, onto respective taxilane, aircraft shall pull forward abeam stands 141 and 149A respectively.

x. During pushback from stand 159 facing North, aircraft taxiing on TWY G towards west, shall hold short of taxilane F3 and aircraft taxiing on TWY G towards east, shall hold short of taxilane F5.

xi. During pushback from stand 150 facing south, aircraft taxiing on TWY D towards west shall hold short of taxilane F3 and aircraft taxiing on TWY D towards east, shall hold short of taxilane F5.

xii. During pushback from stand 159 facing North onto taxilane F3, aircraft shall pull forward abeam stand 158.

xiii. During pushback from stand 150 facing South onto taxilane F3, aircraft shall pull forward abeam stand 151.

xiv. During aircraft taxiing from taxilane F5 to taxilane F3 via taxilane F2 and vice versa, pushback shall not be permitted from aircraft stands 161L, 162, 163R, 155 and 154.

xv. During pushback from stand 132 facing east, aircraft taxiing on taxilane E7 towards south shall hold short of taxilane E3 and aircraft taxiing on taxilane E7 towards north shall hold short of taxilane E2. Similarly, during pushback from stand 123 facing south, aircraft taxiing on TWY D shall hold short of taxilane E3.

xvi. During pushback from stand 132 facing east and from stand 123 facing south, onto respective taxilane, aircraft shall pull forward abeam stands 131 and 124 respectively.

xvii. During pushback from stand 133L facing west and from stand 137 facing east, onto respective taxilane, aircraft shall pull forward abeam stands 133R and 136R respectively.

xviii. During pushback from stands 121 facing east, aircraft taxiing on taxilane E7 towards south shall hold short of taxilane E3 and aircraft taxiing on taxilane E7 towards north shall hold short of taxilane E2.

xix. During pushback from stands 122 facing east, aircraft taxiing on taxilane E7 towards south shall hold short of taxilane E4 and aircraft taxiing on taxilane E7 towards north shall hold short of taxilane E6.

xx. During pushback from stand 137 facing east, aircraft taxiing on taxilane E7 towards south shall hold short of taxilane E4 and aircraft taxiing on taxilane E7 towards north shall hold short of taxilane E6.

xxi. Two aircraft up to code letter C can taxi simultaneously on Taxilane E9 and E7 between the IHPs located short of TWY D and TWY G.

xxii. Two aircraft up to code letter C can taxi simultaneously on Taxilane E2 and E3 between the IHPs located short of TWY D and taxilane E7.

xxiii. Two aircraft up to code letter C can taxi simultaneously on Taxilane E4 and E6 between the IHPs located short of TWY G and taxilane E7.

2.1- B. DUAL TAXILANE OPERATION PROCEDURES IN APRON I

Designation, Location and Compatibility	F5 - Yellow Centre line Marking; between TWY D and TWY G; Compatible for up to Code letter C aircraft F3 - Yellow Centre line Marking; between TWY D and TWY G; Compatible for up to Code letter C aircraft F4 - Blue Centre line Marking; between TWY D and TWY G; Compatible for up to Code letter E aircraft F2 - Yellow Centre line Marking; between taxilanes F3 and F5; Compatible for up to Code letter C aircraft
Marking and lighting	Markings: Centre line Markings, IHP Marking, Non load bearing Markings on curves, Information, Location and Direction Markings, Maximum Wingspan Markings Lightings: Centre line Lights, Taxiway Edge Lights on curves only, Stop Bars, IHP Lights.
Location of IHPs	• On TWY G and TWY D short of taxilane F3 and F5 • On taxilane F3, F4 and F5 short of TWY D and TWY G • On taxilane F3 and F5 short of taxilane F2
Location of Stop Bars	• F5 (south of D/F5 intersection), F5 (north of G/F5 intersection) • F4 (south of D/F4 intersection), F4 (north of G/F4 intersection) • F3 (south of D/F3 intersection), F3 (north of G/F3 intersection)

i. During Code letter D/E aircraft taxiing to/from TWY D and TWY G via Taxilane F4:

a) Stop Bars on Taxilane F4 will be switched off.

b) The taxilane centre line lights of Taxilane F4 from TWY D and TWY G will be switched on automatically.

c) Stop Bars on Taxilane F5 and F3 will be switched on automatically.

d) The taxilane centre line lights of Taxilane F5, F3 and F2 will be switched off automatically.

e) *The stand lead-in lights of Code C aircraft stands will be switched off automatically.

f) ATC to ensure switching off Stop Bar on Taxilane F4 before permitting Code D/E Aircraft to cross RWY 09/27 from TWY A2, as there is no IHP on TWY D3.

g) No aircraft shall taxi or pushback on Taxilane F3/F5.

ii. During up to Code letter C aircraft taxiing to/from TWY D and TWY G via Taxilane F3/F5:

a) Stop Bars on Taxilane F5 and F3 will be switched off.

b) The taxilane centre line lights of Taxilane F5 and F3 from TWY D and TWY G and Taxilane F2 will be switched on automatically.

c) Stop Bars on Taxilane F4 will be switched on automatically.

d) The taxilane centre line lights of Taxilane F4 will be switched off automatically.

e) *The stand lead-in lights of Code C aircraft stands will be switched on automatically.

f) ATC to ensure switching off Stop Bar on Taxilane F5/F3 before permitting Code C Aircraft to cross RWY 09/27 from TWY A2, as there is no IHP at TWY D3.

g) No aircraft shall taxi on Taxilane F4.

2.2 PROCEDURES FOR APRON II

APRON-II		
Arrival Procedures		
Aircraft Stands	**Procedure**	
201-206	The aircraft will taxi in via taxiway N3/N4/N onto respective stand.	
207-211 and 212-218	The aircraft will taxi in via taxi-lane N1 onto respective stand.	
220-231	The aircraft will taxi in via taxi-lane N8 onto respective stand.	
232-246	The aircraft will taxi in via taxiway N onto respective stand.	
Departure Procedure		
Aircraft Stands	**RWY in use 09/27/10/28/11R/29L/11L/29R**	
209, 210, 211 and 212 to 215	The aircraft shall be pushed back onto TWY N1 facing north and to be towed abeam stand 210 and follow further instructions from ATC.	
206, 207, 208 and 216 to 218	The aircraft shall be pushed back onto TWY N1 facing north and to be towed abeam stand 207 and follow further instructions from ATC. (Aircraft taxing towards east on TWY N shall hold at IHP on TWY N-N4 junction and aircraft taxing towards west on TWY N shall hold at IHP on TWY LINK 39/N1/ N junction while aircraft is pushing back from stands 206 Simultaneous pushback from stands 206 and 232, 233, 234 not allowed).	
202, 203, 204 and 205	The aircraft shall be pushed back onto TWY N3 facing north-east and to be towed abeam stand 203 and follow further instructions from ATC. (Aircraft taxing towards east on TWY N shall hold at IHP on TWY N-N4 junction and aircraft taxing towards WEST on TWY N shall hold at IHP on TWY LINK 39/N1/N Junction while aircraft is pushing back from stands 205. Simultaneous pushbacks from stands 205 and 232 to 236 not allowed. Also simultaneous pushback from stands 204 and 235, 234 not allowed.)	
201	The aircraft shall be pushed back onto TWY N4 facing South-east and to be towed abeam stand 203 and follow further instructions from ATC.	
232 to 246	The aircraft shall be pushed back onto Taxiway N facing East or West as per the RWY in use and follow further instructions from ATC. Simultaneous pushback from stands A12, A12R, A14, A14L, A14R and 243 to 246 not allowed.	
	Departure via TWY LINK 39	**Departure via TWY LINK 34**
220 to 223 (Block 1)	The aircraft shall be pushed back onto taxi-lane N8 facing West and towed abeam stand 223 and follow further instructions from ATC	The aircraft shall be pushed back onto taxi-lane N8 facing East and towed abeam stand 220 and follow further instructions from ATC
224 to 227 (Block 2)	The aircraft shall be pushed back onto taxi-lane N8 facing West and towed abeam stand 227 and follow further instructions from ATC	The aircraft shall be pushed back onto taxi-lane N8 facing East and towed abeam stand 224 and follow further instructions from ATC
228 to 231 (Block 3)	The aircraft shall be pushed back onto taxi-lane N8 facing West and towed abeam stand 231 and follow further instructions from ATC	The aircraft shall be pushed back onto taxi-lane N8 facing East and towed abeam stand 228 and follow further instructions from ATC

Note:

i. On stands 209, 210, 211 and 212 to 215 (Block 1), only one pushback allowed at a time.

ii. On stands 206, 207, 208 and 216 to 218 (Block 2), only one pushback allowed at a time

iii. On stands 202, 203, 204, 205 and 201 (Block 3), only one pushback allowed at a time.

iv. On stands 220-223 (Block-1), 224-227 (Block-2) and 228-231 (Block-3), only one pushback allowed at a time.

v. Aircraft taxiing on TWY LINK 34/LINK 39 on to TWY N, shall hold at Intermediate Holding Position on TWY LINK 34/ LINK 39 short of taxi lane 'N8', when an aircraft is pushing back from stands 220 and 231 respectively.

vi. Aircraft taxiing on TWY N on to TWY LINK 34/LINK 39, shall hold at Intermediate Holding Position on TWY N short of TWY LINK 34/LINK 39, during an aircraft is pushing back from stands 220 and 231, respectively.

vii. Aircraft taxiing on N8 shall hold at Intermediate Holding Position on taxi-lane N8, short of TWY LINK 39/LINK 34, when an aircraft is taxiing on TWY LINK 39/LINK 34.

viii. Aircraft Stand Manoeuvring Guidance Lights available at all stands from 201-211 and 232-244.

ix. Refuelling is facilitated with fuel browser at stands 203 to 207,211 to 231,233to 237 and 241 to 242.

x. Fuel hydrant facility is available at all stands 201,202,208 to 210,232,238,240,243 to 246.

2.3 PROCEDURES FOR CARGO APRON

Cargo Apron

Arrival Procedures

Aircraft Stands	**Procedure**
252-265	i.The aircraft up to Code E shall taxi in via TWYLINK 39/LINK 34/LINK 33,TWY N/ Taxilane N and onto respective stand. ii.Code F Aircraft (restricted up to max wingspan 73.30m) shall taxi in via TWY LINK 33 / Link 34 /Taxi lane N and onto respective stand. iii. Code F Aircraft (wingspan greater than 73.30m) shall taxi in via TWY LINK 33/ Link 34.

Departure Procedures

Aircraft Stands	**RWY in use 09/27/10/28/11R/29L/11L/29R**	
	Departure via TWY LINK 34	**Departure via TWY LINK 33**
Block-1 252, 253, 260, 261 and 262	The aircraft shall pushback onto taxilane N facing West, tow up to tug disconnect point CA3 and follow further ATC instructions. Only one pushback is allowed from these stands at a time.	The aircraft shall pushback onto taxilane N facing East, tow up to tug disconnect point CA1 and follow further ATC instructions. Only one pushback is allowed from these stands at a time.
Block-2 254, 255, 256, 257, 263, 264 and 265	The aircraft shall pushback onto taxilane N facing West and tow up to tug disconnect pointCA4, follow further ATC instructions.Only one pushback is allowed from these stands at a time	The aircraft shall pushback onto taxilane N facing East, tow up to tug disconnect point CA2 and follow further ATC instructions. Only one pushback is allowed from these stands at a time.

Note:

i. Code F Aircraft (restricted up to max wingspan 73.30m) shall taxi in/out to/from Cargo apron via TWY LINK 33/TWY LINK 34 or TXL N only, whereas Code F Aircraft (wingspan greater than 73.30m) shall taxi via LINK 33 or LINK 34 only.

ii. Simultaneous pushback from above mentioned block-1 and 2 of aircraft stands are allowed except that only one push back is permitted at a time from stands 253, 254, 262 and 263.

iii. During pushback from stand 257 facing east, the aircraft taxing on TWY N facing east and on TWY LINK 34 facing south shall hold short of TWY N and LINK 34 junction on respective IHP..

iv. Tug Disconnect Points (TDP):

a. CA1: East of stand 252

b. CA2: Behind stand 255

c. CA3: Behind stands 253

d. CA4: Behind stands 256

v. CA1 and CA2 will be used for pushback facing east and CA3 and CA4 will be used for pushback facing west.

vi. All stands are power-in push back.

vii. Aircraft stands 252-257 and 260 are suitable for aircraft with wingspan upto 68.4m.

viii. Fuel hydrant facility is available at all stands in cargo apron for re-fuelling of aircraft.

ix. Aircraft Stand Manoeuvring Guidance Lights(ASMGL), CAT III lighting available at all stands.

x. Visual Docking Guidance System are not available at stands in Cargo apron.

2.4 PROCEDURES FOR GENERAL AVIATION APRON

A. PROCEDURES FOR GA APRON

Arrival Procedures

Aircraft Stands	Procedure
901, 901C, 902, 902C, 903, 904, 904C, 905, 906C, 907, 907C, 908, 909, 909C, 910, 911C	The aircraft shall taxi in via taxilane LINK 13 to respective stand.
951 - 956	The aircraft shall taxi in via taxilane LINK 16 to respective stand.

Departure Procedures

Aircraft Stands	Procedures
901, 901C, 902, 902C, 903, 904, 904C, 905, 906C, 907, 907C, 908, 909, 909C, 910, 911C	The aircraft shall be pushed back onto taxilane LINK 13 facing South and follow further instructions from ATC
912, 912C, 913, 914, 914C, 915, 916C, 916, 917, 917C, 918, 919, 919C, 920	The aircraft shall be pushed back onto taxilane LINK 13 facing South and follow further instructions from ATC

921, 922, 922B, 923, 924B, 924, 925, 926B, 926, 927B, 927, 928	The aircraft shall be pushed back onto taxilane LINK 15 facing North and follow further instructions from ATC
929, 929C, 930, 931C, 931, 932, 932C, 933, 934B, 934, 935, 935B, 936, 937B	The aircraft shall be pushed back onto taxilane LINK 15 facing North and follow further instructions from ATC
938, 939B, 939, 940, 941B, 941, 942, 942B, 943, 944B, 945, 946B, 946, 947	The aircraft shall be pushed back onto taxilane LINK 16 facing South and follow further instructions from ATC
951 - 956	The aircraft shall be pushed back onto taxilane LINK 16 facing South and follow further instructions from ATC

B. PROCEDURES FOR GA HANGARS

Arrival Procedures

Hangar Nos.	**Procedure**
G1 - G7	The aircraft shall initially be allocated to any of the power in stands and after de-boarding, the aircraft shall be towed to the respective Hangar.

Departure Procedures

Hangar Nos.	**Procedure**
G1 and G2	From Hangars G1 and G2 aircraft shall be pushed back or pulled forward facing south abeam stand 907 or stand 951 and further follow ATC instructions.
G3 - G7	From Hangars G3 to G7 aircraft shall be pushed back or pulled forward facing south abeam stand 907 or stand 951 and further follow ATC instructions.

Note:

i. Stands 901, 901C, 902, 902C, 903, 904, 904C, 905, 906C, 907, 907C, 908, 909, 909C, 910, 911C and 951-956 are power-in-push-back stands, remaining all other stands are tow in-push back stands.

ii. Re-fuelling of aircraft is done with fuel bowser.

iii. All arrival and departure procedure will be assisted by Marshaller and wing-walkers.

iv. Aircraft Stand Maneuvering Guidance Lights (ASMGL) not available.

v. Visual Docking Guidance System not available.

vi. High mast lights installed in the apron to provide lux level of 20 to enable night operations.

vii. Apron elevation is 745 feet.

viii. No Aeronautical Ground Lights are available on these taxiways/taxilanes(LINK 13, LINK 14, LINK 15 and Portion of LINK 16 between LINK 14 and TWY C).

ix. Retro reflective edge markers available on taxilane LINK 16 between TWY C and GA Apron and LINK 13 .

x. Helicopter start-up and switch-off location provided as HELI-1 and HELI-2 on taxilane LINK 13 and taxlane LINK 16 respectively at north of IHP .

xi. All arriving helicopters shall switch off their engines at HELI-1 or HELI-2 and further follow ATC instructions.

xii. All departing helicopters shall pushback on respective taxi lanes and tow up to HELI-1 or HELI-2 and further follow ATC instructions.

xiii. ACFT pushing back from stands 912C, 911C, 909C and 907C shall be pulled forward abeam stand 907 for startup and then further follow ATC instructions.

xiv. AOCC will allocate only power- in stands (901, 901C, 902, 902C, 903, 904, 904C, 905, 906C, 907, 907C, 908, 909, 909C, 910, 911C and 951-956) for all incoming aircraft. After the passenger and crew de boarding, aircraft will be towed to central parking stands/GA Hangar

xv.ARRIVAL PROCEDURE FOR GA HANGARS:

Aircraft intended to be parked at hangars G1 and G2 may taxi up to Tug disconnecting point (TDP) GA1 and then be towed up to respective Hangars. Similarly aircraft intended to be parked at hangars G3 to G7 may taxi up to TDP GA2 and then be towed up to respective hangars.

2.5 PROCEDURES FOR T3 APRON

2.5.1 PROCEDURES FOR APRON 31

APRON 31

Arrival Procedures

Aircraft Stands	Procedure
A06, A08L, A08, A08R, A10L, A10, A10R, A12L, A12, A12R, A14L, A14, A14R, R10L, R10, R10R, R11 and R12	The aircraft will taxi in via Taxilane N5 onto respective stand.

Departure Procedures

Aircraft Stands	Procedures

A08L, A08, A08R, A10L, A10, A10R, A12L, A12, A12R, R10L, R10, R10R and R11	The aircraft shall be pushed back onto taxilane N5 facing North West and to be towed abeam respective stand, follow further instructions from ATC
A06 and R12	The aircaft shall be pushed back onto taxilane N5 facing North West and to be towed abeam stand A08, follow further instructions from ATC.
A14, A14R and A14 L	The aircaft shall be pushed back onto taxilane N5 facing North West and to be towed abeam stand A12, follow further instructions from ATC

Note:

i. Simultaneous aircraft pushback from opposite stands is not permitted

ii.Only one aircraft is permitted to push back from stands A12, A12R, A14, A14L, A14R and stands from 243 to 246 at any time.

2.5.2 PROCEDURES FOR APRON 32:

APRON 32

Arrival Procedures

Aircraft Stands	Procedures
A01L, A01, A01R, A03L, A03, A03R, A07L, A07, A07R, A09L, A09, A09R, A11L, A11, A11R , A13, B15L, B15, B15R, B17L, B17, B17R, B19L, B19, B19R, B21, R01, R02L, R02, R02R, R03L, R03, R03R and R04	The aircraft will taxi in via TWY LINK 2 onto respective stand.

Departure Procedures

Aircraft Stands	Procedures
A07L, A07, A07R, A09L, A09, A09R, A11L, A11, A11R and A13	The aircraft shall be pushed back onto taxi lane facing North West and to be towed up to tug disconnecting point A2 , Taxi via LINK 1, follow further instructions from ATC.
A01L, A01, A01R, A03L, A03 and A03R	The aircraft shall be pushed back onto taxilane facing North and to be towed up to tug disconnecting point A3, Taxi via LINK 1, follow further instructions from ATC.

B15L, B15, B15R, B17L, B17, B17R, B19L, B19, B19R and B21	The aircraft shall be pushed back onto taxilane, facing North East and to be towed up to tug disconnecting point A4, Taxi via LINK 1, follow further instructions from ATC.
R01, R02L, R02 and R02R	The aircraft shall be pushed back onto taxilane facing North West and to be towed up to tug disconnecting point A2, Taxi via LINK 1, follow further instructions from ATC.
R03L, R03, R03R and R04	The aircraft shall be pushed back onto taxilane, facing North East and to be towed up to tug disconnecting point A4, Taxi via LINK 1, follow further instructions from ATC.

Note:

i. The blocks in Apron 32 are as below:
   a. Block 1: Stands A07 to A13 and R01 to R02
   b. Block 2: Stands A01 and A03
   c. Block 3: Stands B15 to B21 and R03 to R04.

ii. Only one pushback is allowed within a block at a time.

iii. **Location of TDPs (Tug Disconnect Points)**

A2: BTN stands A09 and A11 B3: Behind stand B17R

A3: Behind stand A03 B4: Behind stand A01

A4: Behind stand B15 B5: Behind stand A07

iv. **Push-Back from B15:**

•To TDP A4: Airraft from stand A01 shall not push-back.

•To TDP B4: Aircraft from stand A01 and A03 shall not push-back.

v. Push-back from A01:

•To TDP A3: Aircrfat from stand B15 and B17 shall not push-back (Aircraft from stand B15 or B17 can push-back to TDP A4 once aircraft on A01 has completed pushed back and pulled to TDP A3)

•To TDP B4: Aircraft from stand B15 and A07 shall not push-back (Aircraft from stand A07 can push back to TDP B5 once aircraft from stand A01 has completed pushed back and pulled to TDP B4)

vi. Push-back from A03:

•To TDP A3: Aircraft from stand A07 and B15 shall not push-back (Aircraft from stand B15 can push-back to TDP A4 once aircraft from A03 has completed pushed back and pulled to A3)

•To TDP B4: Aircraft from stand B15, A07 and A09 shall not push-back (Aircraft from stand A07 and A09 can push-back to TDP B5 once aircraft from stand A03 has completed pushed back and pulled to TDP B4)

vii. Push-back from A07:

•To TDP A2: Aircraft from stand A01 and A03 shall not pushback (Aircraft from stand A01 and A03 can push-back to TDP A3 once aircraft from stand A07 has completed pushed back and pulled to TDP A2)

•To TDP B5: Aircraft from stand A03 shall not push-back

2.5.3 PROCEDURES FOR APRON 33:

APRON 33	
Arrival Procedures	
Aircraft Stands	**Procedures**
B18, B20L, B20, B20R, B22L, B22, B22R, B24L, B24, B24R, B26L, B26, B26R,	The aircraft will taxi in via TWY LINK 3 onto respective stand.

C28L, C28, C28R, C30L, C30, C30R, C32L, C32, C32R, C34L, C34, C34R and C36	The aircraft will taxi in via TWY LINK 6 onto respective stand.

Departure Procedures	
Aircraft Stands	**Procedures**
B22L, B22, B22R, B24L, B24, B24R, B26L, B26, B26R, and B20L	The aircraft shall be pushed back onto taxi lane facing South West and to be towed abeam respective stand, follow further instructions from ATC
B20	The aircraft shall be pushed back onto taxi lane facing South West and to be towed abeam stand B22, follow further instructions from ATC
B20R and B18	The aircraft shall be pushed back onto taxi lane facing South West and to be towed abeam stand B20, follow further instructions from ATC
C32L, C32, C32R, C34L, C34, C34R, C36 C30R and C30L	The aircraft shall be pushed back onto taxi lane facing South West and to be towed abeam respective stand, follow further instructions from ATC
C30	The aircraft shall be pushed back onto taxi lane facing South West and to be towed abeam stand C32, follow further instructions from ATC
C28L and C28R	The aircraft shall be pushed back onto taxi lane facing South West and to be towed abeam stand C30, follow further instructions from ATC
C28	The aircraft shall be pushed back onto taxi lane facing South West and to be towed abeam stand C32, follow further instructions from ATC

Note:

i. Simultaneous pushback of Code E aircraft from opposite stands in Apron 33A and Apron 33B is not permitted. However, two narrow body aircraft from opposite stands can be pushed back simultaneously on Apron 33A and Apron 33B.

ii. Simultaneous push back from stands B18 and C28 or C28L is not permitted.

iii. For aircraft stand B18, pushback limit line is provided for an aircraft to pushback and further pull ahead abeam stand B20.

iv. For aircraft stands C28R and C28L, pushback guidance line is provided for an aircraft to pushback back and further pull ahead abeam stand C30.

v.Aircraft up to and including code C can taxi from TWY LINK 3 to TWY LINK 6 and vice versa i.e. Aircraft up to and including code C can taxi from stands B26 to B18 towards C28 to C36 and from stand C28 to C36 towards B26 to B18. Code D/E aircraft can towfrom TWY LINK 3 to TWY LINK 6 and vice versa i.e. Code D/E aircraft can tow from stands B26 to B18 towards C28 to C36 and from stand C28 to C36 towards B26 to B18.

2.5.4 PROCEDURES FOR APRON 34:

APRON 34		
Arrival Procedures		
Aircraft Stands	**Procedure (RWY in use 27/28/29L/29R)**	**Procedure (RWY in use 09/10/11L/11R)**

C27L, C27, C27R, C29L, C29, C29R, C31L, C31, C31R, C33, D37L, D37, D37R, D39L, D39, D39R, D41, D43, D45, D47, D49, D51, D53, D55, R05, R06, R07L, R07, R07R, R08L, R08, R08R and R09	The aircraft will taxi in via TWY LINK 7 onto respective stand.	The aircraft will taxi in via TWY LINK 8 onto respective stand.

Departure Procedures

Aircraft Stands	**Procedures**	
	RWY IN USE27/28/29L/29R	**RWY IN USE 09/10/11L/11R**
C27L, C27, C27R, C29L, C29, C29R, C31L, C31, C31R and C33	The aircraft shall be pushed back onto taxilane facing North East and to be towed up to tug disconnecting point D5; Taxi via LINK 8, follow further instructions from ATC	The aircraft shall be pushed back onto taxilane facing South West and to be towed up to tug disconnecting point C3; Taxi via LINK 7, follow further instructions from ATC
D37L, D37, D37R, D39L, D39, D39R and D41	The aircraft shall be pushed back onto taxilane facing East and to be towed up to tug disconnecting point D4, Taxi via LINK 8, follow further instructions from ATC	The aircraft shall be pushed back onto taxilane facing West and to be towed up to tug disconnecting point C4, Taxi via LINK 7, follow further instructions from ATC
D43, D45, D47, D49, D51, D53 and D55	The aircraft shall be pushed back onto taxilane facing South East and to be towed up to tug disconnecting point D3, Taxi via LINK 8, follow further instructions from ATC.	The aircraft shall be pushed back onto taxilane facing North West and to be towed up to tug disconnecting point C5, Taxi via LINK 7, follow further instructions from ATC.
R05 and R06	The aircraft shall be pushed back onto taxilane facing North East and to be towed up to tug disconnecting point D5; Taxi via LINK 8, follow further instructions from ATC	The aircraft shall be pushed back onto taxilane facing South West and to be towed up to tug disconnecting point C3; Taxi via LINK 7, follow further instructions from ATC
R07L, R07, R07R, R08L, R08 and R08R	The aircraft shall be pushed back onto taxilane facing East and to be towed up to tug disconnecting point D4, Taxi via LINK 8, follow further instructions from ATC	The aircraft shall be pushed back onto taxilane facing West and to be towed up to tug disconnecting point C4, Taxi via LINK 7, follow further instructions from ATC
R09	The aircraft shall be pushed back onto taxilane facing South East and to be towed up to tug disconnecting point D3, Taxi via LINK 8, follow further instructions from ATC	The aircraft shall be pushed back onto taxilane facing North West and to be towed up to tug disconnecting point C5, Taxi via LINK 7, follow further instructions from ATC

Note:

i. i. The blocks in Apron 34 are as below:

a. Block 1: Stands C27 to C33 and R05 to R06

b. Block 2: Stands D37 to D41 and R07 to R08

c. Block 3: Stands D43 to D55 and R09.

ii. Only one pushback is allowed within a block at a time.

iii. **Location of TDPs (Tug Disconnect Points)**

C3: Behind stand C29L D3: Behind stand D51
C4: Behind stand D37 D4: Behind stand D41
C5: Behind stand D45 D5: Behind stand C27L

iv. Push-back from C27:

- To TDP D5: Aircraft from stand D37 shall not pushback.
- To TDP C3: Aircraft from stand D37, D39 shall not push-back (Aircraft from stand D37 or D39 can push-back to TDP C4 once aircraft from stand C27 has completed pushed back and pulled to TDP C3)

v. Push-back from D37:

- To TDP D4: Aircraft from stands C27, C29 and D43 shall not pushback (Aircraft from stands C27 or C29 can push-back to TDP D5 once aircraft from stand D37 has completed pushed back and pulled to TDP D4)
- To TDP C4: Aircraft from stand C27 and D43 shall not pushback (Aircraft from stand D43 can push back to TDP C5 once aircraft from stand D37 has completed pushback and pulled to TDP C4).

vi. Push-back from D39:

- To TDP D4: Aircraft from stand C27 and D43 shall not pushback (Aircraft from stand C27 can push back to TDP D5 once aircraft from stand D39 has completed pushback and pulled to TDP D4)
- To TDP C4: Aircraft from stand D43 and C27 shall not pushback (Aircraft from stand D43 can push-back to TDP C5 once aircraft from stand D39 has completed pushback and pulled to TDP C4)

vii. Push-back from D41:

- To TDP D4: Aircraft from stand D43 and D45 shall not pushback
- To TDP C4: Aircraft from stand C27, D43 and D45 shall not pushback (Aircraft from stand D43/D45 can pushback to TDP C5 once aircraft from stand D41 has completed pushback and pulled to TDP C4).

2.5.5 PROCEDURES FOR APRON 35:

APRON 35

Arrival Procedures

Aircraft Stands	Procedures
D46, D48, D50, D52, D54, D56, D58, D60 and D62	The aircraft will taxi in via TWY LINK 9 onto respective stand.
E64, E66, E68, E70, E72, E74, E76, E78, E80, E82 and E84	The aircraft will taxi in via TWY LINK 10 onto respective stand.

Departure Procedures

Aircraft Stands	Procedures
D46	The aircraft shall be pushed back onto taxilane facing south east and to be towed abeam stand D48, follow further instructions from ATC
D48, D50 and D52	The aircraft shall be pushed back onto taxilane facing South East and to be towed abeam respective stand, follow further instructions from ATC
D54, D56, D58, D60 and D62	The aircraft shall be pushed back onto taxilane facing South East and to be towed abeam respective stand, follow further instructions from ATC
E64, E66, E68 and E70	The aircraft shall be pushed back onto taxilane facing south east and to be towed abeam stand E70,follow furthe instructions from ATC.

E72, E74, E76, E78, E80, E82 and E84	The aircraft shall be pushed back onto taxilane facing south east and to be towed abeam respective stand ,follow further instructions from ATC

Note:

i. Only one aircraft at a time is permitted to pushback from stands D46, D48,D50 and D52.

ii. Only one aircraft at a time is permitted to pushback from stands E64,E66, E68 and E 70.

iii. When aircraft on stand D46 D48, D50, D52, D54 and D56 will push back, no push back is allowed from E64,E66 and E68 and vice versa.

iv. The aircraft taxiing in/out to/from on stands D46,D48, D50, D52, D54, D56, D58, D60 and D62 shall use taxilane LINK 9

v. Aircraft taxiing in/out stands E68,E70, E72, E74, E76, E78, E80, E82 and E84 shall use taxilane LINK 10

vi. Arriving aircraft to stands E64 and E66 shall use LINK 9.

vii. Departing aircraft from stand E64 and E66 shall use LINK 10.

2.5.6 LIMITATIONS FOR T-3 APRON

2.5.6.1 Push-back of both Wide-body and Narrow-body Aircraft from Bays A14, A12, A13, A11, B19, B21, C33, C31, D53 and D55 onto Aprons infringe the taxi clearance strip for **N** and C taxiways respectively with either Tail or Wingtip. Arriving aircraft will taxi on outside Taxiway A and these aircraft may be required to hold short of adjoining taxiways such as N5, LINK 1, LINK 2, LINK 7 and LINK 8 on A .

Departing aircraft taxiing N and C may be required to hold short of the adjoining taxiways such as N5, LINK 1, LINK 2, LINK 7 and LINK 8 on C respectively until tail / wing-tip of push back aircraft has cleared the intersections.

Additional CCTV feeds to monitors in the tower will assist SMC in ensuring that affected taxiways are clear of obstructions prior to instructing holding aircraft to proceed.

2.5.6.2 For a Wide-body Code D/E/F aircraft pushing back from an aircraft bay in Apron 32 and Apron 34 , there should always be 2 Code E/F stands (respectively) separation from another aircraft pushing back from adjacent bays. AOCC will not allocate an adjacent aircraft bay to aircraft requiring push-back within 20 minutes of an another aircraft push-back within 2 parking stands.

2.5.6.3 With respect to the aircraft taxiing into the corner bays in Apron 32 and Apron 34, the following taxiing limitations are to be observed by ATCOs:

a. ACFT taxiing on TWY C shall hold at intermediate holding point:

i. Short of TWY LINK 8 when an ACFT is taxiing in to stand D53 and D55.

ii. Short of TWY LINK 7 when an ACFT is taxiing in to stand C31, C31L and C33.

b. ACFT taxiing on TWY C shall hold at intermediate holding point:

i. Short of TWY LINK 1 when an ACFT is taxiing in to stand A13.

ii. Short of TWY LINK 2 when an ACFT is taxiing in to stand B21.

3. START-UP PROCEDURE IN A-320 HANGER:

3.1 Aircraft/ Helicopter shall tow up to short of TWY C on TWY LINK 32, facing north then further follow ATC instruction.

4. DETAILS OF STANDS 801, 802, 803

4.1 STAND DETAILS:

S.no	Aircraft Stand No.	Co-ordinates	Surface	PCR	Elevation	Suitability Aircraft	Location
1	801	283417.55N 0770629.59E	Asphalt	862/F/A/W/T	740 FT	Up to Code Letter E	At TWY A2, facing North-West

2	802	283243.53N 0770610.68E	Asphalt	793/F/A/W/T	766 FT	Up to Code Letter F	At TWY C between TWY LINK 16 and TWY LINK 19, facing East
3	803	283311.12N 0770401.54E	Asphalt	862/F/A/W/T	714 FT	Up to Code Letter F	West of TWY R7 (between TWY Q and TWY R), facing West

Note:

I. All the stands are remote stands and power-in push back.

II. Re-fueling can be done with fuel bowser.

III. During Aircraft Parking on stand 802 under unlawful interference following taxiways not available for operations.

a. Portion of TWY C, east of TWY LINK 16 up to TWY P6.

b. TWY LINK 19

c. Portion of TWY A east of TWY LINK 16 up to TWY P6.

d. TWYs P3, P4, P5.

IV. During Aircraft parking on stand 803 under Bomb Threat following taxiways not available for operations.

a. TWY R7

b. Portion of TWY Q between TWY R6 and TWY R7

c. Portion of TWY R between TWY R6 and TWY R7

V. During Aircraft parking on stand 801 under unlawful interference following restrictions shall apply.

a. RWY 09/27 will not be available for arrivals/ departures.

b. Portion of RWY 09/27 from D9 to D4 will be available for taxiing of aircraft.

c. Crossings from TWY D1 or TWY D2 to the technical area and vice-aversa via RWY 27 dumbbell will be available.

5. DETAILS OF ENGINE RUN UP AT HIGH POWER

S.no	Aircraft Stand No.	Coordinates	Surface	PCR	Suitability Aircraft	Location
1	800	283405.55N 0770530.74E	Concrete	1506/R/B/W/T	Up to Code Letter C	Parallel to and south of TWY D, at a distance of 77.5m from TWY D centre line, west of TWY D4, facing west

2	803	283310.74N 0770405.14E	Asphalt	862/F/A/W/T	Up to Code Letter F	West of TWY R7 (between TWY Q and TWY R), facing east
3	High Power Engine Run-up Bay	283405.94773N 0770708.87937E	Asphalt	708/F/B/W/T	Up to Code Letter C	East of TWY D1, Short of TXL E2, Facing south
			Concrete	611/R/B/W/T		

Note:

i. Aircraft from stand 800 shall Push Back on TWY D4 facing North or South.

ii. During RWY 09/27 in use for Arrival and Departure High Power Engine Run restricted on stand 800.

iii. Towing of Aircraft To/From stand 800 will be carried out under Follow Me supervision.

iv. TWY R7 not avbl for ops while aircraft is being repositioned at stand 803 for high power run-up.

VIDP AD 2.21 NOISE ABATEMENT PROCEDURES

I.CONTINUOUS DESCENT ARRIVAL (CDA) PROCEDURES

1.In order to mitigate the aircraft noise within the areas located below the flight path in the vicinity of IGI airport, the concept of continuous descent approach (CDA) is implemented. All the turbojet aircraft landing at IGI Airport between 1630 UTC and 0030 UTC may participate in CDA subject to clearance from ATC.

2.CDA will be operated when radar and ILS facilities are available.

3.Operation of CDA should not impact the airport capacity adversely.

4.Aircraft shall normally be cleared via a STAR. Since, the STARs for IGI Airport are open ended, aircraft may expect radar vectors for turn on base leg and final approach. Profile/crossing restrictions are well depicted in the STARs. Aircraft shall be provided information regarding the distance to touchdown from the beginning of CDA and subsequently any revision thereafter.

5.For track shortening or lengthening for sequencing purposes, the aircraft may be radar vectored off the STAR and subsequently, be re-cleared to a point along the STAR or vectored to intercept the final approach track.

6.When radar vectors are issued, ATC shall provide an estimate of distance to touch down.

7.Pilots participating in CDA should request CDA within 10 minutes of top of descent (TOD) and provide the estimates of VOR (SSB), SAPLO, AKBAN and VOR (SAMPLA) as the case may be. ATC may issue clearance for CDA depending upon the traffic conditions.

8.When following STARs, ATC may add further altitude restrictions, in addition to the promulgated restrictions, if necessary for the purpose of traffic separation.

9.Radar controller will ensure that in case of STARs merging, adequate radar separation exists between the successive aircraft at the conflict point.

10.It is preferable if CDA is commenced from top of descent. If it is not feasible due to ATC constraints, it may be commenced from any intermediate level preferably before crossing FL100. Once CDA is commenced there will be continuous descent to 3600 feet AMSL before intercepting the glide path.

11.Once CDA is cleared by ATC, pilot should report the beginning of the descent to ATC.

12.Radar controller shall provide 20 NM distance to touch down (DTD) information when the speed shall be reduced to 210KT IAS maximum

13.Speed shall be reduced to 180 KT IAS maximum when 10NM from touchdown.

14.During CDA operations Glide path interception and commencement of final approach shall take place at 3600 feet.

15.ATC may suspend or cancel the CDA due to traffic conditions even after CDA is cleared. Alternate instructions will be issued when CDA is suspended or cancelled.

16.All aircraft have to follow the noise abatement procedure during the final approach.

17.Guidance material for CDA operations is available at DGCA website (dgca.nic.in) as Air Navigation Services (ANS) Circular NO.01/2010.

II.RUNWAY USE PLAN FOR NOISE ABATEMENT

1.The Runway use plan for noise abatement is as follows.

1.1	**Schedule for dates 1st ,8th ,17th and 23rd of each month.**			
Time Slot (IST)	**RWY For Westerly flow of traffic**		**RWY For Easterly flow of traffic**	
	Arrivals	**Departures**	**Arrivals**	**Departures**
0000-0100	RWY 27/28	RWY 27/28 and RWY 29L/29R	RWY 09/10 and RWY 11R/11L	RWY 09/10 and RWY 11R/11L
0101-0600	RWY 29L/29R	RWY 27/28 and RWY 29L/29R	RWY 09/10 and RWY 11R/11L	RWY 09/10 and RWY 11R/11L
0601– 2359	RWY 27/28 and RWY 29L/29R	RWY 27/28 and RWY 29L/29R	RWY 09/10 and RWY 11R/11L	RWY 09/10 and RWY 11R/11L
1.2	**Schedule for dates 9th, 16th , 24th and 30th of each month**			
Time Slot (IST)	**RWY For Westerly flow of traffic**		**RWY For Easterly flow of traffic**	
	Arrivals	**Departures**	**Arrivals**	**Departures**
0000-0100	RWY 29L/29R	RWY 27/28 and RWY 29L/29R	RWY 09/10 and RWY 11R/11L	RWY 09/10 and RWY 11R/11L
0101-0600	RWY 27/28	RWY 27/28 and RWY 29L/29R	RWY 09/10 and RWY 11R/11L	RWY 09/10 and RWY 11R/11L

0601– 2359	RWY 27/28 and RWY 29L/29R	RWY 27/28 and RWY 29L/29R	RWY 09/10 and RWY 11R/11L	RWY 09/10 and RWY 11R/11L
1.3	**Schedule for dates from 2nd to 7th and 18th to 22nd of each month**			
Time Slot (IST)	**RWY For Westerly flow of traffic**		**RWY For Easterly flow of traffic**	
	Arrivals	**Departures**	**Arrivals**	**Departures**
0000-0300	RWY 27/28	RWY 27/28 and RWY 29L/29R	RWY 09/10 and RWY 11R/11L	RWY 09/10 and RWY 11R/11L
0301-0600	RWY 29L/29R	RWY 27/28 and RWY 29L/29R	RWY 09/10 and RWY 11R/11L	RWY 09/10 and RWY 11R/11L
0601– 2359	RWY 27/28 and RWY 29L/29R	RWY 27/28 and RWY 29L/29R	RWY 09/10 and RWY 11R/11L	RWY 09/10 and RWY 11R/11L
1.4	**Schedule for dates from 10th to 15th and 25th to 29th and 31st of each month**			
Time Slot (IST)	**RWY For Westerly flow of traffic**		**RWY For Easterly flow of traffic**	
	Arrivals	**Departures**	**Arrivals**	**Departures**
0000-0300	RWY 29L/29R	RWY 27/28 and RWY 29L/29R	RWY 09/10 and RWY 11R/11L	RWY 09/10 and RWY 11R/11L
0301-0600	RWY 27/28	RWY 27/28 and RWY 29L/29R	RWY 09/10 and RWY 11R/11L	RWY 09/10 and RWY 11R/11L
0601– 2359	RWY 27/28 and RWY 29L/29R	RWY 27/28 and RWY 29L/29R	RWY 09/10 and RWY 11R/11L	RWY 09/10 and RWY 11R/11L

2. **Benefits:**

i. Distribution of aircraft noise to all the areas located below the flight path as far as possible without impacting the maintenance work.

ii. The noise level will be distributed to all the localities as the arrivals and departures are distributed to both the runways.

iii. During the night time (0000 to 0600 Hrs IST), the residents will have not have discomfort from the aircraft noise during the specified time when a particular runway is used only for departures on the rotation basis.

3.The plan will not be applicable when one of the runways is either under maintenance or not available for operations due to any other reasons. In such cases, ATC may plan the available runway in order to maintain safe and efficient flow of traffic.

4.Every effort will be made to follow the published runway use plan. However, ATC may implement the alternate plan, as and when, traffic situation warrants in the interest to safety, regularity and efficiency of the aircraft movements to reduce the congestion and delays.

5.This is applicable for use of two runways at a time. As and when traffic demands necessitate the use of all three runways at a time, separate procedures shall be formulated.

6.Consistent with safety of aircraft operations and in consideration of high intensity runway operations, pilots should minimize the use of, reverse thrust after landing to reduce disturbance in areas adjacent to the aerodrome.

III.OPERATIONS OF CHAPTER-2 AIRCRAFT

Between 1630-0030 UTC daily, operations of chapter-2 aircraft as contained in annex-16 vol-1 are not permitted to operate at IGI Airport Delhi for noise abatement. However, ministry of defense (Govt. of India) chapter-2 aircrafts is allowed to operate on Runway 27/09 only.

IV. AIRCRAFT NOISE LIMIT AS Lmax (MAXIMUM SOUND LEVEL) in dB(A)

Aircraft noise level limits in Lmax dB(A) for Delhi Airport is as follow:

Lmax	**Limits in dB(A) at Noise Monitoring Terminals**	
	Day time (0600 to 2200) (IST)	**Night time (2200 to 0600) (IST)**
	105	95
These Lmax values shall be adhered by airlines operating at Delhi Airport and to be monitored and communicated by Airport Operator to the Directorate General of Civil Aviation.		

VIDP AD 2.22 FLIGHT PROCEDURES

I.RADIO COMMUNICATION FAILURE PROCEDURE
1.INTRODUCTION:

Radio communication failure procedures are described in Para 15.3 of ICAO PANS ATM DOC 4444, 15th edition 2007. Based on these provisions, following radio communication failure procedures are established, to standardize the actions to be taken by the pilot of arriving and departing aircraft at IGI airport.

2.GENERAL:

2.1.All Transponder equipped aircraft experiencing Radio Communication failure shall set transponder to Mode A/C code 7600 as soon as practicable.

(Note: This Requirement of Setting transponder to Mode A/C code 7600 in no way imposes any restriction on the pilot's decision to set transponder to Mode A/C code 7500 or 7700, whenever required).

2.2.Pilot shall not overfly VIP 89 (Prohibited area) situated 5.8 NM North East of Delhi Airport under any circumstances.

3.ASSIGNED RUNWAY AND ITS AVAILABILITY FOR RCF AIRCRAFT:

3.1.In case of arriving aircraft, when Runway for landing has already been advised to the aircraft by ATC, such runway shall be considered as assigned runway, except as given in para 3.3 below. In case arriving aircraft has not been advised any runway, Runway 28 shall be considered as assigned runway for such arrival. During the notified periods of maintenance/closure of Runway 28, Runway 29L shall be considered primary assigned Runway.

3.2.In case of departures from Delhi returning on account of RCF, the departure runway of such aircraft shall be considered as assigned runway for landing, except as given in para 3.3 below.

3.3.Runway 09 or 27 is not to be used by RCF aircraft except as under para 3.4 and 3.5 below. Runway 28, therefore, shall be treated as assigned Runway for aircraft to which Runway 27 had been assigned previously as per para 3.1 and departure from Runway 27 as per para 3.2 above. Similarly, Runway 10 shall be treated as assigned Runway for aircraft to which Runway 09 had been assigned previously as per para 3.1 and departure from Runway 09 as per para 3.2 above.

3.4.Notwithstanding anything contained in para 3.3 above, if Radio communication failure occurs after the final/interception turn, the aircraft may continue its approach to land on such Runway.

3.5.It is reiterated for clarity, aircraft experiencing RCF shall not make an approach for Runway 27 or Runway 09 unless final turn/interception heading has already been given to aircraft for Runway 27 or Runway 09 (see para 3.4).

3.6.Runway lights and Approach lights in 'SWITCHED ON' position shall indicate the availability of such Runway for aircraft experiencing RCF.

3.7.Irrespective of visibility/weather conditions, Runway and Approach lights in 'SWITCHED OFF' position shall indicate non- availability of such Runway for aircraft experiencing RCF. In such cases, the primary alternate runway and if primary laternate RWY is not available, the secondary alternate RWY, as indicated in the table below, shall be made available for such aircraft..

Runway (RCF Affected Aircraft)	Primary Alternate Runway	Secondary Alternate Runway
RWY 28	RWY 29 L	RWY 29 R
RWY 29 L	RWY 28	RWY 29 R
RWY 29 R	RWY 28	RWY 29 L
RWY 10	RWY 11 R	RWY 11 L
RWY 11 R	RWY 10	RWY 11 L
RWY 11 L	RWY 10	RWY 11 R

3.8.In strong tail wind conditions for assigned Runway, aircraft experiencing RCF will carry out a missed approach on assigned runway and after following complete missed approach for such approach, aircraft will carry out published ILS/VOR DME ARC approach procedure for the runway which is in opposite direction to the assigned Runway. Runway 10 in case of Runway 28 and Runway 11R in case of Runway 29L and vice versa. RWY 11L in case of Runway 29R and vice versa.

4.PROCEDURE FOR ARRIVALS:

4.1.ARRIVING AIRCRAFT – STAR-ASSIGNED

When STAR has been assigned to arrival:

4.1.1Except when descend clearance has already been received from ATC, pilot shall not commence descend before 100 NM from 'DPN VOR'. Aircraft shall continue on assigned STAR following all level and speed restrictions applicable to STAR, as far as practicable. This allows ATC to resolve any possible traffic conflict.

4.1.2At the end of the STAR, descend to 2600 ft and take a convenient turn to intercept localizer or final approach track of the published procedure for the assigned runway.

4.2.ARRIVING AIRCRAFT – STAR NOT ASSIGNED

Aircraft to which a STAR has not been assigned by ATC, on experiencing radio communication failure, shall

4.2.1.Continue on ATS route, (Re-join ATS route if given heading or flying offset), maintaining/descending to cleared flight level or FL 80 whichever is higher. At 40 miles to 'DPN VOR', take a turn (avoiding VIP 89) to proceed direct to 'SKA' VOR. If higher, descend to FL 80 in 'SKA' hold.

4.2.2.If RCF takes place within 40 NM of 'DPN VOR', continue to 'DPN VOR' and at 10 miles to 'DPN VOR' take a turn (avoiding VIP 89) to proceed direct to 'SKA' VOR maintaining last assigned level or FL 80 whichever is higher. If below FL 80, climb and reach FL 80 before crossing 25 NM outbound from 'DPN VOR'. If higher, descend to FL 80 in 'SKA' hold.

4.2.3.Leave 'SKA VOR' at FL 80 to carry out published ILS/VOR DME ARC approach procedure for assigned runway. Maintain FL 80 untill 20 DME DPN VOR.

4.3.ARRIVING AIRCRAFT BEING RADAR VECTORED

Aircraft being radar vectored for approach, on experiencing RCF, shall maintain last assigned level and heading for 3 minutes after detecting RCF or selecting MODE A/C 7600, whichever is later, and then take a convenient turn (Avoiding VIP 89) to proceed direct to 'SKA' climbing/maintaining last assigned level, or FL 80 whichever is higher.

If below FL 80, climb and reach FL 80 before crossing 25 NM outbound from 'DPN VOR'. If higher, descend to FL 80 in 'SKA' hold

4.3.1.Leave 'SKA' VOR at FL80 to carry out published ILS/VOR DME ARC approach procedure for assigned runway. Maintain FL 080 untill 20 DME DPN VOR.

5.PROCEDURE FOR DEPARTURES:

5.1.DEPARTURE INTENDING TO CONTINUE TO DESTINATION

Any departing aircraft experiencing RCF and intending to continue to its filed plan destination shall:

5.1.1.Continue on assigned SID or heading climbing to or maintaining cleared level or FL80, whichever is higher.

5.1.2.Three minutes after setting Mode A/C code 7600 or reaching FL 80 or cleared Flight level (if higher than FL80) whichever is later:

i.If following 'SID', continue on SID to join ATS route and climb to filed Flight level and continue as per the filed flight plan.

ii.If following Radar heading, turn (avoiding VIP 89) to join ATS route by shortest route maintaining cleared flight level/FL 80 as per para 5.1.1. After joining ATS route, climb to filed Flight level and Continue as per the filed flight plan. Departure following ATS Route V18 and G452/L760 shall pass FL160 or above by 50NM DPN and aircraft filed flight level below FL150 shall contact Jewar Approach (Frequency: APP/ 120.825 MHZ, 122.175 MHZ & TWR/ 118.675 MHZ, 119.125 MHZ) after ascertaining that aircraft is RCF with Delhi route.

5.2DEPARTURE INTENDING TO LAND BACK AT IGI Airport:

Any departing aircraft experiencing RCF after departure and intending to land back at IGI Airport shall

5.2.1Continue on assigned SID or heading climbing to or maintaining cleared level or FL80, whichever is higher.

5.2.2Three minutes after setting Mode A/C code 7600 or reaching FL80 or cleared Flight level (if higher than FL80) whichever is later, take a turn (avoiding VIP 89) to proceed direct to 'SKA' VOR.

5.2.3If higher, descend to FL80 in 'SKA' hold.

5.2.4If required, Jettison fuel while in 'SKA' hold, taking all necessary precautions.

5.2.5Leave 'SKA' VOR at FL80 to carry out published ILS/VOR DME ARC approach procedure for assigned runway. Maintain FL080 until 20 DME DPN VOR.

NOTE :-
1.In Case Of Non-availability Of SKA VOR aircraft required to hold at SKA VOR may hold at following FIX/WAY POINT:

A. R186/43 D (116.1 DPN)
B. 275056N 0770030E

II.ARRIVAL, DEPARTURE PROCEDURE AND RUNWAY CAPACITY ENHANCEMENT MEASURES

The traffic has been growing at Delhi Airport at a faster rate which necessitates the optimum use of resources to meet our requirement of safe, efficient and expeditious handling of air traffic. Runway Occupancy Time has been one of the important factors affecting such requirement. The following procedures and important Aerodrome data are published (TORA/LDA from different entry/exit TWYs) which will facilitate the Airlines/Pilots in achieving minimum Runway Occupancy Time for common goal of enhancing runway capacity utilizing the existing available resources.

1. <u>DELHI DEPARTURE PROCEDURE</u>

1.1 Speed Restrictions:

1.1.1 All aircraft following RNAVSIDs shall comply with level and speed limitations as published in the procedures unless explicitly cancelled by ATC.

1.2 **To reduce RT congestion, flight crew is encouraged:**

1.2.1 NOT to request direct routings in Delhi TMA. ATC will provide direct routings as and when feasible subject to traffic conditions.

1.2.2 NOT to enter into unnecessary arguments on R/T. Communication on R/T must be restricted to necessary transmissions only, using standard phraseology. Pilots must adhere to ATC instructions unless they feel that the, safety is being jeopardized or they are unable to comply with the same due aircraft performance limitation or operational reasons. In such cases the pilot may ask for revised / alternate clearance / instructions.

1.3. Pre Departure Procedures:

1.3.1 The aircraft should be in a position to commence its taxi in not more than five minutes after the issue of pushback and start-up clearance, failing which the start-up clearance will be cancelled.

1.3.2 Taxiing aircraft should maintain a minimum taxiing speed of not less than 15 KT on the straight portion of taxiways and between 8-12Kts during turning manoeuvres.

1.3.3 Based on the aircraft type and its performance characteristics, ATC will issue taxiing instructions so as to depart from the nearest runway intersection from where adequate take-off run is available for departure. Pilots unable to accept departure from intersection may request ATC for alternate take-off position. Pilots requiring departure from the beginning of runway should make such request at the time of Pushback/Startup.

1.3.4 Pilot shall complete all mandatory pre-departure checks before entering the active runway for departure so that the aircraft is in a position to take-off immediately upon receipt of take-off clearance in one continuous movement.

1.3.5 When the aircraft is issued with take-off clearance after lining up on the runway it shall commence take-off roll immediately.

2. <u>DELHI ARRIVAL PROCEDURES</u>

2.1 **Speed Restrictions:**

2.1.1 All aircraft following RNAV STARs shall comply with speed limitations as published in the procedures.

2.1.2 Arriving aircraft at or below F290 to maintain speed 270 KT and thereafter follow the speed restrictions as published in STAR, unless otherwise superseded by ATC instructions.

2.1.3 All aircraft entering Delhi TMA, except aircraft navigating under conditions of RNAV STARs, shall follow IAS as per the following, unless otherwise instructed by ATC:

2.1.3.1 250 KT – 230 KT: 60 NM from DPN;

2.1.3.2 210 KT: from downwind leg to base leg;

2.1.3.3 190 KT: on base leg and closing heading to final approach;

2.1.3.4 180 KT: 20 NM to 10 NM from touchdown on final approach;

2.1.3.5 160 KT: 10NM to 5 NM from touchdown.

2.1.4 These speeds are applied for ATC separation purposes and are mandatory.

2.1.5 All speed restrictions are to be flown as accurately as possible. Aircraft unable to comply with these speeds must inform ATC and state what speeds can be used. Pilots should also advise ATC if circumstances necessitate a change of speed due to aircraft performance limitation.

2.2. To reduce RT congestion, flight crew is encouraged:

2.2.1 NOT to request direct routings in Delhi TMA. ATC will provide direct routings as and when feasible subject to traffic conditions.

2.2.2 NOT to enter into unnecessary arguments on R/T. Communication on R/T must be restricted to necessary transmissions only, using standard phraseology. Pilots must adhere to ATC instructions unless they feel that the safety is being jeopardized or they are unable to comply with the same due to aircraft performance limitation or operational reasons. In such cases the pilot may ask for revised / alternate clearance / instructions.

2.3. Runway Occupancy Time:

2.3.1 To ensure maximum runway utilization, arriving flights, on completion of landing roll are expected to vacate expeditiously at first suitable taxiway exit, or as instructed by ATC. Aircraft should not slow down below normal taxi speed or stop on exit taxiway unless otherwise approved by ATC.

2.3.2 Pilots are reminded that rapid exit from the runway enables ATC to apply minimum spacing on final approach that will achieve maximum runway utilization and will minimize the occurrences of 'go around'.

2.3.3 Arriving aircraft to use preferred exit taxiway as per the table below. Aircraft that cannot comply with these requirements, shall notify ATC, as soon as, possible.

RWY	Aircraft Code	Preferred exit Taxiway	Distance from threshold (M)
09	A,B,C	D3	2099
	D,E	D2	2773
10	A, B, C	J1 (exiting towards South)	1972
	D,E	J2(exiting towards South)	2436
	F	J3(exiting towards South)	2961
	A,B,C	G1(exiting towards North)	1972
	D,E	G2(exiting towards North)	2436
11R	A,B,C	Y1	1898
	D, E	Y2	2247
	F	Y4	2648
27	A,B,C	D5	1862
	D,E	D6	2162

28	A,B,C,D	K1(exiting towards South)	2212
	E	K2(exiting towards South)	2839
	F	K6(exiting towards South)	3813
	A,B,C	H1(exiting towards North)	2312
	D,E	H2(exiting towards North)	2752
29L	A,B and all turbo prop aircraft	Z2	2067
	C	Z2	2067
	D,E	Z3	2316
	F	Z4	2700
29R	A,B	R1	1779
	C	R2	2010
	D,E	R3	2460
	F	R4	2950
11L	A,B,C	P1	1991
	D,E	P3	2441
	F	P4	2992

III) DELHI AIRPORT COLLABORATIVE DECISION MAKING (DA-CDM)

1 BACKGROUND

1.1The Delhi Airport - Collaborative Decision Making (DA-CDM) undertaken at IGI Airport is a joint programme among all airport partners –

•Air Navigation Service Provider (ATC)

•Airline Operators (AO)

•Air Traffic Flow Management Unit (ATFMU)*

•Delhi International Airport Pvt. Limited (DIAL)

•Ground Handlers (GH)

•Support services (Police, Customs and Immigration, Air Force Movement Liaison Unit, etc.)

All the partners are required to work in close collaboration to ensure the successful operation of DA-CDM.

1.2The efficiency of the Air Transport System is highly dependent on traffic predictability. DA-CDM effectively enhances predictability (this reduces buffer times for resource planning and flight times), overall efficiency and punctuality by linking and sharing of accurate and timely information amongst Airlines, Airport Operator, ATC, etc.

1.3The primary objective of DA-CDM is to facilitate the sharing of operational data for a better informed, well planned and transparent decision making to ensure more precise overall operational processes. It leads to an optimized utilization of resources, an efficient turn round process and everyone has a common awareness of the situation.

1.4Through implementation of DA-CDM, long-lasting benefits can be reaped.

Benefits of DA-CDM are:

•Improved overall efficiency of the operation

•Reduced delays and increased punctuality

•Reduced ATFM slot wastage*

•Optimized en-route capacity

•Optimized use of the airport infrastructure

•Reduced apron and taxiway congestion

•Cost savings

•Reduction of carbon emission and noise pollution

•Enhanced predictability

•Optimized use of ground handling resources

•Optimized use of stands, gates and terminals

•Enables flexible pre-departure planning

2 TERMS AND ABBREVIATIONS

All Acronyms are time parameters which have a standard length of four characters and have been arranged as per aircraft movement sequence.

[While developing the DA-CDM procedures, the guidance material published by Euro control, ICAO and other Organizations has been used].

S. NO.	ACRONYMS	DEFINITION	EXPLANATION
2.1	ELDT	Estimated Landing Time	The estimated time that an aircraft will touchdown on the runway. (Equivalent to ATC ETA = Estimated Time of Arrival = landing).
2.2	TLDT	Target Landing Time	Targeted Time from the Arrival management process at the threshold, taking runway sequence and constraints into account. It is not a constraint but a progressively refined planning time used to coordinate between arrival and departure management processes. Each TLDT on one runway is separated from other TLDT or TTOT to represent vortex and/or SID separation between aircraft.
2.3	ALDT	Actual Landing Time	The time that an aircraft lands on a runway. (Equivalent to ATC ATA = Actual Time of Arrival = landing, ACARS = ON).
2.4	EXIT	Estimated Taxi-In Time	The estimated taxi time between landing and in-block.
2.5	AXIT	Actual Taxi-In Time	AXIT = AIBT – ALDT
2.6	SIBT	Scheduled In-Block Time	The time that an aircraft is scheduled to arrive at its first parking position.
2.7	EIBT	Estimated In-Block Time	The estimated time that an aircraft will arrive in-blocks. (Equivalent to Airline/Handler ETA = Estimated Time of Arrival).
2.8	AIBT	Actual In-Block Time	The time that an aircraft arrives in-blocks. (Equivalent to Airline/Handler ATA = Actual Time of Arrival, ACARS = IN).
2.9	ACGT	Actual Commence of Ground Handling Time	The time when ground handling on an aircraft starts, can be equal to AIBT *(to be determined locally)*
2.10	ASBT	Actual Start Boarding Time	Time passengers are entering the bridge or bus to the aircraft.
2.11	AEGT	Actual End of Ground handling Time	The time when ground handling on an aircraft ends, can be equal to ARDT *(to be determined locally)*
2.12	AGHT	Actual Ground Handling Time	The total duration of the ground handling of the aircraft. AGHT = ACGT – AEGT
2.13	STTT	Scheduled Turn-round Time	STTT = SOBT – SIBT
2.14	ETTT	Estimated Turn-round Time	The time estimated by the AO/GH on the day of operation to turn-round a flight taking into account the operational constraints.
2.15	MTTT	Minimum Turn-round Time	The minimum turn-round time agreed with an AO/GH for a specified flight or aircraft type.
2.16	ATTT	Actual Turn-round Time	ATTT = AOBT – AIBT
2.17	ASRT	Actual Start Up Request Time	Time the pilot requests start up clearance.
2.18	TSAT	Target Start Up Approval Time	The time provided by ATC taking into account TOBT, *CTOT** and/or the traffic situation that an aircraft can expect start-up / push back approval. Note: The actual start up approval (ASAT) can be given in advance of TSAT.
2.19	ASAT	Actual Start Up Approval Time	Time that an aircraft receives its start-up approval.
2.20	SOBT	Scheduled Off-Block Time	The time that an aircraft is scheduled to depart from its parking position.
2.21	EOBT	Estimated Off-Block Time	The estimated time at which the aircraft will start movement associated with departure (ICAO).

2.22	TOBT	Target Off-Block Time	The time that an Aircraft Operator or Ground Handler estimates that an aircraft will be ready, all doors closed, boarding bridge removed, push back vehicle available and ready to start up / push back immediately upon reception of clearance from the SMC (GROUND).
2.23	AOBT	Actual Off-Block Time	Time the aircraft pushes back / vacates the parking position. (Equivalent to Airline / Handlers ATD = Actual Time of Departure and ACARS = OUT)
2.24	ARDT	Actual Ready Time (for Movement)	When the aircraft is ready for start-up/push back or taxi immediately after clearance delivery, meeting the requirements set by the TOBT definition.
2.25	EXOT	Estimated Taxi-Out Time	The estimated taxi time between off-block and take off. This estimate includes any delay buffer time at the holding point prior to take off.
2.26	AXOT	Actual Taxi-Out Time	AXOT = ATOT – AOBT
2.27	ETOT	Estimated Take Off Time	The estimated take off time taking into account the EOBT plus EXOT.
2.28	CTOT*	Calculated Take Off Time*	A time calculated and issued by the appropriate Central Management unit, as a result of tactical slot allocation, at which a flight is expected to become airborne. (ICAO Doc 7030/4 – EUR, Table 7)*
2.29	TTOT	Target Take Off Time	The Target Take Off Time taking into account the TOBT/TSAT plus the EXOT. Each TTOT on one runway is separated from other TTOT or TLDT to represent vortex and/or SID separation between aircraft.
2.30	ATOT	Actual Take Off Time	The time that an aircraft takes off from the runway. (Equivalent to ATC ATD = Actual Time of Departure, ACARS = OFF).

3 TWO KEY PARAMETERS OF DA-CDM: TOBT and TSAT.

3.1TARGET OFF-BLOCK TIME (TOBT)

3.1.1Definition: The time that an Aircraft Operator or Ground Handler estimates that an aircraft will be ready, all doors closed, boarding bridge removed, push back vehicle available and ready to start up/push back immediately upon reception of clearance from the SMC (GROUND).

3.1.2Purpose: It is a fact that Air Traffic Management (ATM), based on FPL-data only, is insufficient to guarantee smooth and efficient operations on the ground and in the air. The main reason for this is the incorrect updating/no updating at all, of the Estimated Off-Block Time (EOBT) of the flight plan. The result is that airport partners have an incorrect idea about the actual status of that specific flight.

Target Off-Block Time (TOBT) is the dynamic target on the operational readiness of a flight thus, more accurate than the static departure time of a flight plan and is therefore, an ideal milestone to be used by all airport partners.

3.1.3When/How to calculate TOBT: In order to create an initial time reference, at EOBT - 45 minutes, TOBT will be calculated by adding Estimated In-block Time (EIBT) + Minimum Turn-round Time (MTTT) for distribution and to derive further updates.

When the aircraft arrives on stand at Actual In-Block Time (AIBT), TOBT is updated by AO/GH based on MTTT or the Scheduled Off-Block Time (SOBT). These MTTT’s are regularly updated and Aircraft Operators are consulted for the most accurate times.

Refining of TOBT (Example)

TOBT based on EOBT [At EOBT - 45 minutes]

TOBT = EIBT (ELDT issued at ELDT-20 minutes + EXIT) + MTTT

TOBT = EIBT (ELDT at 10NM from touchdown + EXIT) + MTTT

TOBT = EIBT (ALDT + EXIT) + MTTT

TOBT = AIBT + MTTT

3.1.4Originator(s): Airlines Operator/Ground Handler.

3.1.5Communication of Target Off Block Time (TOBT):

(i)AO/GH shall update their TOBTs via DA-CDM portal.

(ii)If there is no input in DA-CDM portal, EOBT itself will be treated as TOBT.

3.1.6When to update: The aim of the Target Off-Block Time (TOBT) is to provide a timely, accurate and reliable estimate to the DA-CDM partners, of an aircraft's off block time. Accurate TOBTs enhance operations on the ground as they provide all airport partners with a clear picture of the intentions of aircraft on the ground.

TOBTs only need to be updated if the time that the aircraft will be ready to leave stand changes.

Once Target Start-up Approval Time (TSAT) is allocated by ATC at TOBT- 20 minutes, the TOBT may be updated thrice (equal to or less than TSAT).

3.1.7Accuracy: TOBTs must be updated to an accuracy of +/- 5 minutes.

3.2TARGET START UP APPROVAL TIME (TSAT)

3.2.1Definition: The time provided by ATC taking into account TOBT, CTOT* and/or the traffic situation that an aircraft can expect start-up/push back approval.

Note: The actual start up approval (ASAT) can be given in advance of TSAT.

3.2.2Purpose: TSATs provide an optimized start-up sequence, leading to Pre-Departure sequence. TSATs reduce queuing times at the runway holding points, while maintaining a high degree of runway utilization. ATC will continue to maximize departure rates and optimize the departure order for creating the right mix of traffic.

3.2.3How to calculate/generate TSAT: The TSAT is calculated by taking into account TOBT, Calculated Take-Off Time (CTOT)*, wake vortex, Standard Instrument Departure (SID) routing, variable taxi times, and any capacity constraints such as Low Visibility Procedures.

ELDT + EXIT = EIBT
EIBT + MTTT = TOBT
TOBT + EXOT + Constraints = TTOT
TTOT - EXOT = TSAT
ELDT: Estimated Landing Time
EXIT: Estimated Taxi-In Time
EIBT: Estimated In-Block Time
MTTT: Minimum Turn-round Time
TOBT: Target Off-Block Time
EXOT: Estimated Taxi-Out Time
TTOT : Target Take Off Time
TSAT: Target Start Up Approval Time

Note: A re-assessment of a flight on the network (re-calculation of CTOT) will be done if the TTOT is outside the Slot Tolerance Window (STW = CTOT-5 till CTOT+10 minutes). The Calculated Take-Off Time becomes tailor-made where possible.*

3.2.4 Originator(s): ATC.

3.2.5 When to issue TSAT: At TOBT - 20 minutes by ATC in DA-CDM portal.

3.2.6 Communication of Target Start-up Approval Time (TSAT): The Target Start-up Approval Time (TSAT) will be displayed in the DA-CDM portal. Pilots will be informed of their Target Start-up Approval Time (TSAT) through DA-CDM portal and any changes to it by ATC. The AO/GH shall advise flight crew of TSAT, displayed in the DA-CDM portal.

3.2.7When to update: The factors that may cause a revision of TSAT include but not limited to:

(i) Change of runway

(ii) Change in taxi time

(iii) Revised TOBT

(iv) Change of route (SID)

(v) Change of aircraft type

(vi) Application of ATFM regulation or ATC restriction resulting in new CTOT or start up delay*

(vii) Inclement Weather

(viii) Priority handling

(ix) Bird activity

(x) Unusual Occurrences, etc.

In case of TSAT revision, ATC will re-calculate a new TSAT and an alert in the DA-CDM portal will notify the AO/GH. The pilot will be informed by ATC of any revised TSAT if there is a delay to TSAT in excess of 5 minutes.

3.2.8Accuracy: TSAT +/- 5 minutes.

3.3 Co-ordination with the Air Traffic Flow Management Unit (ATFMU)*

3.3.1 A permanent and fully automatic data exchange with the ATFMU will be established. The data transfer will enable highly accurate early predictions of landing and departure times.*

3.3.2 Furthermore, it allows for more accurate and efficient CTOT calculations due to the use of more accurate local target take-off times.*

4 TOBT and TSAT based DA-CDM Start-up/Push-back procedures

4.1A Flight Plan Check shall be performed by DA-CDM Cell, to the extent possible, two hours prior to SOBT/EOBT to verify the consistency between SOBT and EOBT. The EOBT must correspond to the SOBT. If the EOBT deviates from the SOBT, the relevant contact person will be informed by DA-CDM Cell and advised to adjust the times accordingly. AO/GH shall be responsible to obtain ADC number from IAF-MLU.

4.2Estimated Landing Time (ELDT): ATC will input Estimated Landing Time (ELDT) at ELDT - 20 minutes on DA-CDM portal. At 10 NM to touchdown, report is generated automatically by ATC automation system [AutoTrac III (AT3)] and sent to DIAL through AODB link.

4.3Estimated In-block Time (EIBT): This estimate will be calculated by adding Variable Taxi Time (VTT) [Estimated Taxi-In Time (EXIT)] from the RWY to the Stand/Gate automatically by DA-CDM portal using VTT matrix.

4.4Pre-Departure Sequence: Pre-Departure sequence is the order, in which the ATC plans the aircraft to depart from their Gate/Stands (Push-off blocks). It should not be confused with the Pre-Take-off sequence where ATC decides on the order in which the aircraft at holding points of the runway will depart.

4.5. Data link clearance facility is operational at IGI airport,DELHI. In accordance with best equipped best served policy the flights receiving clearance through data link may contact relevant SMC/GROUND frequency directly as per the TSAT(TArget Start-up Approval Time) issued.

4.6 Start-Up and Push-Back Procedure

i.The pilot should contact Clearance Delivery (CLD) to request en-route clearance and SID between TOBT - 15 minutes to TOBT - 5 minutes.

ii.The aircraft must be ready to Start-up/Push-back at TOBT and request SMC (GROUND) for Start-up/Push-back at TOBT.

(Note: TSAT can be preponed up to TOBT)

iii.If at TOBT +5 minutes, ATC has not received Start-up/Push-back request, the aircraft may lose its position in sequence.

a.ATC will advise the pilot that a new TOBT is required.

b.On receipt of new TOBT, the flight will be re-sequenced according to new TOBT and a new TSAT with a subsequent delay will be issued.

iv.ATC should normally be able to issue start-up/push-back at TSAT. Pilots will be informed by ATC of any revised TSAT if there is a delay to TSAT in excess of 5 minutes.

v.Taxi clearance must be requested within 5 minutes of Start-up/Push-back approval time.

(Note: If this has not occurred, SMC (GROUND) must be notified of the extent of delay. In such cases, aircraft may lose its departure slot and a new TOBT may be required in DA-CDM portal. AO/GH shall be responsible to obtain new ADC number from IAF-MLU).

vi.The Pre-Departure (Off- block) Sequence will be determined in accordance with Target Start up Approval Time (TSAT) and NOT in accordance with the Start-up Request. Pre-Departure Sequence will not have any bearing on Actual Departure sequence.

(Note: Actual Departure sequence may differ from pre-departure sequence in order to optimize the Runway and Airspace utilization.)

* [Future provision kept for incorporating with India – Central Air Traffic Flow Management]

IV. INORMATION REGARDING INTERSECTION DEPARTURES

1. Details of TORA for intersection departure taxiways:

Runway	Taxiway	TORA from Taxiway given in Meters
28	J5	3356
	J6	3568
	G3	3157
	G4	3259
	G5	3356
10	H3	3265
	K2	2839
	H2	2752
	K3	3235
	K5	3723
27	D2	2773
	D3	2099
09	D8	2765
	D7	2683
	D4	1842

29L	Y5	3932
	Y6	4334
	Y7/Y8	4430
11R	Z5	3933
	Z4	3834
29R	P8	4152
	S8	4152
	P7	4027
	S7	4027
	P6	3929
	S6	3929
	P5	3523
	P4	3432
	P2	2725
	R1	1684
11L	R7	3670
	T7	3670
	R6	3572
	T6	3572
	R5	3307
	R4	2950
	R1	1779

2. Location of different Exit TWYs from THR:

Runway	Exit Taxiway	Distance of exit TWY from THR(M)
29L	Z2(RET)	2067
	Z3(RET)	2316
	Z4	2700
	Z5	2800
	Z6	2970
11R	Y1(RET)	1898
	Y2(RET)	2247
	Y4(RET)	2648
	Y5	3294

28	K1 (RET)	2212
	H1 (RET)	2312
	K2 (RET)	2839
	H2	2752
	H3	3265
	H6	3813
	K3	3235
	K5	3723
	K6	3813
10	G1(RET)	1972
	G2(RET)	2436
	J1 (RET)	1972
	J2 (RET)	2436
	J3(RET)	2961
	G5	3356
	G4	3259
	G3	3157
	J5	3356
	J6	3568
	J7	3813
27	D4	1691
	D5(RET)	1862
	D6 (RET)	2162
	D7	2532
	D8	2614
	D9	2665
09	D3	2099
	D2	2773
	D1	2816

29R	R1	1779
	R2(RET)	2010
	R3(RET)	2460
	R4	2950
	R5	3307
11L	P1(RET)	1991
	P2	2285
	P3(RET)	2441
	P4	2992

3. Preferred Exit Taxiways

Refer table under sub para 2.3 Runway Occupancy Time

V. Speed control Procedure

a)Speed control Procedure under non Radar environment shall be as specified in ENR1.1 para 2.5 of eAIP India.

b)Arriving ACFT passing F290, to maintain speed 270KTS and thereafter follow the speed restrictions as published in STAR,unless otherwise superceded by ATC instructions.

VI. IMPLEMENTATION OF TACTICAL AIR TRAFFIC FLOW PROCEDURES (T-ATFP)

1 Introduction

1.1 In order to reduce airspace congestion, arrival delays and holding in Delhi TMA, Tactical Air Traffic Flow Control Procedure (T-ATFP) is introduced. For this purpose, sequenced entry into airspaces is established and maintained.

1.2 T-ATFP is applied to aircraft landing at Delhi airport, classified as follows:

1.2.1 Arrivals from airports within Delhi FIR;

1.2.2 Arrivals from airports outside Delhi FIR but within 10 min flying time from Delhi FIR boundary;

1.2.3 Arrivals from airports other than those in 1.2.1 and 1.2.2

2 Acronyms used in the document and associated expansion

AAO	Airline and aircraft operators
CTO	Calculated Time Over
ETO	Estimated Time Over
RTO	Required Time Over
T-ATFP	Tactical Air Traffic Flow Procedures
TM	Traffic Manager (for the purpose of T-ATFP)

3 Terminology and their meaning

3.1 AAO (Airline and aircraft operators) – includes all aircraft operators including defence, State Govt., paramilitary etc.

3.2 Block Time – Time required for arriving over runway THR following a STAR flight path and adhering to STAR speed restrictions.

3.3 Calculated Time Over (CTO) – Time at which departure from airports at 1.2.1 and 1.2.2 should arrive over the arrival holding fix to fit into established arrival sequence. By back calculation, the time of takeoff can be computed.

3.4 Required Time Over (RTO) – Time at which arrivals in 1.2.3 should arrive over the arrival holding fix to avoid joining the holding.

4 Traffic Manager (TM)

4.1 The Traffic Manager (TM) position is established within Delhi ATC for the implementation of T-ATFP. TM will have access to flight plans and estimates and revisions to estimates of all arrivals landing at Delhi airport.

4.2 Functions of TM

4.2.1 Establishing a sequence of arrivals landing at Delhi airport based on touchdown time.

4.2.2 Assign landing runways to arrivals; whenever more than one runway is available for landing.

4.2.3 Advise CTO/RTO to ACC Sector controller.

4.2.4 Advise CTO/RTO, arrival sequence and assigned landing runway to Approach (Arrival) Controller.

4.3 For effective and efficient management of arrival sequence, TM shall:

4.3.1 Make available a CTO arrival holding fix for the departures from airports as in 1.2.1 and 1.2.2.

4.3.2 Provide information to arrivals as in 1.2.3, regarding RTO arrival holding fix.

4.4 Based on the ETO at arrival holding fix, sequence of arrivals will be determined by TM.

5 Essential requirements

5.1Arrivals landing at Delhi airport shall flight plan via arrival route (STAR) appropriate to the ATS route. To maintain arrival sequence, non-RNAV arrivals to expect routing/vectors along RNAV STAR.

5.2All flight-plans for landing at Delhi airport shall provide ETO arrival holding fix - AKBAN, SAM, SSB and SAPLO as appropriate, for determination of arrival sequence based on time required to reach runway touchdown point from the arrival holding fix.

5.3Departures from airports within Delhi FIR and within 10 min flying time from Delhi FIR shall coordinate with TM for CTO. Uncoordinated departure from such airports shall be refused entry into Delhi TMA or will be subject to extended delays.

6 Tactical Flow Management Process

6.1For ensuring efficient flow during T-ATFP, speed management of sequenced aircraft is a critical factor. Approach (Arrival) Controller shall enforce adherence to speed restrictions.

6.2Delhi ACC Sector controllers will clear all arrivals via STAR. Delhi Radar shall vector Non-RNAV arrivals along appropriate STAR route segment with speed control instructions consistent with STAR segment, so as to ensure that the arrival sequence computed by TM is maintained.

6.3The arrival sequence is calculated based on the Block Time plus separation at touchdown point between successive arrivals. During the initial phases of T-ATFP implementation; the separation over touchdown point will be 2 min.

6.3.1The separation over touchdown point is dependent on dynamic factors such as:

•Weather

•Equipment outage

•Runway closure

•Airspace closure

•Contingency such as RCF, Unlawful interference etc

6.4TM shall advise Delhi ACC sector controller and Approach (Arrival) controller information on RTO. Accordingly, arrivals will be advised of RTO thereby facilitating in gaining/losing time so as to arrive over arrival holding fix at RTO.

6.5For ensuring efficient flow based on the arrival sequence, speed management is critical. Approach (Arrival) controller may enforce speed control based on the STAR restrictions and on final approach.

6.6Safety being primary, minimal radar vectoring may be resorted thereby permitting aircraft to adhere to FMS managed flight profile/path of RNAV STAR.

6.7The arrival runway allocation and sequence established by TM, may not be altered by Delhi Approach; as far as practicable, except during emergency, RCF, unlawful interference or during events requiring priority handling.

6.8When adverse conditions such as weather, equipment outage, runway closure and/or airspace closure etc affect Delhi airport/Delhi TMA, the capacity of the airport and TMA could be severely degraded, resulting in unoptimised traffic flows. Therefore, in order to avoid excessive holding, airborne delay and congestion during such events; following T-ATFP measures will be implemented:

6.8.1Departure airport ATS units shall be advised of the restricted arrival capacity at Delhi and CTO to be provided for all departures before start-up based on flight-plan ETD. Any delay beyond flight-plan ETD will require revised CTO arrival holding fix to be obtained.

6.8.2During events enumerated in Para 6.8, departures from airports within Delhi FIR and within 10 min flying time from Delhi FIR do not have the option to take-off and absorb delay enroute or by holding within Delhi TMA.

7) AAO responsibility

7.1AAOs are to ensure they are fully informed on the following procedures:

7.1.1AAOs shall adhere to the speed control restrictions of the STAR or speed control instructions issued by Delhi ATC.

7.1.2AAOs shall arrange the departure of their flights to comply with the CTO issued.

7.1.3Non-adherence to CTO may result re-sequencing. TM will issue RTO which may result in disproportionate arrival delay.

8 ATS Unit responsibility

8.1ATS Units shall make every effort to enable departing flights to comply with CTO.

8.2ATC shall provide all possible assistance to AAOs to meet CTO/RTO.

9 Arrivals exempted from T-ATFP

9.1The following flights are exempted from T-ATFP:

9.1.1Arrival in a state of emergency, including flights subject to unlawful interference

9.1.2Arrival conducting search and rescue operations

9.1.3Arrival carrying notified VIP

9.1.4Arrival with medical emergencies

VII. TRANSPONDER OPERATING PROCEDURES ON GROUND

1. Introduction:

Advanced Surface Movement Guidance and Control System (A-SMGCS) using Mode-S is commissioned at Indira Gandhi International Airport, Delhi. The Aircraft Transponder Operating Procedures applicable in movement area of the airport are as given below:

2. DEPARTURE

2.1.At the Gate/Stand:

•Select STBY

•Enter the discrete SSR code received from Clearance Delivery/Surface Movement Control. Enter the three letter ICAO designator followed by the flight identification number (e.g. AIC748) through the FMS or the Transponder control panel, depending on the avionics.

2.2On requesting Pushback/Taxi (whichever is earlier):

• Select Transponder or equivalent and AUTO if available

•This action will enable the aircraft ID, used as the Call sign by ATC, to be displayed on the surveillance display of ATC. ATC can verify the data and use it for necessary identification.

2.3 When Lining up:

•Select TCAS

•Select TCAS only after receiving the clearance to line up, to ensure that the performance of systems based on SSR frequencies (including airborne TCAS units, SSR and A-SMGCS) is not compromised.

3. ARRIVAL

3.1.When on the Runway:Keep TCAS selected

3.2.After vacating the Runway:

•Select Transponder or equivalent and AUTO if available

•There is a need that the Transponder remains able to exchange data with the A-SMGCS system. However, to ensure that the performance of systems based on SSR frequencies (including airborne TCAS Unit, SSR and A-SMGCS) is not compromised, TCAS shall be deselected when vacating the Runway.

3.3.Parked on Stand:

•Select STBY

•The Transponder will not reply to interrogation. The discrete SSR Code given to that particular flight can now be recycled for other flights.

4. SHIFTING (TOWING/TAXING) OF AIRCRAFT:

4.1 At the gate/stand: Call Surface Movement Control for shifting after getting approval for shifting from AOCC,

- select STBY,
- enter the SSR code 1400,
- enter the aircraft registration (e.g. VTABC) through the FMS or the transponder control panel, depending on the avionics.

4.2 On requesting Pushback/Startup/Tow or Taxi (whichever is earlier):

- Select transponder or equivalent and Auto if available
- This action will enable the aircraft ID, used as the call sign ATC to, to be displayed on the surveillance display of ATC for identification and verification of data.

4.3 When taxi, towing is completed and aircraft is parked of destination stand:

- Select STBY
- The transponder will not reply to interrogation.

Note: When on ground the aircraft must squawk Mode C, in order to provide the altitude information to the surveillance system, and thereby prevent

i)clutter on Terminal Approach Radar Display (and)

ii)false automatic detection of departure for aircraft still on ground.

VIII. LOW VISIBILITY PROCEDURES (LVP)

Every year due to low visibility conditions during the period From 15 DEC to 15 FEB, aircraft operator are not permitted to file IGI airport as alternate airport, if the latest met forecasted general visibility is expected to be below 550 Meter, at estimated time of use.

1. DEFINITIONS and ABBREVIATIONS

1.1 Instrument Approach Operations. An approach and landing using instruments for navigation guidance based on an instrument approach procedure. There are two methods for executing instrument approach operations:

a) a two-dimensional (2D) instrument approach operation, using lateral navigation guidance only; and
b) a three-dimensional (3D) instrument approach operation, using both lateral and vertical navigation guidance.

Note. — Lateral and vertical navigation guidance refers to the guidance provided either by:

a) a ground-based radio navigation aid; or
b) computer-generated navigation data from ground-based, space-based, self-contained navigation aids or a combination of these.
Instrument approach operations shall be classified based on the designed lowest operating minima below which an approach operation shall only be continued with the required visual reference as follows:

a) Type A: a minimum descent height or decision height at or above 75 m (250 ft); and
b) Type B: a decision height below 75 m (250 ft).

Type B instrument approach operations are categorized as:
1) Category I (CAT I): a decision height not lower than 60 m (200 ft) and with either a visibility not less than 800 m or a runway visual range not less than 550 m;
2) Category II (CAT II): a decision height lower than 60 m (200 ft), but not lower than 30 m (100 ft) and a runway visual range not less than 300 m;
3) Category III (CAT III): a decision height(DH) lower than 30 m (100 ft), or no decision height and a runway visual range less than 300 m or no runway visual range limitations.

Note 1.— Where decision height (DH) and runway visual range (RVR) fall into different categories of operation, the instrument approach operation would be conducted in accordance with the requirements of the most demanding category
Note 2. — The required visual reference means that section of the visual aids or of the approach area which should have been in view for sufficient time for the pilot to have made an assessment of the aircraft position and rate of change of position, in relation to the desired flight path. In the case of a circling approach operation the required visual reference is the runway environment.
Note 3.— Guidance on approach classification as it relates to instrument approach operations, procedures, runways and navigation systems is contained in the All Weather Operations Manual (Doc 9365).

1.2 Decision Height: A specified altitude or height in a 3D instrument approach operation at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.
Note. — Decision altitude (DA) is referenced to mean sea level and decision height (DH) is referenced to the threshold elevation.

1.3 ILS Critical Area: An area of defined dimensions about the localizer and glide path antennas where aircraft and vehicles are excluded during all ILS operations. The critical area is protected because the presence of vehicles / or aircraft inside its boundaries will cause unacceptable disturbance to the ILS signal-in-space.

1.4 ILS Sensitive Area: An area extending beyond the ILS critical area where the parking and/or movement of vehicles, including aircraft, are controlled to prevent the possibility of unacceptable interference to the ILS signal during ILS operations. The sensitive area is protected to provide protection against interferences cause by large moving objects outside the critical area but still normally within the airfield boundary.

1.5 Low Visibility Procedures: Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs.

1.6 Low Visibility Take-Off (LVTO): A term used in relation to flight operations referring to a take-off on a runway where the RVR is less than 400m.

1.7 Obstacle Free Zone: The airspace above the inner approach surface, inner transitional surface and balked landing surface and that portion of the strip bounded by these surfaces, which is not penetrated by any fixed obstacle other than of low mass and frangible mounting, required for air navigation purposes.

1.8 Runway Visual Range: The range over which the pilot of an aircraft on the centerline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centerline.

1.9 Safeguarding Procedures: Safeguarding Procedures (SP) are instructions for relevant airport departments and airside operators to prepare ground services and facilities for low visibility operations, in order that when LVP are implemented all SP are complete and airport is configured for Low Visibility Procedures and Low Visibility Take-offs.

1.10 The abbreviations used in descriptions of Low Visibility Procedures have the following meanings:

ADC	Aerodrome Control
ARFF	Airport Rescue and Fire Fighting
AGL	Aeronautical Ground Light
ASMGCS	Advance Surface Movement Guidance Control System
AOCC	Airport Operations Control Centre
ATC	Air Traffic Control
ATIS	Automatic Terminal Information Service
GP	Glide Path
IHP	Intermediate Holding Position
ILS	Instrument Landing System
LOC	Localizer
LSA	Localizer Sensitive Area
LVP	Low Visibility Procedure
MET	Meteorology
MID	Mid Point
ODM	Operations Duty Manager
RVR	Runway Visual Range
SMC	Surface Movement Control
SP	Safeguarding Procedures
SSO	Shift Supervisory Officer
TDZ	Touch-down Zone
WSO	Watch Supervisory Officer

2. INTRODUCTION

2.1 General

2.1.1 RWY 10,RWY 28, RWY 11R, RWY 29L, RWY 11L and RWY 29R at IGI Airport are equipped with the CAT II/ CATIII operations.
2.1.2 The following equipment shall be serviceable to the required standard to support CAT II/CAT III operations: -

a) ILS localizer, Glide path and ILS DME or Outer Marker.
b) Aeronautical ground lighting (AGL) System
c) RVR system
d) Standby power for ILS and aeronautical ground lighting system
e) SMR for CAT II/CAT III operations (for RVR below 350M) /LVTO (for RVR below 350M).
(DGCA CAR Section 4 Series B Part 1, Para 9.8.7. — Surface movement radar for the manoeuvring area shall be provided at an aerodrome intended for use in runway visual range conditions less than a value of 350 m.)
The pilot shall ensure that he is suitably qualified and certified to carry out the required category of ILS approach.
2.1.3 RWY 09/27 and RWY 11L/29R also available for Low Visibility Take-off (LVTO) in addition to availability of RWY 10/28 and RWY 11R/29L for Low Visibility Take-off (LVTO).

2.2 Safeguarding Procedures (SP)

2.2.1 Safeguarding Procedures are the necessary actions to prepare the airport for Low Visibility Procedures. They include:
a. inspection of aeronautical ground lighting system,
b. termination of all work in progress in the localizer and glide path sensitive area and the manoeuvring area,
c. removal of all equipment/material from localizer and glide path sensitive area and the manoeuvring area,
d. restrictions on the movement of vehicles on the manoeuvring area and aprons.
2.2.2 Watch Supervisory Officer at IGI Airport will co-coordinate with all the concerned agencies for implementation of Low Visibility Procedures.
2.2.3 SP shall be initiated whenever ATC considers the introduction of Low Visibility Procedure is necessary.

2.3 Low Visibility Procedures (LVP)

2.3.1 Low Visibility Procedures are the actions to ensure the safe operation of aircraft during periods of reduced visibility or low cloud base.
2.3.2 LVP shall only be implemented when Safeguarding Procedures (SP) has been completed and the airport is configured for low visibility operations.

2.4 ILS Sensitive and Critical Areas

2.4.1 **Note:** Signage indicating the limits of localizer and glide paths sensitive areas are provided.
2.4.2 Diagram indicating the Critical and Sensitive area of ILS for all runways shall be available with apron control, control tower and WSO.
2.4.3 Refer VIDP AD 2.24 for ILS critical and sensitive areas of RWY 11L, RWY 29R, RWY 29L, RWY 11R and RWY 28

2.5 Reporting of RVR Values

2.5.1 Touch-down zone RVR needs to be reported for CAT I operations, touch-down and mid zone RVR for CAT II operations, touch-down, mid and roll-out zone RVR for CAT III operations. In all cases, touch-down zone will always be controlling, however if any other RVR is reported and is relevant (operator shall not define relevant depending on runway length/aircraft stopping distance unless approved by FSD, DGCA) it also becomes controlling. The mid zone and roll-out zone can be lower than the touch-down zone provided conditions enumerated in Note 1 below are met. The following table is to be used
for reference;

Type of Operation	RVR		
	Touch-down Zone	Mid zone	Roll-out zone
CAT I	550 M	125 M	125 M

CAT II	300 M	125 M	125 M
CAT III	175 M	125 M	125 M
CAT III (with roll-out guidance)	75/50 M	75/50 M	75/50 M

Note 1: The use of minimum RVRs in the table above is subject to:
* operator authorization;
* aeroplane authorization;
* flight crew training and qualification; and
* aerodrome facilities.

Note 2: The use of minimum RVR of 75m or 50m depends on value approved for operators and aeroplanes with roll-out guidance system.
Note 3: The values in bold font are required for the type of operation.

2.5.2 When reporting RVR to pilots, the TDZ RVR shall always be passed for the concerned Landing RWY.
2.5.3 In addition to Para 2.5.2

a) For CAT II Operations - If TDZ RVR is below 550m then MID RVR shall also be passed.
b) For CAT III Operations - If TDZ RVR is below 300m, MID and END RVR reading shall also be passed. TDZ, MID and END RVR shall be available.

3. IMPLEMENTATION OF SAFEGUARDING PROCEDURES (SP) AND LOW VISIBILITY PROCEDRUES (LVP)

3.1 Safeguarding Procedures (SP)

3.1.1 Criteria for Safeguarding Procedures:
Safeguarding Procedures shall be initiated when at either of the runways in use:
3.1.1.1 The RVR is less than 1200 m or visibility is forecast to deteriorate to 800m or less;and/or
3.1.1.2 The cloud ceiling is 400 ft and forecast to fall to 200 ft or less.

3.1.2 Actions to be taken for initiation of Safeguarding Procedures by WSO:
3.1.2.1 When meteorological conditions meet the criteria for initiation of SP as stipulated in Para 3.1.1, the Watch Supervisory Officer will inform AOCC, the Communication/Technical Supervisor and the Aerodrome Tower Supervisor.

3.1.2.2 When all the concerned agencies have completed their necessary actions they shall report to WSO (ATC) that their Safeguarding Procedure (SP) is completed and the airport is safeguarded for LVP operations.

3.2 Low Visibility Procedures (LVP)
3.2.1 Criteria for implementation of Low Visibility Procedures:
Low Visibility Procedures shall be implemented when –
a) Either, TDZ, MID or END RVR is less than 800 m; and/or Cloud ceiling is less than200 ft; and
b) Safeguarding Procedures (SP) have been completed and the airport is safeguarded.

(Note: Though LVP is implemented when RVR is less than 800m, ILS CAT I operation will continue till TDZ RVR is not less than 550m)
3.2.2 Actions to be taken upon Implementation of Low Visibility Procedures
3.2.2.1 When meteorological conditions meet the criteria for LVP as stipulated in Para 3.2.1, WSO shall implement Low Visibility Procedure and inform:
a) Aerodrome Tower Supervisor.
b) Approach Radar Controller
c) Communication/Technical Shift Supervisory Officer (SSO)
d) Duty Met. Officer

WSO will also ensure that **“LOW VISIBILITY PROCEDURE IN FORCE”** is included in ATIS Broadcast.

3.2.2.2 Upon implementation of LVP, Tower Supervisor will inform:
a) Fire station
b) AOCC
c) include “LOW VISIBILITY PROCEDURE IN FORCE” in ATIS Broadcast.

3.3 Cancellation of Safeguarding Procedures (SP) and Low Visibility Procedures (LVP)

3.3.1 WSO may terminate LVP when –
a) Meteorological conditions improve and TDZ, MID and END RVR are 800 m or more and the cloud ceiling is 200 ft or higher, and trend is for improvement for both runways.
b) Facilities and equipment (listed in Para 2.1.2) necessary for CAT II/CAT III operations are degraded and/or the prevailing conditions are considered unsafe for such operations.
3.3.2 WSO should consult Meteorological Office for forecast before cancelling SP and LVP.
3.3.3 On cancelling LVP, Aerodrome Control shall include it in the subsequent two ATIS broadcasts that "LOW VISIBILITY PROCEDURES ARE CANCELLED". Aerodrome Tower Controller will inform all the concerned agencies.
3.3.4 If SP are completed and LVP are not subsequently implemented and meteorological conditions improve and the visibility/RVR is more than 1200 m and the cloud ceiling is 400 ft or higher and both are forecast to remain above the required SP criteria, WSO may cancel SP.

4. LOW VISIBILITY PROCEDURE OPERATIONS

4.1 Approach/ Radar Control Procedures
4.1.1 During LVP the Approach Radar Controller shall have the following information
a) Status of ILS
b) Serviceability of visual aids
c) RVR information of TDZ, MID and END.
4.1.2 In addition to the information normally transmitted by Approach Radar Controller, the information specified in 2.5.2 and 2.5.3 must be passed to the arriving aircraft on first contact or as soon as possible along-with the unserviceability, if any, of any component parts of CAT II/CAT III facilities not previously broadcast on ATIS.
4.1.3 Approach/Radar Controller should vector the arriving aircraft to intercept the localizer at a distance not less than 10NM from touchdown.
4.1.4 Suitable spacing between the arriving aircraft may be provided to ensure that the arriving aircraft can be given a landing clearance by 2 NM from touchdown. If there is a departure between the two arrivals, the spacing between the arriving aircraft may be suitably increased to ensure that the departing aircraft passes overhead the Localizer before the inbound aircraft reaches 2NM from touchdown.
4.1.5 Approach Radar Controller shall not subject an aircraft carrying out CAT II/CAT III approaches to any speed control when within 20NM from touchdown.

4.2 Aerodrome Control Procedures
4.2.1 Arriving aircraft shall be issued landing clearance not later than 2NM from touchdown. If landing clearance cannot be issued when the aircraft is 2NM from touchdown it shall be instructed to carry out a missed approach.
4.2.2 Arriving aircraft should be given unimpeded taxi route to allow it to clear the localizer sensitive area expeditiously.
4.2.3 Landing clearance shall not be issued until: -
A preceding landing aircraft has vacated Localizer Sensitive Area [LSA] AND confirmed by the pilot.
A preceding departing aircraft is airborne and has passed over the localizer antenna.
4.2.4 The LSA in front of an arriving aircraft shall not be infringed from the time it is 2NM from the touchdown until it has completed its landing roll.
4.2.5 ADC shall initiate emergency action if an aircraft is not seen on the CWP/Radar Display and is not in radio contact as expected.

4.3 Surface Movement Control Procedures

4.3.1 Pilots need additional guidance and information when taxiing during periods of reduced visibility. The view from the cockpit of the aircraft is very limited. Therefore, taxi instructions and essential traffic information should be passed in a clear and concise manner.
4.3.2 Taxiing aircraft should be routed in accordance with the prescribed Low Visibility Routes to ensure a simple one-way traffic flow is maintained, however it may be necessary for operational reasons to sometimes route aircraft via alternative taxiways.
4.3.3 AGL team / Follow Me vehicles shall monitor the status of taxiway lights and immediately advise ATC Tower of any unserviceability affecting the taxiways during LVP operations.

During the period of LVP, the lights on taxiways that are not being used may be switched off.

4.3.4 Surface Movement Controller should monitor the progress of arriving aircraft on ASMGCS Display as they vacate the runway after landing and ensure that they do not stop within the Localizer Sensitive Area [LSA] thereby degrading ILS integrity for subsequent landing aircraft. Pilots shall report runway vacated on RTF when the aircraft has reached the color coded part of the exit taxiway centerline light after making allowance for aircraft size to ensure that the entire aircraft is clear of the ILS sensitive area.

4.3.5 Vehicle movement, when RVR is less than 550m, should be restricted. Only operationally essential vehicle duly authorized by Apron Control should be permitted to operate on the manoeuvring area. These vehicles shall remain outside the Localizer Sensitive Area [LSA].

Any movement of vehicle on the manoeuvring area shall be coordinated with ATC. During CAT II/ CAT III operations, vehicles fitted with transponders (vehicle locators) should only be permitted on the manoeuvring area. However, other vehicles crossing taxiways on service road and Terminal-3 taxiways shall be regulated by official of apron control deploying manpower.

4.4 Low Visibility Procedure Taxi Route

4.4.1 Once LVP is declared, AGL team will ensure that the lead-in lights of all CATII/III compatible stands are switched ON.

4.4.2 (a) During CAT III conditions “FOLLOW ME” service will be provided to arriving /departing aircraft on request. “FOLLOW ME” service can be put into use up to a minimum RVR of 50m.

(b) If any immediate portion of exit taxiway is not available, ATC will guide the aircraft to the nearest exit through ASMGCS.

4.4.3 Person providing “FOLLOW ME” service shall be trained and fully familiar with the taxi routes intersections and other maneuvering area/apron/parking stands.

4.4.4 During CAT II/CAT III ILS operations, the preferred taxi routing as per runway in use is provided below:

Table: 2

WESTERLY FLOW		
ARRIVAL		
RWY 29L	To Terminal 1	(Vacate via TWY Z2/Z3/ Z6/Z7 then taxi via Z) OR (Vacate via TWY Z4/Z5), Taxi via (T6, CROSS RWY 11L/29R, R6, (R/Q)) / (T7, CROSS RWY 11L/29R, R7, (R/Q)) then further taxi via A, N7, K6 CROSS RWY 10/28 taxi via H6, D and further taxi via i. F6 for stands 170 to 178 ii. F5 for stands 160R to 165 iii. F3 for stands 150 to 159 iv. E9 for stands 140 to 149B. v. E2 for stands 101, 103, 105, 107, 109, 111, 113, 115, 117, 119 and 121. vi. E7, E4 for stands 102, 104, 106, 108, 110, 112, 114, 116, 118, 120 and 122 vii. E7, E6 for stands 133L to 137 viii. E3 for stands 123-132 ix. D3, CROSS RWY 09/27 A2 to Technical Area for up to code E aircraft

RWY 29R	To Terminal 1	Vacate via TWY R2/R3/R4 then taxi via A OR Vacate via TWY R5 then taxi via Q then turn left on TWY A, N7, K6, CROSS RWY 10/28 taxi via H6, D further taxi via i. F6 for stands 170 to 178 ii. F5 for stands 160R to 165 iii. F3 for stands 150 to 159 iv. E9 for stands 140 to 149B v. E2 for stands 101, 103, 105, 107, 109, 111, 113, 115, 117, 119 and 121. vi. E7, E4 for stands 102, 104, 106, 108, 110, 112, 114, 116, 118, 120 and 122 vii. E7, E6 for stands 133L to 137 viii. E3 for stands 123-132 ix. D3, CROSS RWY 09/27, A2 to Technical Area for up to code E aircraft
RWY 29L	To Terminal 2 and Terminal 3	(Vacate via TWY Z2/Z3/ Z6/Z7 then taxi via Z) OR (Vacate via TWY Z4/Z5), Taxi via (T6, CROSS RWY 11L/29R, R6,(R,A/ Q))/(T7, CROSS RWY 11L/29R, R7, (R,A/Q)) and further taxi via : i. LINK 4, C, LINK 9/LINK10 for Apron 35 ii. LINK 4, C, LINK 7 for Apron 34 iii. LINK 4, C, LINK 6 for Apron 33B iv. A, LINK 3 for Apron 33A v. A, LINK 2 for Apron 32 vi. A, N, N5 for Apron 31 vii. A, N, N4/N3/N1 for stands 201-211 viii. A, N for 232-244 ix. A, N for Cargo Apron (upto code F aircraft restricted up to max wingspan 73.30m) x. A, N7, K,LINK 34, N for Cargo Apron (up to code F aircraft)

RWY 29R	To Terminal 2 and Terminal 3	For Apron 35: (Vacate via TWY R2 Taxi via TWY LINK 7) OR (Vacate via TWY R3, LINK 6) OR (Vacate via TWY R4 Taxi via A, LINK 4) OR Vacate via (TWY R5 Taxi via TWY Q, LINK 4) then Taxi via TWY C, LINK 9/ LINK10 For Apron 34: (Vacate via TWY R2 Taxi via TWY LINK 7) OR (Vacate via TWY R3 Taxi via, LINK 6) OR (Vacate via TWY R4 Taxi via A, LINK 4) OR Vacate via (TWY R5 Taxi via TWY Q, LINK 4) then Taxi via TWY C, LINK 7 For Apron 33B: (Vacate via TWY R2 Taxi via TWY A) OR (Vacate via TWY R3 Taxi via LINK 6) OR (Vacate via TWY R4 Taxi via A, LINK 4) OR Vacate via (TWY R5 Taxi via TWY Q, LINK 4, C) then further Taxi via TWY LINK 6 For Apron 33A: (Vacate via TWY R2/R3/R4 Taxi via TWY A) OR Vacate via (TWY R5 Taxi via TWY Q, A) then further Taxi via TWY LINK 3 For Apron 32: (Vacate via TWY R2/R3/R4 Taxi via TWY A) OR Vacate via (TWY R5 Taxi via TWY Q, A) then further Taxi via TWY LINK 2 For Apron 31: (Vacate via TWY R2/R3/R4 Taxi via TWY A) OR Vacate via (TWY R5 Taxi via TWY Q, A) then further Taxi via TWY N/N5 **For** Apron II and Cargo Apron: (Vacate via TWY R2/R3/R4 Taxi via TWY A) OR Vacate via (TWY R5 Taxi via TWY Q, A) then further Taxi via i. N, N4/N1/N3 for stands 201-211 ii. N for 232-244 iii. N for Cargo Apron (upto code F aircraft restricted up to max wingspan 73.30m) iv. N7, K, LINK 34, N for Cargo Apron (up to code F aircraft)

RWY 28	To Terminal 1	Vacate via TWY H1/H2, then taxi via TWY D4, D OR Vacate via TWY H3/H6, D i. F6 for stands 170 to 178 ii. F5 for stands 160R to 165 iii. F3 for stands 150 to 159 iv. E9 for stands 140 to 149B. v. E2 for stands 101, 103, 105, 107, 109, 111, 113, 115, 117, 119 and 121. vi. E7, E4 for stands 102, 104, 106, 108, 110, 112, 114, 116, 118, 120 and 122 vii. E7, E6 for stands 133L to 137 viii.E3 for stands 123-132 ix. D3 cross RWY 09/27 then A2 to Technical Area for up to Code-E aircraft
RWY 28	To Terminal 2 and Terminal 3	Vacate via TWY K1/ K2/K5 then taxi via K, N6 or vacate via TWY K6 then taxi via N6 and further taxi via i. C, LINK 9/LINK 10 for Apron 35 ii. C, LINK 7 for Apron 34 iii. C, LINK 6 for Apron 33B iv. C, LINK 3 for Apron 33A v. C, LINK 2 for Apron 32 vi. N, N5 for Apron 31 vii. N, N4/N3/N1 for stands 201-211 viii. N for stands 232-244 ix. N for Cargo Apron (upto code F aircraft restricted up to max wingspan 73.30m) x. Vacate via TWY K2/ K5/K6 taxi via K, LINK 34, N OR vacate via TWY K1 then taxi via LINK 34, N for Cargo Apron (up to code F aircraft)
DEPARTURE		
RWY 29L	From Terminal 1	Taxi Via F3/F5/F6/E9/E6/E4/(E2,E7)/(E3, E7) OR (A2 cross RWY 09/27 TWY (D3, D, F6/E9/E7)/(D3, F5/F3)), G,G5, CROSS RWY 10/28, J5, K, A, (P5, CROSS RWY 11L/29R, S5, Y5)/(P5, CROSS RWY 11L/29R, S5, Z, Y6/Y7/Y8) OR Taxi via (TWY P6 CROSS RWY 11L/29R, S6, Y6)/(TWY P7 CROSS RWY 11L/29R, S7, Y7)/(TWY P8 CROSS RWY 11L/29R, S8, Y8). For Code D/E: A2 cross RWY 09/27 Taxi via TWY D3, F4,G then folow above mentioned taxi route.

RWY 29R	From Terminal 1	Taxi Via F3/F5/F6/E9/E6/E4/(E2,E7)/(E3, E7) OR (A2 cross RWY 09/27 TWY (D3, D, F6/E9/E7)/(D3, F5/F3)), G, G5, CROSS RWY 10/28, J5, K, A, P2/P4/P5/P6/P7/P8. For Code D/E: A2 cross RWY 09/27 Taxi via TWY D3, F4, G then folow above mentioned taxi route.
RWY 29R	From Terminal 2 and Terminal 3	i. Apron 35: LINK 9/LINK 10, C, (LINK 16, P2)/(LINK 19, (A,P4/P5))/ (P6/P9, A, P7/P9, P8) ii. Apron 34: LINK 8, C, (LINK 16, P2)/(LINK 19, (A,P4/P5))/(P6/P9, A, P7/P9, P8) iii. Apron 33A: LINK 3, C, (LINK 16, P2)/(LINK 19, (A,P4/P5))/(P6/P9, A, P7/P9, P8) iv. Apron 33B: LINK 6, C, (LINK 16, P2)/(LINK 19, (A,P4/P5))/(P6/P9, A, P7/P9, P8) v. Apron 32: LINK 1, C, (LINK 16, P2)/(LINK 19, (A,P4/P5))/(P6/P9, A, P7/P9, P8) vi. Apron 31: N5, N, C, (LINK 16, P2)/(LINK 19, (A,P4/P5))/(P6/P9, A, P7/P9, P8) vii. Stands 201-211: (N3/N1), N, LINK 33, K, A, (P2/P4/P5/P6/P7/P8) viii. Stands 232-244: N, LINK 33, K, A, (P2/P4/P5/P6/P7/P8) ix. Stands 252-257, 260-265: N, LINK 33, K, A, (P2/P4/P5/P6/P7/P8)

RWY 29L	From Terminal 2 and Terminal 3	i. Apron 35: LINK 9/LINK 10, C, ((LINK 19, (P5, CROSS RWY 11L/ 29R, S5, Y5)/(P5, CROSS RWY 11L/29R, S5, Z, Y6/Y7/Y8)) OR ((P6, CROSS RWY 11L/29R S6, Y6)/(P9, A, P7, CROSS RWY 11L/ 29R, S7, Y7)/(P9, P8, CROSS RWY 11L/29R , S8, Y8)) ii. Apron 34: LINK 8, C, ,((LINK 19, (P5, CROSS RWY 11L/29R, S5, Y5)/(P5, CROSS RWY 11L/29R, S5, Z, Y6/Y7/Y8)) OR ((P6, CROSS RWY 11L/29R S6, Y6)/(P9, A, P7, CROSS RWY 11L/29R, S7, Y7)/ (P9, P8, CROSS RWY 11L/29R , S8, Y8)) iii. Apron 33A: LINK 3, C, ((LINK 19, (P5, CROSS RWY 11L/29R, S5, Y5)/(P5, CROSS RWY 11L/29R, S5, Z, Y6/Y7/Y8)) OR ((P6, CROSS RWY 11L/29R S6, Y6)/(P9, A, P7, CROSS RWY 11L/29R, S7, Y7)/(P9, P8, CROSS RWY 11L/29R , S8, Y8)) iv. Apron 33B: LINK 6, C, ((LINK 19, (P5, CROSS RWY 11L/29R, S5, Y5)/(P5, CROSS RWY 11L/29R, S5, Z, Y6/Y7/Y8)) OR ((P6, CROSS RWY 11L/29R S6, Y6)/(P9, A, P7, CROSS RWY 11L/29R, S7, Y7)/(P9, P8, CROSS RWY 11L/29R , S8, Y8)) v. Apron 32: LINK 1, C, ((LINK 19, (P5, CROSS RWY 11L/29R, S5, Y5)/(P5, CROSS RWY 11L/29R, S5, Z, Y6/Y7/Y8)) OR ((P6, CROSS RWY11L/29R S6, Y6)/(P9, A, P7, CROSS RWY 11L/29R, S7, Y7)/ (P9, P8, CROSS RWY 11L/29R , S8, Y8)) vi. Apron 31: N5, N, C, ((LINK 19, (P5, CROSS RWY 11L/29R, S5, Y5)/(P5, CROSS RWY 11L/29R,S5, Z, Y6/Y7/Y8)) OR ((P6, CROSS RWY 11L/29R S6, Y6)/(P9, A, P7, CROSS RWY 11L/29R, S7, Y7)/ (P9, P8, CROSS RWY 11L/29R , S8, Y8)) vii. Stands 201-211: (N3/N1)/N, LINK 33, K, A, ((P5, CROSS RWY 11L/29R, S5, Y5)/(P5, CROSS RWY 11L/29R, S5, Z, Y6/Y7/Y8)) OR Taxi via (TWY P6 CROSS RWY 11L/29R, S6, Y6)/(TWY P7 CROSS RWY 11L/29R, S7, Y7)/(TWY P8 CROSS RWY 11L/29R, S8, Y8) viii. Stands 232-244: N, LINK 33, K, A, (P5, CROSS RWY 11L/29R, S5, Y5)/(P5, CROSS RWY 11L/29R, S5, Z, Y6/Y7/Y8) OR Taxi via (TWY P6 CROSS RWY 11L/29R, S6, Y6)/(TWY P7 CROSS RWY 11L/29R, S7, Y7)/(TWY P8 CROSS RWY 11L/29R, S8, Y8) ix. Stands 252-257, 260-265: N, LINK 33, K, A, (P5, CROSS RWY 11L/29R, S5, Y5)/(P5, CROSS RWY 11L/29R, S5, Z, Y6/Y7/Y8) OR Taxi via (TWY P6 CROSS RWY 11L/29R, S6, Y6)/(TWY P7 CROSS RWY 11L/29R, S7, Y7)/(TWY P8 CROSS RWY 11L/29R, S8, Y8).
RWY 28	From Terminal 1	Taxi Via F3/F5/F6/E9/(E2, E7)/(E3, E7) OR (A2 cross RWY 09/27 TWY (D3, D, F6/F/E9/E7)/(D3, F5/F3)), G, G3/G4/G5 OR Taxi via E6/E4, G4/(G, G3/G5) OR For Code E: A2 cross RWY 09/27 Taxi via TWY D3, F4, G, G3/G4/G5.

RWY 28	From Terminal 2 and Terminal 3	For aircraft upto Code F: i. Apron 35: LINK 9/LINK 10, C, M1, K, J5/J6/J7 ii. Apron 34: LINK 8, C, M1, K,J5/J6/J7 iii. Apron 33B: LINK 6, C, M1, K, J5/J6/J7 iv. Cargo 252-257, 260-265:N, LINK 33, K, J5/J6/J7 For aircraft upto Code F(maximum wingspan 73.30 m) v. Apron 33A: LINK 3, A, N, LINK 33, K, J5/J6/J7 vi. Apron 32: LINK 1, A, N, LINK 33, K, J5/J6/J7 vii. Apron 31: N5, N, LINK 33, K, J5/J6/J7 viii. Stands 201-211: N3/N1, N, LINK 33, K, J5/J6/J7 ix. Stands 232-244: N, LINK 33, K, J5/J6/J7 **NOTE:** For aircraft upto Code F from APRON 32: LINK 1, C, M1, K, J5/J6/J7
RWY 27	From Terminal 1	Taxi Via F3/F5/F6/E9/(E6, E7)/(E4, E7)/(E2, E7)/(E3, E7), G, F, D, D3/ D2/D1 OR A2
RWY 27	From Terminal 2 and Terminal 3	**From Terminal 3** i. Apron 35: (LINK 10, C, LINK 9)/ LINK 9, A, N7, K6, CROSS RWY 10/28 taxi via H6, D, D3/D2/D1. ii. Apron 34: LINK 8, A, N7, K6, CROSS RWY 10/28 taxi via H6, D, D3/D2/D1. iii. Apron 33B: LINK 6, A, N7, K6, CROSS RWY 10/28 taxi via H6, D, D3/D2/D1. iv. Apron 33A: LINK 3, A, N7, K6, CROSS RWY 10/28 taxi via H6, D, D3/D2/D1. v. Apron 32: LINK 1, A, N7, K6, CROSS RWY 10/28 taxi via H6, D, D3/D2/D1. vi. Apron 31: N5, N, N7, K6, CROSS RWY 10/28 taxi via H6, D, D3/ D2/D1. **From Terminal 2** i. Stands 201-211: N3/N1, N, LINK 39, K, K6, CROSS RWY 10/28 taxi via H6, D, D3/D2/D1. ii. Stands 232-244: N, , LINK 39, K, K6, CROSS RWY 10/28 taxi via H6, D, D3/D2/D1. Cargo 252-257, 260-265:N, LINK 33, K, K6, CROSS RWY 10/28 taxi via H6, D, D3/D2/D1.

Table: 3
EASTERLY FLOW
ARRIVAL

RWY 11R	For Terminal 1	(Vacate via TWY Y1, Y2, Y4 then taxi via Z) OR (Vacate via TWY Y5), S5, CROSS RWY 11L/29R, P5, LINK 19 OR Vacate via TWY Y5, Z, S6/S7/S8, CROSS RWY 11L/29R then taxi via TWY P6/(P7, A, P9)/(P8, P9) then taxi via C, M1, K AND further taxi via J5 CROSS RWY 10/28 and then taxi via TWY G5, G and taxi via i. F6 for stands 170 to 178 ii. F5 for stands 160R to 165 iii. F3 for stands 150 to 159 iv. E9 for stands 140 to 149B v. E4 for stands 102, 104, 106, 108, 110, 112, 114, 116, 118, 120 and 122 vi. E7, E2 for stands 101, 103, 105, 107, 109, 111, 113, 115, 117, 119 and 121. vii. (E7, E3) for stands 123-132 viii. E6 for stands 133L to 137 ix. (F5/F3) OR (F6/F/E9/E7, D), D3 to Technical Area. x. F4/(F,D), D3 CROSS RWY 09/27 A2 to Technical Area for Code D/E aircraft.
RWY 11L	For Terminal 1	Vacate via (TWY P1/P2 taxi via LINK 16)/(P3, A, LINK 16)/(P4,A, LINK 19) then taxi via C, M1, K AND further taxi via J5 CROSS RWY 10/28 and then taxi via TWY G5, G and further taxi via i. F6 for stands 170 to 178 ii. F5 for stands 160R to 165 iii. F3 for stands 150 to 159 iv. E9 for stands 140 to 149B v. E4 for stands 102, 104, 106, 108, 110, 112, 114, 116, 118, 120 and 122 vi. E7, E2 for stands 101,103, 105, 107, 109, 111, 113, 115, 117, 119 and 121 vii. (E7, E3) for stands 123-132 viii. E6 for stands 133L to 137 ix.(F5/F3) OR (F6/F/E9/E7,D), D3 to Technical Area. x. F4/(F,D), D3 CROSS RWY 09/27 A2 to Technical Area for Code D/E aircraft.

RWY 11R	For Terminal 2 and Terminal 3	**For Terminal 3** (Vacate via TWY Y1, Y2, Y4 then taxi via Z) OR (Vacate via TWY Y5), S5, CROSS RWY 11L/29R, P5 OR (Vacate via TWY Y5), Z, S6/S7/S8, CROSS RWY 11L/29R then taxi via TWY P6/P7/P8, A further taxi via For Apron 35: LINK 9/(LINK 9, C, LINK 10). For Apron 34: LINK 8. For Apron 33A: LINK 3 For Apron 33B: LINK 6 For Apron 32: LINK 2 For Apron 31: N, N5 **For Terminal 2** (Vacate via TWY Y1, Y2, Y4 then taxi via Z) OR (Vacate via TWY Y5), S5, CROSS RWY 11L/29R, P5, LINK 19 OR Vacate via TWY Y5, Z, S6/S7/S8, CROSS RWY 11L/29R then taxi via TWY (P6/(P7, A, P9)/(P8, P9) i. further taxi via C, M1, K, LINK 39, N1/ (N, N3) For stands 201-211 ii. further taxi via C, M1, K, LINK 39, N for stands 232- 244 iii.further taxi via C, M1, K, LINK 33, N For Cargo Apron
RWY 11L	For Terminal 2 and Terminal 3	For Terminal 3 Vacate via TWY P1/P2/P3/P4 then taxi via A then further taxi via i. For Apron 35: LINK 9 / (LINK 9, C, LINK 10) ii. For Apron 34: LINK 8 iii. For Apron 33B: LINK 6 iv. For Apron 33A: LINK 3 v. For Apron 32: LINK 2 vi. N, N5 For Apron 31: N, N5 For Terminal 2 Vacate via (TWY P1/P2 taxi via LINK 16)/(P3, A, LINK 16)/(P4, A, LINK 19) then taxi via C, M1, K then further taxi via: i. LINK 39, N1/ (N, N3) for stands 201-211 ii. LINK 39, N for stands 232- 244 iii. LINK 33, N for Cargo Apron

RWY 10	For Terminal 1	i. For stands 170-178 Vacate via TWY G1 then taxi via TXL F6 to stands 170-178. OR Vacate via TWYs G2, G3, G4 and G5 then taxi via TWY G and further taxi via TXL F6 to stands 170-178. ii. For stands 160R-165 Vacate via TWYs G1, G2, G3, G4 and G5 then taxi via TWY G and further taxi via TXL F5 to stands 160R-165. iii. For stands 150-159 Vacate via TWYs G1, G2, G3, G4 and G5 then taxi via TWY G and further taxi via TXL F3 to stands 150- 159. iv. For stands 140-149B Vacate via TWYs G1, G2, G3, G4 and G5 then taxi via TWY G and further taxi via TXL E9 to stands 140-149B. v. For stands 102, 104, 106, 108, 110, 112, 114, 116, 118, 120 and 122 Vacate via TWYs G1, G2, G3, and G5 then taxi via TWY G and further taxi via TXL E4 to stands 102, 104, 106, 108, 110, 112, 114, 116, 118, 120 and 122. OR Vacate via TWY G4 then further taxi via TXL E4. vi.For stands 101, 103, 105, 107, 109, 111, 113, 115, 117, 119 and 121. Vacate via TWYs G1, G2, G3, G4 and G5 then taxi via TWY G and further taxi via TXL E7, E2 to stands 101, 103, 105, 107, 109, 111, 113, 115, 117, 119 and 121 vii. For stands 123-132 Vacate via TWYs G1, G2, G3, G4 and G5 then taxi via TWY G and further taxi via TXL E7, E3 to stands 123-132

		viii. For stands 133l-137 Vacate via TWYs G1, G2, G3 and G5 then taxi via TWY G and further taxi via TXL E6 to stands 133L-137. OR Vacate via TWY G4 and further taxi via TXL E6 to stands 133L-137 ix. For Technical Area Vacate via TWYs G1, G2, G3, G4 and G5 then taxi via TWY G and further taxi via F4/(F,D), D3 CROSS RWY 09/27 A2 to Technical Area for Code D/E aircraft
RWY 10	For Terminal 2 and Terminal 3	For Terminal 3 Vacate via TWY J1 taxi via TWY K OR vacate via TWY J2 then taxi via TWY LINK 32, C, M or Vacate via TWY J5/J6/J7 then taxi via TWY K or vacate via TWY J3 Then further taxi via TWY A For Apron 35: TWY LINK 9/(LINK 9, C, LINK 10) For Apron 34: TWY LINK 8 For Apron 33A: TWY LINK 3 For Apron 33B: TWY LINK 6 For Apron 32: TWY LINK 2 For Apron 31: TWY N, N5 For Terminal 2 Vacate via TWY J1/J2/J3/J5/J6/J7 then taxi via TWY K i. Further taxi via LINK 39, N1/ (N, N3) for stands 201-211 ii. Further taxi via LINK 39, N for stands 232- 244 iii. Further taxi via TWY LINK 33, N for Cargo Apron
DEPARTURE		
RWY 11R	From Terminal 1	Taxi Via F3/F5/F6/E9/(E6, E7)/(E4,E7)/E2/E3) OR (A2 cross RWY 09/27 TWY D3), D,H3, CROSS RWY 10/28, K3,K, N6, C and further taxi via LINK 4, Q OR LINK 5, A, R And further taxi via (R6, CROSS RWY 11L/29R, T6, Z4)/ (R7, CROSS RWY 11L/29R, T7, Z5)/(R6/ R7, CROSS RWY 11L/ 29R, T6/T7, Z, Z6/ Z7)
RWY 11L	From Terminal 1	Taxi Via F3/F5/F6/E9/(E6, E7)/(E4,E7)/E2/E3 OR (A2 cross RWY 09/27 TWY D3), D, H3, CROSS RWY 10/28, K3,K, N6, C and further taxi via LINK 5, A, (R4)/(R, R5/R6/R7) OR LINK 4, Q, (R5/R6/R7)

RWY 11R	From Terminal 2 and Terminal 3	i. Apron 35: LINK 9/ (LINK 10, C, LINK 9), A, R, (R6, CROSS RWY 11L/29R, T6, Z4)/ (R7, CROSS RWY 11L/29R, T7, Z5)/ (R6/ R7, CROSS RWY 11L/ 29R, T6/T7, Z, Z6/ Z7) ii. Apron 34: LINK 7, A, R, (R6, CROSS RWY 11L/ 29R, T6, Z4)/ (R7, CROSS RWY 11L/29R, T7, Z5)/ (R6/R7, CROSS RWY 11L/29R, T6/T7, Z, Z6/ Z7) iii. Apron 33B: LINK 6, A, R, (R6, CROSS RWY 11L/29R, T6, Z4)/ (R7, CROSS RWY 11L/29R, T7, Z5)/ (R6/ R7, CROSS RWY 11L/29R, T6/T7, Z, Z6/ Z7) iv. Apron 33A: LINK 3, C,(LINK 4, Q)/(LINK 5, A, R), (R6, CROSS RWY 11L/29R, T6, Z4)/ (R7, CROSS RWY 11L/29R, T7, Z5)/ (R6/ R7, CROSS RWY 11L/29R, T6/T7, Z, Z6/ Z7) v. Apron 32: LINK 1, C, (LINK 4, Q)/(LINK 5, A, R), (R6, CROSS RWY 11L/29R, T6, Z4)/ (R7, CROSS RWY 11L/29R, T7, Z5)/ (R6/ R7, CROSS RWY 11L/29R, T6/T7, Z, Z6/ Z7) vi. Apron 31: N5, N, C, (LINK 4, Q)/(LINK 5, A, R), (R6, CROSS RWY 11L/29R, T6, Z4)/ (R7, CROSS RWY 11L/29R, T7, Z5)/(R6/ R7, CROSS RWY 11L/ 29R, T6/T7, Z, Z6/ Z7) vii. Stands 201-211: N1/ N3/N4, N, C, (LINK 4, Q)/ (LINK 5, A, R), (R6, CROSS RWY 11L/29R, T6, Z4)/ (R7, CROSS RWY 11L/29R, T7, Z5)/ (R6/ R7, CROSS RWY 11L/29R,T6/T7, Z, Z6/ Z7) viii. Stands 232-244: N, C, (LINK 4, Q)/(LINK 5, A, R), (R6, CROSS RWY 11L/29R, T6, Z4)/ (R7, CROSS RWY 11L/29R, T7, Z5)/ (R6/ R7, CROSS RWY 11L/29R, T6/T7, Z, Z6/ Z7) ix. For upto code F aircraft (restricted up to max wingspan 73.30m) from stands 252-257, 260- 265: N, C, (LINK 4, Q) / LINK 5, A, R, (R6, CROSS RWY 11L/29R, T6, Z4)/ (R7, CROSS RWY 11L/29R, T7, Z5)/ (R6/ R7, CROSS RWY 11L/29R, T6/T7, Z, Z6/ Z7)x. For code F aircraft from stands 252-257, 260 and 265: LINK 34, K, N6, C, (LINK 4, Q) / LINK 5, A, R, (R6, CROSS RWY 11L/29R, T6, Z4)/ (R7, CROSS RWY 11L/29R, T7, Z5)/ (R6/ R7, CROSS RWY 11L/29R, T6/T7, Z, Z6/ Z7).

RWY 11L	From Terminal 2 and Terminal 3	i. Apron 35: LINK 9/ (LINK 10, C, LINK 9), A,(R4/ (R/R5/R6/R7) ii. Apron 34: LINK 7, A, (R4/ (R/R5/R6/R7) iii. Apron 33B: LINK 6, A, (R4/ (R/R5/R6/R7) iv. Apron 33A: LINK 3, C, (LINK 5, A,(R4/(R/R5/ R6/R7))OR LINK 4, (A, R4)/(Q, (R5/R6/R7)). v. Apron 32: LINK 1, C, (LINK 5, A, (R4/ (R, R5/ R6/R7)) OR LINK 4, (A, R4)/(Q, (R5/R6/R7)) vi. Apron 31: N5, N, C,(LINK 5, A, (R4/(R,/R5/R6/ R7)) OR LINK 4, (A, R4)/(Q, (R5/R6/R7)) vii. Stands 201-211: N1/ N3/N4, N, C,(LINK 5, A, (R4 /(R, R5/R6/R7)) OR LINK 4, (A, R4)/(Q, (R5/ R6/R7)) viii. Stands 232-244: N, C, (LINK 5, A, (R4/(R, R5/ R6/R7)) OR LINK 4, (A, R4)/(Q, (R5/R6/R7)) ix. For upto code F aircraft (restricted up to max wingspan 73.30m) from stands 252-257, 260- 265: N, C,(LINK 5, A, (R4/(R,R5/R6/R7)) OR LINK 4, (A, R4)/(Q, (R5/R6/R7)) x. For code F aircraft from stands 252-257, 260 and 265: LINK 34, K, N6, C(LINK 5, A, (R4/(R, R5/R6/ R7)) OR LINK 4, (A, R4)/(Q, (R5/R6/R7))
RWY 10	From Terminal 1	Taxi Via F3/F5/F6/E9/(E6, E7)/(E4,E7)/E2)/E3 OR (A2 cross RWY 09/27 TWY D3), D, H3
RWY 10	From Terminal 2 and Terminal 3	i. Apron 35: LINK 9/(LINK 10, C, LINK 9), A, N7, K6/(K,K5) ii. Apron 34: LINK 7, A, N7, K6/(K,K5) iii. Apron 33B: LINK 6, A, N7, K6/(K,K5) iv. Apron 33A: LINK 3, A, N7, K6/(K,K5) v. Apron 32: LINK 1, A, N7, K6/(K,K5) vi. Apron 31: N5, N, N7, K6/(K,K5) vii. Stands 201-211: (N1/N3/N4), N, N7, K6/(K,K5) viii. Stands 232-244: N, N7, K6/(K,K5) ix. For up to code F aircraft (restricted up to max wingspan 73.30m) from Cargo Apron: N, N7, K6/ (K,K5) x. For up to code F aircraft from Cargo Apron: N, LINK 34, K,K5/K6.
RWY 09	From Terminal 1	Taxi Via F3/F5/F6/ E9/(E6,E7)/(E4,E7)/E2/E3, D, OR (A2 cross RWY 09/27 TWY D3), D then taxi via D4/D7/D8/D9

RWY 09	From Terminal 2 and Terminal 3	i. Apron 35: LINK 9/(LINK 10, C, LINK 9), A, N7, K, K3, CROSS RWY 10/28 H3, D, D4/D7/D8/D9 ii. Apron 34: LINK 7, A, N7, K, K3 CROSS RWY 10/28, H3, D, D4/D7/D8/D9 iii. Apron 33B: LINK 6, A, N7, K, K3 CROSS RWY 10/28, H3, D, D4/D7/D8/D9 iv. Apron 33A: LINK 3, A, N7, K, K3 CROSS RWY 10/28,H3, D, D4/D7/D8/D9 v. Apron 32: LINK 1, A, N7, K, K3 CROSS RWY 10/28, H3, D, D4/D7/D8/D9 vi. Apron 31: N5, N, N7, K, K3 CROSS RWY 10/28, H3, D, D4/D7/D8/D9 vii. Stands 201-211: (N1/N3/N4), N, N7, K, K3 CROSS RWY 10/28, H3, D, D4/D7/D8/D9 viii. Stands 232-244: N, N7, K, K3 CROSS RWY 10/28, H3, D, D4/D7/D8/D9 ix. from Cargo Apron: N, N7, K, K3 CROSS RWY 10/28, H3, D, D4/D7/D8/D9

4.5 Lighting Inspection Procedures

4.5.1 The appropriate aeronautical ground lights must have been inspected during the hour preceding implementation of LVP. DIAL AGL team to carry out close monitoring of serviceability of AGL through monitoring system.
4.5.2 Operations Duty Manager is responsible for organizing lighting inspections. He shall arrange an inspection of the relevant aeronautical ground lighting. To ensure minimum delay in completing the inspection, separate teams may inspect the landing runway, associated taxiways and apron area and confirm that they are clear of runway/taxiway/apron to ATC.
4.5.3 For SP and LVP only the lighting for the active runway and associated taxiways are inspected.

5. DESCRIPTION OF EQUIPMENT AND SERVICES

5.1 Runway Visual Range (RVR)

5.1.1 There are three transmissometers for RWY 27/09 to record RVR values. One transmissometer is located each at touchdown zone, runway midpoint, and end of runway. RVR values always refer to Touchdown RVR (TDZ), Midpoint RVR (MID) and Stop End RVR (END).

5.1.2 There are three Main and three Standby transmissometers installed at touchdown zone, runway midpoint and end of runway for RWY 28/10 to record RVR values.

5.1.3 RWY 29L/11R has Five RVR (transmissometers) at following locations:

a) Physical beginning of RWY-29L, termed as 'A'

b) Near displaced threshold of RWY-29L, termed as 'B'
c) Mid of RWY 11R/29L, termed as 'C'
d) Near displaced threshold of RWY-11R, termed as 'D'
e) Physical beginning of RWY-11R, termed as 'E.

5.1.4 RWY 29R/11L has 5 RVR (Forward Scatter meter) at following locations:

a) Physical beginning of RWY-29R, termed as 'A'

b) One at near displaced threshold of RWY-29R, termed as 'B'

c) One at Mid of RWY 11L/29R, termed as 'C'

d) One at near displaced threshold of RWY-11L, termed as 'D'

e) Physical beginning of RWY-11L, termed as 'E'

5.1.5 The RVR information available for Rwy – 29L/11R and RWY 29R/11L should be used as following:

Table: 4

RWY	Dep/Arr	Beginning	TDZ	MID		END	
				Primary	Standby	Primary	Standby
29L	Dep	A	-	B	C	E	D
29L	Arr	-	B	C	-	D	E
11R	Dep	E	-	C	-	A	B
11R	Arr	-	D	C	-	B	A
29R	Dep	A	-	B	C	E	D
29R	Arr	-	B	C	-	D	E
11L	Dep	E	-	C	-	A	B
11L	Arr	-	D	C	-	B	A

5.1.6 RVR is reported in the following scales:
a) Above 800m till 2000m,increments of 100m
b) From 400m to 800m, increments of 50m
c) m Below 400m, increments of 25m.

5.1.7 Equipment serviceability for CAT II/CAT III operations:
a) For CAT II operations TDZ and MID RVR shall be available.
b) For CAT III operations TDZ, MID and END RVR shall be available.

5.2 Aeronautical Ground Lighting (AGL) System

5.2.1 The Precision Approach lighting system for CATII/CAT III operations are installed on RWY 10,RWY 28, RWY 29L, RWY11R, RWY 29R and RWY 11L at IGI Airport.
5.2.2 During LVP operation irrespective of the RVRs, the state electricity board (BSES) power supply will be the primary source through online UPS for the systems which require a maximum of one second switchover time and the DGs shall be in auto operation mode to meet the requirements in case of power failure. The power changeover for the other systems which are not covered with UPS backup shall be getting the power supply through DGs within 15 seconds from the time of power failure. For the systems covered with UPS back up, the changeover will be zero seconds. There is no specific requirement for keeping the DG source as primary source when BSES power source is available during the LVP operations.

5.2.3 All taxiways except following taxiways have CAT III standard taxiway lighting: TWYs LINK 32,LINK 30, Portion of LINK 16 between LINK 14 and TWY C, LINK 14, LINK 15 and LINK 13.

5.2.4 The following parking stands have been provided with the CAT III Aircraft Stand Manoeuvring Guidance Lighting system:

a) All Parking stands in Apron I.

b) Parking stands 201 to 211, 232 to 244 in Apron II and 252 to 257 and 260 to 265 in Cargo Apron.

c) Following Table provides the details of parking stands in Apron 31 to Apron 35:

Table 5:

APRON	AIRCRAFT STAND NUMBER	TOTAL AIRCRAFT STANDS
APRON 31	A06, A08, A10, A12, A14	5
APRON 32	A01, A03, A07, A07L, A07R, A09, A09L, A09R,A11, A11L, A11R, A13 B15, B15L, B15R, B17, B17L, B17R, B19, B19L, B19R, B21R02, R03	24
APRON 33A	B18, B20, B20L, B20R, B22, B22L,B22R, B24, B24L, B24R, B26, B26L, B26R	13
APRON 33B	C28L, C28R, C30L, C30R, C32L, C32R, C34L, C34R, C36	9
APRON 34	C27L, C27R, C29L, C29R, C31L, C31R,C33, D37, D37L, D37R, D39, D39L, D39R , D41, D43, D45, D47, D49, D51, D53,D55	21
APRON 35	D46, D48, D50, D52, D54, D56, D58, D60, D62, E64, E66, E68, E70, E72, E74, E76, E78, E80, E82, E84	20
Note: Remote Stands in T3 apron (except R02, R03) do not have stand manoeuvring guidance lights.		

5.2.5 The alternate yellow and green lights have been provided on the under mentioned taxi tracks to denote their presence in ILS sensitive area:

a) RWY 28/10: H2,H1, H3, H6,K6,K5, K2, K3, K1, J1,J2, J3, G1,G2, G3, G4,G5, J5,J6 and J7

b) RWY 29L/11R: Y8, Y7, Y6, Y5, Y4, Y2, Y1, Z2, Z3, Z6, Z4, Z5 and Z7.

c) RWY 27/09: D1, D2, D3, D4,D5, D6,D7,D8, D9 and A2

(d) RWY 29R/11L: P1, P2, P3, P4, P5, P6, P7, P8, R1, R2, R3, R4, R5, R6, R7, S1, S2, S5, S6, S7, S8, T4, T5, T6 and T7

Table 6:

5.2.6 The preventive maintenance system for AGL have its objective that minimum level of serviceability as mentioned in below table is maintained during CAT II/ III operations:

AGL Facility	CAT II/CATIII	Restrictions
Approach Lights (Inner 450m)	95%	Suspend CAT II / CAT III operations
Approach Lights (Beyond 450m)	85%	
Runway Center-line lights	95%	Suspend CAT II / CATIII operations
	No Two adjacent lamps be unserviceable	
Threshold lights	95%	Suspend CAT II / CATIII operations
	No Two adjacent lamps be unserviceable	
Runway Edge lights	95%	Suspend CAT II / CATIII operations
	No Two adjacent lamps be unserviceable	
Touchdown Zone lights	90%	Suspend CAT II / CATIII operations
	No Two lamp in Barrette be unserviceable.	
Runway End lights	75%	Suspend CAT II / CATIII operations
	No Two adjacent lamps be unserviceable	
Taxiway Centre-line lights	No Two adjacent lamps be unserviceable (Runway visual range conditions less than a value of 350 m)	Close affected taxiways, use alternate taxi route.
Standby Generators	Generators in any one substation.	Suspend CAT II / CATIII operations in the affected Runway.

NOTE 1:
Under CAT II/III operations, for stop bar:

a) Not more than two lights will remain unserviceable; and

b) Two adjacent lights will not remain unserviceable unless the light spacing is significantly less than that specified

NOTE 2:

Unserviceability of any of the following facilities does not affect CAT II/ CAT III Operations.

a) PAPI.

b) Taxiway Edge Lights on curves.

c) Taxiway clearance lights wherever provided.

5.3 Non-Visual Ground Surveillance System:

5.3.1 IGI Airport has been equipped with Surface Movement Radar. The system provides nonvisual electronics surveillance of manoeuvring area and facilitates the controllers to identify potential ground conflict and runway incursions.

5.3.2 For CAT III and CAT II operations below 350m availability of Non-Visual Surveillance System such as Surface Movement Radar is mandatory.

5.4 Navigational Aids

5.4.1 RWY 10,RWY 28, RWY 11R, RWY 29L, RWY 11L and RWY 29R have been equipped with Instrument Landing System (ILS) for CAT II/CAT III.

5.4.2 The ILS Category Monitor Panel at the Control Tower console indicates the ILS category availability by monitoring the following equipment:

a) Main and standby localizer transmitters

b) Main and standby glide path transmitters

5.4.3 The status of the following facilities is monitored and displayed by a separate nav-aid status indicator panel:

a) ILS DME

b) Outer Marker

5.4.4 ILS equipment serviceability required for CAT II/CAT III operations: -

a) Both main and standby localizer transmitters;

b) Both main and standby glide path transmitters;

c) One standby power generator in each unit.

d) Outer marker/ ILS DME

NOTE 1 - Unserviceable ILS DME will not change the status of ILS provided OM and Glide Path are serviceable for runway 28 only.

NOTE 2 - Unserviceable Outer Marker will not change the status of ILS provided ILS DME is operational for runway 28 only.

5.5 Airport Rescue and Fire Fighting Service (ARFF)

5.5.1 During LVP following predetermined positions will be taken by ARFF vehicles

a)RWY 09/27: One CFT from FS-3 south of RWY 09/27 on the approach road holding point RWY 09/27 east of TWY D4. (PDP 1)

b) RWY 10/28: One CFT from FS-3 north of RWY 10/28 abeam TWY H1 in-front of FS 3. (PDP 2)

c)) RWY 11L/29R: One CFT from FS-1 on the approach road holding point RWY 11L/29R and TWY A. (PDP 5)

d)) RWY 11R/29L: One CFT from FS-5 on the approach road holding point RWY 11R/29L (PDP 7)

5.5.2 In the event of an incident when LVP are in force, ADC and SMC should provide the maximum assistance in directing ARFF to required location.

6. LOW VISIBILITY TAKE-OFF (LVTO)

6.1 LVTO pertains to take off when the RVR is below 400 m and is applicable whenever the reported RVR in any zone (touch-down/mid/roll-out RVR) is below 400 m. The facilities and conditions of the undermentioned Table 7 required will be as per the lowest RVR reported in any zone (e.g. if the RVR is 400/300/300 representing the three zones, then the 300 m will be the RVR for reckoning facilities and conditions of Table 7. If the RVR is 300/150/Not Reported, then 150 m will be the reckoning RVR and as the RVR is below 200 m all three RVRs are required).

Table: 7

Take-Off RVR/Visibility	
Facilities	RVR/Visibility* CAT A, B,C and D
Adequate visual reference** (Day only)	500 M
Runway Edge Lights or Runway Center Line Markings***	400 M
Runway Edge Lights and Center Line Marking	300 M
Runway Edge Lights and Runway Center Line Lights	200 M
Runway Edge Lights and Runway Center Line Lights and relevant RVR information****	150 M
High Intensity Runway Edge Lights and Runway Center Line Lights (Spacing 15 m or less) and relevant RVR information****	125 M
High Intensity Runway Edge Lights and Runway Center Line Lights (Spacing 15 m or less), approved lateral guidance system and relevant RVR information****	75 M

*The TDZ RVR/VIS may be assessed by the pilot

**Adequate Visual reference means, that a pilot is able to continuously identify the take-off surface and maintain directional control.

***For night operations at least runway edge lights or center line lights and runway end lights are available.

**** The required RVR must be achieved for all relevant RVR reporting points (touchdown, mid- point and stop-end/roll-out). The governing RVR shall be the lowest of the reported RVRs.

6.2 An operator shall not conduct low visibility take-offs in less than 400 m RVR unless approved by DGCA.

6.3 An operator shall not conduct take-off with visibility/RVR less than Category I conditions unless low visibility procedures are enforced.

7. SUMMARY OF THE LOW VISIBILITY PROCEDURES

7.1 Subject to completion of Safeguarding Procedures, LVP comes into operation:

a) When either TDZ, MID or END RVR below 800m, and/or

b) cloud ceiling below 200ft

7.2 Vehicular movement

a) Vehicles shall not be cleared to cross the runway, once an inbound aircraft is 8NM from touchdown.

b) Vehicular movement on the manoeuvring area shall be restricted to essential vehicles.

c) During CATII /CAT III operations only vehicles equipped with transponder (Vehicle Locator) shall operate in manoeuvring area. However, other vehicles crossing taxiways on service road shall be regulated by official of Apron Control by deploying man power.

d) Vehicles shall not be held at any point closer to the runway than the CAT II/ CATIII holding point/stop-bar.

7.3 Aircraft Movement

a) Aircraft shall not be cleared to cross the runway once an inbound aircraft is 8NM from touchdown.

b) Aircraft shall not be held at any point closer to the runway than the CAT II/CATIII holding point/stop-bar.

c) Aircraft which have landed are not to be instructed to hold on a runway turnoff

8. ACTIONS TO BE TAKEN BY VARIOUS AGENCIES

8.1. Before commencement of winter season, a meeting will be held by Chief Operating Officer/ Deputy Chief Operating Officer, Delhi International Airport Private Ltd. and Regional Executive Director (Northern Region)/ GM (ATM), IGI Airport, AAI in the month of November every year to inform all airlines and agencies operating at airport about their roles/ responsibilities and create awareness to ensure cooperation for safe airport operations during periods of low visibility.
8.2. All the agencies shall ensure that staff and drivers are suitably trained during CAT III operations.

8.3. Action by Watch Supervisory Officer (WSO), AAI

8.3.1 Implementing Safeguarding Procedures - When RVR is less than 1200m and visibility is forecast to deteriorate 800m or below and/or the cloud ceiling is 400ft and is forecast to fall to 200ft or less, WSO will inform -
a) AOCC,
b) Communication/ Technical supervisor (SSO), AAI
c) Tower Supervisor, AAI -for implementation of Safeguarding Procedures.

8.3.2 Implementing LVP

8.3.2.1 WSO shall implement Low Visibility Procedures when either
a) TDZ, MID or END RVR is less than 800m; and/or
Cloud ceiling is less than 200ft.
AND
c) Safeguarding Procedures (SP) have been completed and the airport is safeguarded.

8.3.2.2 WSO shall inform Aerodrome Tower Controller and Approach/Radar Controller,Duty Met. Officer and Communication/Technical Shift Supervisor Officer (SSO) and to ensure that "**LOW VISIBILITY PROCEDURE IN FORCE**" is included in ATIS broadcast.

8.3.3 WSO may terminate LVP when;

a) Meteorological conditions improve and TDZ, MID and END RVR are 800 m or more and the cloud ceiling is 200 ft or higher, and trend is for improvement for both runways.
b) Facilities, equipment and services necessary for CAT III operations are degraded and/or the prevailing conditions are considered unsafe for such operations.

8.3.3.1 WSO will intimate Aerodrome Tower Controller, Approach/Radar Controller and Communication Technical Shift Supervisory Officer (SSO) regarding the Termination of LVP operation.

8.4. Action by Tower Supervisor:

8.4.1 On being notified by WSO that ILS CAT II/CAT III Low Visibility Procedures are to commence, the Aerodrome Tower Controller will:

a) inform Aerodrome Rescue and Fire Fighting Services and AOCC,
b) check ILS status
c) check lighting is correctly selected and operating properly
d) check transmissometer displays
e) check ASMGCS (for operations below RVR 350 m)

8.4.2 After the commencement of ILS CAT II/CAT III operations, the Aerodrome Tower Controller shall
a) Check ATIS broadcast and include the message that "ILS CAT II/CAT III Low Visibility Procedures in Force".
b) Give landing clearance to aircraft not later than 2NM from touchdown.
c) Inform changes in RVR readings to the landing aircraft.
d) Give an unimpeded taxi route to arriving aircraft to allow it to clear the Localizer Sensitive Area expeditiously.
e) Inform pilots of all failures of ILS, lighting system, transmissometers relevant to ILS CAT II/ CAT III Low Visibility Operations.
f) Initiate emergency action if aircraft on CAT II/CAT III ILS is not seen (on radar display or otherwise) or is not in radio contact, as expected.

8.4.2.1 Record of the above actions with time be maintained and signed by the officer taking action.

8.5. Action by Approach/Radar Controller

8.5.1 On being advised by WSO that ILS CAT II/CAT III Low Visibility Procedures are in force, the Approach/Radar Controller shall:-
a) Inform the arriving aircraft "ILS CAT II/CAT III "Low VisibilityProcedures in Force".
(**Note:** Ensure that Pilot acknowledges of being cleared for ILS CAT II/CAT III approach.)
b) Inform TDZ RVR to arriving aircraft and in addition:
(i) For CAT II operations - If TDZ RVR is below 550m then MID RVR shall also be passed.

(ii) For CAT III operations – If TDZ is below 300m, then MID and End RVR readings shall also be passed.

NOTE: After an aircraft is 8NM from Touch Down or has passed outer marker, RVR observations need not be passed unless there is changes in RVR values.
c) vector the aircraft to intercept the localizer not less than 10NM from touchdown.
d) Not subject an aircraft to any speed control when within 20NM from touchdown.
e) issue landing clearance to arriving aircraft not later than 2NM from touchdown.

8.6. Action by Surface Movement Controller

During the period the Low Visibility Procedures are effective the Surface Movement Controller shall:
a) Monitor all surface movement of aircraft and vehicles on the manoeuvring area.
b) Inform all taxiing aircraft of the preceding taxiing or holding aircraft.
c) Hand over only one aircraft at a time to Tower Controller.
d) Permit only the vehicles equipped with transponder in the manoeuvring area during Cat III operations. However, other vehicles crossing taxiway crossings on service road shall be regulated by official of Airside Operations, DIAL by deploying man power.

8.7. Action by Communication/Technical/Electronics Shift Supervisory Officer (SSO)

8.7.1 On receipt of "Outlook for LVP" from the WSO, the Communication/Technical/Shift Supervisory Officer [SSO] will inform the Duty Officer, Equipment Room and have the ILS equipment and its status indicators in ATC units checked up. He will inform WSO of any unserviceability in the equipment which is likely to affect ILS CAT II/CAT III operation.

8.7.2 On receipt of "Advisory Message" from WSO that LVP are to be made effective SSO will maintain continuous watch on the performance of ILS equipment and will inform WSO of any un-serviceability which may affect ILS CAT II/CAT III operation.

9. Action by AOCC and Airside Operations, DIAL:

9.1 Action by Shift Head, AOCC
9.1.1 On receipt of advice from WSO to implement Low Visibility Procedures, the Shift Head AOCC, will immediately inform the following:

a) Operations Duty Manager on company channel / ATG.
b) Shift Engineer (electrical) on company channel / ATG.
c) Manager (Civil), in Operational Area Maintenance to ensure that all civil works in progress in manoeuvring area, are stopped and that the work area is restored in complete serviceable condition and confirm to ODM/Apron Control accordingly. During ILS CAT II/CAT III Low Visibility Operations, no equipment, manpower or material shall be present in sensitive areas of localizer and glide path.
d) Central Industrial Security Force (CISF) official positioned in AOCC.
e) Domestic and International airlines or GHA officials positioned in AOCC.
f) Terminal Duty Manager - Terminal I, Terminal II and Terminal III.

9.2 Action by Operations Duty Manager/Apron Control

9.2.1 During CAT II/CAT III operations, ODM/ Apron Control may authorize operations of vehicles, equipped with VELO, on manoeuvring area, including the crossing of taxiways, with the prior permission from ATC Tower. However, the vehicles, not equipped with VELO, crossing taxiway crossings on service road shall be regulated by official of Airside Operations, DIAL by deploying man power.

9.3 Follow-me vehicles:

9.3.1 Such vehicles shall ensure that
a) No vehicle/person enters or is present in the sensitive/critical areas of localizer and glide path.
b) All civil/electrical works in progress are to be stopped in the manoeuvring area immediately and men/material/equipment to be removed from the sensitive/critical areas of localizer and glide path.
c) They shall ensure that no civil maintenance work is carried out during LVPeither in power house or sub-stations or near any other electrical facilities usedduring LVP operations.
9.3.2 After ensuring above, Follow-me vehicles will confirm the same to the ODM.
9.3.3 Subsequently, follow-me vehicle shall remain available in apron and will maintain listening watch on their respective frequencies, i.e., R/T 121.625 MHz, 121.90 MHz and 121.75 MHz & 119.575 MHz frequency.
9.3.4 On receipt of information about unserviceability of any of the runway visual aids or power supply system, ODM through AOCC will immediately inform ATC Tower accordingly.
9.3.5 No vehicles shall enter/cross in the vicinity of runway without permission from ATC Control Tower. Such vehicles shall be equipped with VELO and able to communicate with ATC. The vehicles of the ARFF services, civil, electrical division or of any other agency which are not equipped with VELO and/ or RT but has to enter the runway or taxi track for urgent operational requirement shall be escorted by the follow me jeep all the time.
9.3.6 None of the workers/vehicles of grass cutting contractor, garbage removal contractor, and electrical/civil contractor shall enter the manoeuvring area during the operations of ILS CAT II/CAT III Low Visibility Procedures.
9.3.7 All coordination with DIAL units and ATC shall be carried out by AOCC.
9.3.8 ODM would advise ATC Tower/Watch Supervisory Officer when all actions are completed for commencement of LVP.

9.4 Action by Shift Engineers (Electrical)

9.4.1 On receipt of advice to implement Low Visibility Procedures from AOCC, respective Shift Engineer of T1, T2 and T3 (Electrical) along with ODM, will check that following visual aids associated with RWY10/28/29L/11R/29R/11L are serviceable and can be operated at full intensity.

a) Approach lighting system
b) Approach supplementary lighting
c) Runway edge lights
d) Runway threshold and end lights
e) Runway centre line lights
f) Runway touchdown zone lights
g) Stop Bar Lights
h) Intermediate holding position lights
i) RWY guard lights
j) TWY C/L lights
k) Stand manoeuvring lights

(**NOTE -** No adjustment in light intensity shall be made without permission from ATC Tower during LVP)

9.4.2 They shall ensure that no electrical maintenance works is carried out during LVP either in power house or on any other electrical facilities used during CAT II/CAT III operations.
9.4.3 They will ensure that runway switch room is manned and position themselves at CCR halls for standby power supply requirements and will maintain a listening watch on company frequency. They will ensure availability of power supply to meet CAT III requirement.

9.4.4 They will inform the un-serviceability or any change in status of any facility/systems to OMD immediately.

9.5 Action by Fire Station-5, Fire Station-3 and Fire Station-1

9.5.1 **Operations Commander** will ensure that ARFF (Airport Rescue and Fire Fighting) vehicles equipped with serviceable transponder take predetermined positions. Following predetermined positions will be taken by ARFF vehicles:-

a) One CFT from FS-3 south of Rwy 09/27 on the approach road holding point Rwy 09/27 east of Twy D4. (PDP 1)
b) One CFT from FS 3 North of RWY 28/10 abeam TWY H1 in front of FS 3. (PDP 2).
c) One CFT from FS-1 on the approach road holding point Rwy 11L/29R between Rwy 11L/29R and Twy A. (PDP 5).
d) One CFT from FS-5 on the approach road holding point Rwy 11R/29L, south of RWY 29L/11R abeam TWY S1. (PDP 7)

10. Aviation Security Group (ASG-CISF)

10.1 Chief Airport Security Officer shall ensure by deploying adequate manpower that power house and other vital electrical installations are properly secured and protected against any unauthorized intrusion.

10.2 The Inspector in charge, Control Room on receipt of advice to implement Low Visibility Procedures, will immediately inform all access gates and CISF posts under their respective controls not to permit labour to enter manoeuvring area while LVP are in force unless cleared by apron control.

11. Action by Duty Officer (Meteorological Office)

11.1 Duty Meteorological Officer would issue an 'Outlook for Low Visibility Procedures' to the Watch Supervisory Officer [WSO] of air traffic services whenever he expects that the RVR and/or cloud ceiling will fall below 800 m and/or 200ft or less respectively.

11.2 Whenever Duty Meteorological Officer visualizes that RVR is likely to fall below 800 m and/or cloud ceiling to 200ft or less within next 2 hours, he will issue an 'Advisory Message' to WSO to this effect.

11.3 When the RVR and/or cloud ceiling are 800m and/or 200ft respectively and the trend is towards improvement in these elements of weather Conditions the Duty Met Officer may, when requested by WSO, advise him about such improving weather conditions for the purpose of termination of LVP.

11.4 The Duty Met Officer would ensure that the RVR displays in ATC units in the Control Tower and Approach Control are serviceable. He would also ensure that RVR/visibility recorders of Touch-down zone, Mid-point and end positions are serviceable.

NOTE: Due to high variability of meteorological elements in space and time and the limitations of forecasting techniques available, it may not be always possible to issue a precise forecast of RVR particularly in case of transient weather phenomenon within two hours.

12. Action by Other agencies at airport such as - Airlines, Re-fuelling Companies,Catering Agencies, Airport Police, Customs, Immigration, Health

12.1 All agencies operating in the operational area shall ensure that minimum number of their vehicles, as are absolutely essential for aircraft operations, operate in the operational area. The drivers of these vehicles should keep a look out for taxing aircraft and other vehicles to prevent incident/ accidents. During CAT III operations only vehicles equipped with serviceable transponder shall be permitted to operate in manoeuvring area.

12.2 All the vehicles must have their obstruction/ anti-collision lights “ON” during operation of low visibility procedures.

12.3 Follow all instructions/sign boards provided on vehicular movement area/ service roads.

12.4 No vehicle/equipment/personnel shall enter in and around the vicinity of the runways or taxi-tracks except with prior permission of ODM, who in turn shall coordinate with aerodrome control tower.

13. TERMINATION OF LOW VISIBILITY PROCEDURES

13.1 When met conditions improve and TDZ, MID and END RVR are 800m or moreand the cloud ceiling is 200ft or higher and trend shows improvement, the WSO would terminate operation of LVP. He may obtain advice from Duty Met Officer, as provided at Para 10.3, about the improving weather conditions for the purpose of termination of LVP operations.

13.2 The WSO will intimate the following regarding termination of LVP operations

a) Aerodrome Tower Supervisor
b) Approach/Radar Controller
c) Communication/Technical Shift Supervisory Officer (SSO)

13.3 Aerodrome Tower Supervisor will in turn inform AOCC, who will advise all thepreviously notified personnel to resume normal operations.

IX. MODES OF OPERATION

1. INTRODUCTION:

IGI Airport, New Delhi has four runways i.e. 09/27, 10/28 ,11R/29L and 11L/29R. IGI Airport, New Delhi comprises of Two Runway Systems, viz Northern System and Southern System. Northern Runway System comprises of Runway 09/27 and Runway 10/28 whereas Southern Runway System comprises of Runway 11L/29R and Runway 11R/29L. Two Runways within the Runway system (Northern or Southern) are dependent on each other.
These runways are used in various modes for optimal use of available ground and air resources. Particular mode of operation to be implemented is decided by ATC considering various factors including the prevalent weather conditions, visibility/ RVR, serviceability of navigation and landing aids, ground infrastructure, traffic density and its composition, noise abatement requirements etc.
One/Two/ Three/Four runway modes of operations shall be tactically utilized during low/high density traffic situations so as to attain the optimal capacity at all times.

2. OPERATING PRINCIPLES

a. The runway assignment to a particular flight is primarily done based on the concept of Proximity to the arrival/departure terminal. However, ATC may deviate from this principle of near terminal arrival/departure due to ground infrastructure constraints and/or load balancing to meet the objective of least average delay.
b. Change of runway to arrivals if any, will be intimated before the aircraft is at least 20 NM from touchdown. Within 20 NM of touchdown, change of runway will be affected only with the consent of Pilot-in-Command.
c. Independent arrivals on any two runways are not permitted. The sequence of arrivals on any two runways is considered as one stream for the purpose of separation and sequencing.
d. In case of dependent modes, the operation on two runways will be in the form of in-trail operations wherein a movement on arrival runway is considered as movement on departure runway and vice-versa for the purpose of applying relevant separation.

e. In independent mode for departures, simultaneous departures are permitted from any Runway of Northern Runway System and another Runway of Southern Runway System, when the departures follow the published Runway SID or Non RNAV instructions.

f. Simultaneous departures are not permitted from RWY 27 and RWY 28. Similarly, simultaneous departures are not permitted from RWY 09 and RWY 10.

g. Simultaneous departures are not permitted from RWY 29R and RWY 29L. Similarly, simultaneous departures are not permitted from RWY 11L and RWY 11R.

3. PREFERRED MODES OF OPERATIONS

Four runways available at IGI airport are used in various modes. There are many modes of operations A particular mode of operation is employed depending upon various factors such as the prevalent weather conditions, visibility/RVR, serviceability of navigation and landing aids, ground infrastructure and airspace constraints, traffic density and its composition, noise abatement requirements, VVIP movements, etc.

The available Runways can be used in following combination.

a. **Westerly Flow**: Major modes of operations in westerly flow are as under.

Mode of operation	Remarks

Two RWY operations	27- DEP and ARR 29R- DEP and ARR	a. Departures from RWY 27/28 are independent of position of arrival/departure on RWY 29L/RWY 29R. b. Departures from RWY 29L/29R are independent of position of arrival/departure on RWY 27/28. c. Departures from RWY 28 are dependent upon position of arrival on RWY 27 d. Departures from RWY 29R are dependent upon position of arrival on RWY 29L.
	28- DEP and ARR 29L- DEP and ARR	
	28- DEP and ARR 29R- DEP and ARR	
	27- DEP and ARR 29L- DEP and ARR	
	27- ARR 28- DEP and ARR	
	29L ARR 29R DEP	
Three RWY operations	27 –ARR 28– DEP 29L- ARR and DEP	
	27- ARR 28- DEP 29R- ARR and DEP	
	27- ARR and DEP 29L- ARR 29R- DEP	
	28- ARR and DEP 29L- ARR 29R- DEP	
Four RWY operations	27- ARR 28- DEP 29L- ARR 29R- DEP	

b. Easterly Flow: Major modes of operations in Easterly flow are as under	
Mode of operation	Remarks

Two RWY operations	09- DEP and ARR 10- DEP	a. Departures from RWY 09/10 are independent of position of arrival/departure on RWY 11R/ RWY 11L. b. Departures from RWY 11R/ RWY 11L.are independent of position of arrival/departure on RWY 09/10. c. Departures from RWY 09 are dependent upon position of arrival on RWY 10. d. Departures from RWY 10 are dependent upon position of arrival on RWY 09. e. Departures from RWY 11L are dependent upon position of arrival on RWY 11R.
	09- DEP and ARR 10- ARR	
	09- DEP 10- DEP and ARR	
	09-ARR 10- DEP and ARR	
	09- DEP and ARR 11L- DEP and ARR	
	09-DEP and ARR 11R- DEP and ARR	
	10- DEP and ARR 11L- DEP and ARR	
	10- DEP and ARR 11R- DEP and ARR	
	11L- DEP 11R-ARR	
Three RWY operations	09 – Dep 10 – ARR 11R – ARR and DEP	
	09- DEP 10- ARR 11L- DEP and ARR	
	09- ARR 10- DEP 11R- ARR and DEP	
	09-ARR 10- DEP 11L- DEP and ARR	
	10- ARR and DEP 11L- DEP 11R- ARR	
	09- ARR and DEP 11L - DEP 11R- ARR	
Four RWY operations	09- DEP 10- ARR 11L- DEP 11R- ARR	
	09- ARR 10- DEP 11L- DEP 11R- ARR	

4. SEPARATION MINIMA:

a. 3 NM surveillance based separation at or below FL 140 within 60 NM of DPN subject to availability of ASR. In case ASR is not available, the separation minima shall be 5 NM.
b. 5 NM surveillance based separation above FL 140 within 60 NM of DPN.
c. 10 NM surveillance based separation beyond 60 NM of DPN.

5. SID and/or Radar Departure Instructions:

a. RNAV SIDs for IGI Airport as published in eAIP India/AIP Supplement are used as departure instructions to RNAV 1 compliant aircraft.
b. Departure instructions to NON-RNAV-1 departures are given as below.

RWY	Departure instructions
27	After departure RWY 27, turn right heading 295 climbing to FL 70, expect further instructions with radar.
28	After departure RWY 28, MNTN RWY TR 284 DEG (M), CMB TO FL70. EXP further instructions with radar
29R	After departure RWY 29R, climb to 4000 FT, passing 1700 FT turn left heading 210, expect further instructions with radar.
29L	After departure RWY 29L, climb to 4000 FT, passing 1700 FT turn left heading 210, expect further instructions with radar.
09	After departure RWY 09, MNTN RWY TR 091 DEG (M), CMB TO FL70. EXP further instructions with Radar.
10	After departure RWY 10, MNTN RWY TR 104 DEG (M), CMB TO FL70. EXP further instructions with radar.
11L	After departure RWY 11L, climb to 4000 FT, passing 1700 FT turn right heading 170, expect further instructions with radar.
11R	After departure RWY 11R, climb to 4000 FT, passing 1700 FT turn right heading 170, expect further instructions with radar.

6. LIMITATIONS, SUSPENSION AND TERMINATION:

ATC may suspend, terminate or implement any mode of operation in the interest of Flight safety and efficiency of operations considering various factors including weather/ VVIP movement/special conditions/equipment degradation/ emergency handling / aircraft accident /closure of any part of maneuvering area/manpower etc.

6.1 Holding procedure at VOR/DME fix are defined to facilitate management of air traffic flow within Delhi TMA. The holding procedure shall be available only under radar surveillance parameters common to all holding procedure.
Outbound time - 1.5 MIN
Holding speed - 265K IAS MAX
MIN/MAX Holding level - F220/F280
KALOL (280802.38N0782414.16E, SSB-R112/40D) Right Hand Pattern,
Inbound track 292 DEG M.
BAXID (270750.38N0770937.42E, DPN - R177/86D) Right Hand Pattern,
Inbound track 357 DEG M.
UKASO (273815N 0755151E, CHI -R224/60D) Left Hand Pattern,
Inbound track 044 DEG M.
VIKOT (292312.95N0760720.57E, SAM-R312/50D Left Hand Pattern,
Inbound track 132 DEG M.

X. RNAV 1(GNSS) SIDs & STARs

1. GENERAL

1.1. Delhi comprises of Two Runway Systems, viz Northern Runway System and Southern Runway System. Northern Runway System comprises of Runway 09/27 & Runway 10/28 whereas Southern Runway System comprises of Runway 11L/29R & Runway 11R/29L.

1.2. For RNAV 1(GNSS) SIDs and STARs operations, the aircraft shall be GNSS equipped, and the navigation systems shall meet ICAO RNAV1 standard of accuracy or equivalent.

1.3. Operators/pilots who are not approved to fly the RNAV1(GNSS) SIDs and STARs shall inform ATC and expect vectors.

1.4. Pilots shall comply with the ATC assigned level. Pilots shall also adhere to the vertical and speed restrictions of the cleared RNAV 1(GNSS) SID/STAR.

1.5. ATC clearance will always take precedence when the ATC clearance does not allow the pilots to adhere to the vertical and speed restrictions of the RNAV 1(GNSS) SID/ STAR.

1.6. Aircraft may be vectored off an RNAV 1(GNSS) SID or STAR. Such aircraft may subsequently be given instructions to join the appropriate RNAV1 SID/STAR.

2. ARRIVALS

2.1. STARs are presented in diagrammatic format on a chart. STAR begins at a specified waypoint on the ATS route.

2.2. RNAV1 equipped aircraft to Delhi Airport will be cleared on the appropriate RNAV1 STAR by ATC. All arriving aircraft are required to follow the appropriate RNAV1 STARs as follows:

2.2.1 Westerly flow STAR Scheme				
STARs When VIR155A Hindan Airspace is available for Civil Use				
ATS ROUTE	**27**	**28**	**29R**	**29L**
W35/39	SP6A	SP6B	SP6C	SP6D
A466	ELKUX 6A	ELKUX 6B	ELKUX 6C	ELKUX 6D
G452	LKA 6A	LKA 6B	LKA 6C	LKA 6D

A347/Q1	UKASO 6A	UKASO 6B	UKASO 6C	UKASO 6D
W65	ATRIN 6A	ATRIN 6B	ATRIN 6C	ATRIN 6D
W20/Q24	BAVOX 6A	BAVOX 6B	BAVOX 6C	BAVOX 6D
W10N/L760	POSIG 6A	POSIG 6B	POSIG 6C	POSIG 6D
R594	JAL 6A	JAL 6B	JAL 6C	JAL 6D
W37	PUMOT 6A	PUMOT 6B	PUMOT 6C	PUMOT 6D
L626	HW 6A	HW 6B	HW 6C	HW 6D
Conditional STARs When VIR155A Hindan Airspace is NOT available for Civil Use				
W35/39	SP6L	SP6M	SP6N	SP6P
A466	ELKUX 6L	ELKUX 6M	ELKUX 6N	ELKUX 6P
Conditional STARs for CDR Routes When CDR routes are available through AUP/UUP or Approved by ATC and VIR155A Hindan Airspace is available for Civil Use				
J6	APUDI 6A	APUDI 6B	APUDI 6C	APUDI 6D
J21	UPTEX 6A	UPTEX 6B	UPTEX 6C	UPTEX 6D
J24	MUPID 6A	MUPID 6B	MUPID 6C	MUPID 6D
Conditional STARs for CDR Routes When CDR routes are available through AUP/UUP or Approved by ATC and VIR155A Hindan Airspace is NOT available for Civil Use				
J6	APUDI 6L	APUDI 6M	APUDI 6N	APUDI 6P
J21	UPTEX 6L	UPTEX 6M	UPTEX 6N	UPTEX 6P
J24	MUPID 6L	MUPID 6M	MUPID 6N	MUPID 6P
2.2.2 Easterly flow STAR Scheme				
STARs When VIR155A Hindan Airspace is available for Civil Use				
ATS ROUTE	**09**	**10**	**11L**	**11R**
W35/39	SP6E	SP6F	SP6G	SP6H
A466	ELKUX 6E	ELKUX 6F	ELKUX 6G	ELKUX 6H
G452	LKA 6E	LKA 6F	LKA 6G	LKA 6H
A347/Q1	UKASO 6E	UKASO 6F	UKASO 6G	UKASO 6H
W65	ATRIN 6E	ATRIN 6F	ATRIN 6G	ATRIN 6H
W20/Q24	BAVOX 6E	BAVOX 6F	BAVOX 6G	BAVOX 6H
W10N/L760	POSIG 6E	POSIG 6F	POSIG 6G	POSIG 6H
R594	JAL 6E	JAL 6F	JAL 6G	JAL 6H
W37	PUMOT 6E	PUMOT 6F	PUMOT 6G	PUMOT 6H
L626	HW 6E	HW 6F	HW 6G	HW 6H

Conditional STARs When VIR155A Hindan Airspace is NOT available for Civil Use				
R594	JAL 6Q	JAL 6R	JAL 6S	JAL 6T
W37	PUMOT 6Q	PUMOT 6R	PUMOT 6S	PUMOT 6T
L626	HW 6Q	HW 6R	HW 6S	HW 6T
Conditional STARs for CDR Routes When CDR routes are available through AUP/UUP or Approved by ATC				
J6	APUDI 6E	APUDI 6F	APUDI 6G	APUDI 6H
J21	UPTEX 6E	UPTEX 6F	UPTEX 6G	UPTEX 6H
J24	MUPID 6E	MUPID 6F	MUPID 6G	MUPID 6H

2.2.3 Additional elements on the STAR chart include the following:

2.2.3.1 Vertical restrictions – designed to contain aircraft in controlled airspace and to separate aircraft from obstacles and to avoid, to the degree possible, conflict with departing aircraft.

2.2.3.2 Speed restrictions – designed for flow control purposes.

2.2.4 At the end of STAR, arrivals to expect vectors to intercept the localizer for an ILS approach or Final approach Track for other available instrument approach.

3. TERMINAL HOLDINGS

3.1. For the purpose of holding, holding waypoints and their associated RNAV1 holding Tabular description are listed below:

Holdings										
Path Descriptor	Waypoint Identifier	Fly-Over	Course °M(°T)	Distance (Nm)	Turn Direction	Upper Limit Altitude ft	Lower Limit Altitude ft	Speed limit (Kt)	VPA/ TCH	Navigation Specification
HM	AKBAN	Y	067.00° Mag / 067.75° True	10	L	- FL160.00	+ FL110.00	-250		RNAV 1 (GNSS)
HM	KAPTA	Y	195.00° Mag / 195.75° True	10	R	- FL140.00	+ FL070.00	-250		RNAV 1 (GNSS)
HM	SAM	Y	132.00° Mag / 132.75° True	10	L	- FL160.00	+ FL100.00	-250		RNAV 1 (GNSS)
HM	SAPLO	Y	010.00° Mag / 010.75° True	10	R	- FL200.00	+ FL110.00	-250		RNAV 1 (GNSS)

HM	SUGOK	Y	291.00° Mag / 291.75° True	10	L	- FL140.00	+ FL070.00	-250		RNAV 1 (GNSS)

3.2. For RNAV1 systems without holding functionality or for manual holding, the outbound time is:

a) 1.0 min up to FL140 and

b) 1.5 min above FL140

4. DEPARTURES

4.1. RNAV1 SIDs have been promulgated for all the runways at Delhi. These RNAV 1(GNSS) SIDs can also be assigned during simultaneous operations from any runway of Northern runway system and another runway of Southern runway system.

4.2. All departing aircraft will normally be cleared on the appropriate RNAV1(GNSS) SID as below:

4.2.1 Westerly flow SID Scheme

SIDs (Westerly Flow)					
ATS ROUTE	**27**	**28**	**29R**	**29L**	**Remarks**
W35/39	SP 6J	SP6K	-	-	
A589	OROTI 6A	OROTI 6B	OROTI 6C	OROTI 6D	
G333	SURAM 6A	SURAM 6B	SURAM 6C	SURAM 6D	
W19/Q23	AKRIB 6A	AKRIB 6B	AKRIB 6C	AKRIB 6D	
A474/Q2	DUDUM 6A	DUDUM 6B	DUDUM 6C	DUDUM 6D	
W10S/L759	ITBAN 6A	ITBAN 6B	ITBAN 6C	ITBAN 6D	
V18	ALI 6A	ALI 6B	ALI 6C	ALI 6D	
SIDs (Easterly Flow)					
ATS ROUTE	**09**	**10**	**11L**	**11R**	**Remarks**
W35/39	SP 6U#	SP6V#	-	-	When VIR155A Hindan Airspace is available for Civil Use
A589	OROTI 6E	OROTI 6F	OROTI 6G	OROTI 6H	
G333	SURAM 6E	SURAM 6F	SURAM 6G	SURAM 6H	
W19/Q23	AKRIB 6E	AKRIB 6F	AKRIB 6G	AKRIB 6H	
A474/Q2	DUDUM 6E	DUDUM 6F	DUDUM 6G	DUDUM 6H	
W10S/L759	ITBAN 6E	ITBAN 6F	ITBAN 6G	ITBAN 6H	
V18	ALI 6E	ALI 6F	ALI 6G	ALI 6H	
NOTE: SID# denotes Conditional RNAV1 SIDs and are available only when VIR 155(A) Hindan-I is not active and when specifically authorized by Delhi ATC.					

5. RADIO COMMUNICATIONS FAILURE PROCEDURE

5.1. Aircraft to follow the promulgated Radio Communication Failure Procedure.

6. WAYPOINTS (WGS 84 DATUM)		
S.NO	**WAYPOINT**	**COORDINATES**
1.	AKBAN	281353.58N 0762626.95E
2.	AKRIB	272501.00N 0764756.00E
3.	AKSAM	282540.47N 0781154.23E
4.	ALI	274945.50N 0781042.00E
5.	APUDI	290250.00N 0771225.00E
6.	ASUKO	291558.34N 0761512.64E
7.	ATRIN	274906.00N 0751908.00E
8.	ATSOL	283337.61N 0780945.32E
9.	AVPIN	281722.12N 0770908.62E
10.	BAVOX	260831.00N 0771256.00E
11.	BIPAN	283633.83N 0765401.56E
12.	DAGUL	281443.42N 0771148.26E
13.	DONAT	285331.19N 0771903.14E
14.	DP403	283710.38N 0764211.09E
15.	DP405	285429.73N 0770054.63E
16.	DP414	283110.93N 0762735.55E
17.	DP419	281139.09N 0772256.90E
18.	DP420	280516.95N 0772208.06E
19.	DP421	282021.53N 0771709.26E
20.	DP424	284929.68N 0771729.80E
21.	DP501	283712.10N 0765422.99E
22.	DP502	282811.44N 0765759.10E
23.	DP503	282645.06N 0770441.14E
24.	DP504	282339.31N 0771856.75E
25.	DP505	280009.49N 0771218.89E
26.	DP507	280309.92N 0771556.98E
27.	DP509	284923.59N 0771526.76E
28.	DP510	282051.20N 0763957.70E
29.	DP512	284938.97N 0770431.08E
30.	DP608	282528.14N 0771035.04E
31.	DP609	282707.36N 0770255.55E

32.	DP612	282140.73N 0771629.86E
33.	DP613	282248.28N 0765006.33E
34.	DP614	282224.14N 0764956.25E
35.	DP645	281502.56N 0770934.11E
36.	DP659	283246.30N 0764244.93E
37.	DP666	281452.31N 0765459.80E
38.	DP667	281400.25N 0765650.40E
39.	DP672	275825.35N 0773335.95E
40.	DP673	281443.23N 0765829.77E
41.	DP676	280958.73N 0770845.62E
42.	DP680	282321.95N 0765125.45E
43.	DP681	282050.86N 0765736.06E
44.	DP682	281616.23N 0770847.21E
45.	DP683	282004.47N 0765707.03E
46.	DP684	281250.38N 0770807.50E
47.	DP685	281849.66N 0765054.37E
48.	DP686	281711.82N 0764357.23E
49.	DP688	283907.62N 0764241.70E
50.	DUDUM	274801.00N 0763121.00E
51.	ELKUX	295030.00N 0753041.00E
52.	EPDON	290236.15N 0763213.99E
53.	ERVIX	281338.40N 0760348.53E
54.	FE522	282753.68N 0773129.95E
55.	FN614	284320.26N 0772702.62E
56.	FN617	284247.29N 0773024.33E
57.	FN911	284058.26N 0765327.59E
58.	FN914	284140.49N 0765008.80E
59.	FN917	284222.63N 0764649.95E
60.	FN920	284304.69N 0764331.07E
61.	FS711	282405.83N 0771658.92E
62.	FS714	282320.33N 0772019.33E
63.	FS717	282235.45N 0772337.12E
64.	FS720	282150.11N 0772654.78E
65.	FS811	282949.33N 0765021.05E
66.	FS814	283031.56N 0764702.23E
67.	FS817	283113.70N 0764343.37E
68.	FS820	283156.91N 0764025.33E

69.	GUGEX	281725.05N 0772950.50E
70.	HW	290151.56N 0792825.95E
71.	IDLAM	275843.33N 0761139.52E
72.	ISREL	292543.63N 0770801.26E
73.	ITBAN	274943.00N 0774248.00E
74.	JAL	274139.60N 0793920.20E
75.	KAPTA	284214.33N 0773346.02E
76.	LELIT	281358.22N 0780952.43E
77.	LKA	281120.42N 0740640.68E
78.	MUBKO	292917.13N 0764802.50E
79.	MUNUV	283042.05N 0765215.78E
80.	MUPID	294922.00N 0755848.00E
81.	OBDAG	283128.88N 0771608.38E
82.	OMKAL	290914.49N 0764836.84E
83.	OROTI	291700.00N 0752956.00E
84.	PABGU	283357.00N 0771703.05E
85.	PEGVI	292134.02N 0762214.05E
86.	POSIG	271301.00N 0773455.00E
87.	PUMOT	282800.00N 0784655.00E
88.	SAM	284911.79N 0764910.96E
89.	SAPLO	273854.00N 0770815.00E
90.	SP	295959.68N 0772555.23E
91.	SUGOK	282336.20N 0774229.20E
92.	SURAM	284500.00N 0755256.00E
93.	SURGO	272901.00N 0770846.00E
94.	TOLEV	280615.87N 0760531.29E
95.	UKASO	273815.00N 0755151.00E
96.	UPTEX	294440.00N 0764736.00E
97.	UXABA	281851.09N 0770533.00E
98.	VIDID	290523.60N 0763544.61E
99.	VILUT	290728.07N 0765834.45E

VIDP AD 2.23 ADDITIONAL INFORMATION

I. SPECIAL RESTRICTIONS

1. Closure of Airspace in and around Delhi for Air Force Day fly past

Indian Air Force will carry out fly past rehearsal on 3rd, 4th, and 6th October, and final day fly past on 8th October every year on the occasion of Air Force Day.In view of the above, the airspace of following dimensions will remain closed between 0230-0430 UTC on 3rd, 4th, 6th and 8th October every year for all aircraft except those participating in the rehearsal/final day fly past:

i.Within a radius of 20NM from IGI Airport, New Delhi: Ground level to FL130.

ii.Within a circular area between 20NM and 40NM radius of IGI Airport, New Delhi: Ground Level to FL150.

2.Airspace Closure and flying restrictions in Delhi FIR during celebrations connected with Independence Day and Republic Day

i.The following flying restrictions shall be applied within Delhi FIR during the under mentioned dates and time every year in connection with the celebrations of Independence Day and Republic Day.

ii.Dates and Timings of Airspace Restrictions are as under:

Date	**Time**
22nd January	1400 - 1830 IST
26th January	0700 - 1230 IST, and 1400 - 1830 IST
29th January	1400 - 1900 IST
15th August	0600 - 1000 IST, and 1600 - 1900 IST

iii.No flight shall be permitted to take off / land / over fly at Delhi and other airports within 300 KMS zone around it except:

a.Routine Commercial flights,

b.IAF, BSF and ARC flights,

c.State owned aircraft/helicopters flying Governor or Chief Minister within the state.

iv.Flights of Micro light aircraft and hang gliders from flying clubs and other airports shall not be permitted in the above mentioned zone and period around Delhi.

v.Safdarjung airport shall remain closed during the above-mentioned dates and period except for IAF helicopters, which may be deployed on emergency duty or VVIP duty.

3.Surface Movement Radar-I and II available and ASMGCS EQUIPMENT available. Coverage of ASMGCS has been extended to cover the additional area TWYs C,Q,LINK 4, Y,R, Y5, LINK 19 ,P5,S5, S2,LINK 8,LINK 9,LINK 7,LINK 6,D, and TERMINAL-III aprons in operational Area.

4.RWY 11R/29L and associated taxiways are compatible for operations of code F aircraft.

5.MODE-S capability is operational with DELHI ATC. All the aircraft operators are advised to keep flight id and six digit hexadecimal ICAO 24-bit address updated in the aircraft avionics to avoid misleading information to ATC

II. DETAILS OF THE TAXIWAYS :

S. No.	Designation	Taxiway Width(M)	Length (M)	Surface	PCR	Remarks
1	D1	23	209	Asphalt	862/F/A/W/T	• BTN RWY 09/27 up to TWY D. • Suitable for up to Code letter C Aircraft.
2	D2	23	174.5	Asphalt	862/F/A/W/T	•BTN RWY 09/27 up to TWY D. •Suitable for up to Code Letter E Aircraft
3	D3	23	180	Asphalt	943/F/B/W/T	•BTN RWY 09/27 up to TWY D. •Suitable for up to Code Letter E Aircraft
4	D4	23	473	Asphalt	721/F/A/W/T	• BTN RWY 09/27 up to junction of TWY G and TWY H2 • . Portion of TWY D4 between RWY 09/27 and TWY D including TWY D-D4 junction suitable for up to Code letter F aircraft. . Portion of TWY D4 between TWY D and TWY G suitable for up to Code letter E Aircraft
5	D5	23	330	Asphalt	862/F/A/W/T	•BTN RWY 09/27 up to TWY D. •Suitable for up to Code Letter E Aircraft.
6	D6	23	320	Concrete	1506/R/B/W/T	•BTN RWY 09/27 up to TWY D. •Suitable for up to Code Letter E Aircraft
7	D7	23	154	Concrete	1175/R/A/W/T	•BTN RWY 09/27 up to TWY D. •Suitable for up to Code Letter C Aircraft
8	D8	23	150	Concrete	1175/R/A/W/T	•BTN RWY 09/27 up to TWY D. •Suitable for up to Code Letter F Aircraft
9	D9	23	191	Concrete	1175/R/A/W/T	•BTN RWY 09/27 up to TWY D. •Suitable for up to Code Letter F Aircraft.
10	A2	23	537	Asphalt	862/F/A/W/T	• North of RWY 09/27 • Suitable for up to Code Letter E Aircraft

11	D	23	3115	Asphalt	729/F/B/W/T	• Portion west of TWY D4 including TWY D-D4 junction suitable for up to Code Letter F aircraft. •Portion of TWY D between TWY D2 and TWY D4 including TWY D-D2 junction suitable for up to Code Letter E aircraft. •Portion east of TWY D2 suitable for up to Code Letter C aircraft
12	F	23	640	Concrete	1506/R/B/W/T	• BTN TWY G/G2 junction and TWY D. • Suitable for up to Code Letter E aircraft.
13	Taxilane F2	23	186.5	Concrete	1266/R/B/W/T	•Connecting Taxilanes F5 and TWY F. •Suitable for up to Code letter C Aircraft
14	Taxilane F3	23	607	Concrete	1506/R/B/W/T	•Connecting TWY D and TWY G •Suitable for up to Code letter C Aircraft
15	Taxilane F4	23	602	Concrete	1506/R/B/W/T	•Connecting TWY D and TWY G •Suitable for up to Code letter E Aircraft
16	Taxilane F5	23	597	Concrete	1506/R/B/W/T	•Connecting TWY D and TWY G •Suitable for up to Code letter C Aircraft
17	Taxilane F6	23	548	Concrete	1506/R/B/W/T	•Connecting TWY D and TWY G •Suitable for up to Code letter C Aircraft
18	Taxilane E2	23	819	Concrete	1506/R/B/W/T	•Connecting TWY D and Taxilane E9. •Suitable for up to Code letter C Aircraft
19	Taxilane E3	23	717.5	Concrete	1506/R/B/W/T	•BTN TWY D and Taxilane E9. •Suitable for up to Code Letter C aircraft.
20	Taxilane E4	23	751	Concrete	1506/R/B/W/T	•BTN TWY G and Taxilane E9. •Suitable for up to Code Letter C aircraft.
21	Taxilane E6	23	671	Concrete	1175/R/A/W/T	•Between TWY G and Taxilane E9. •Suitable for up to Code Letter C aircraft.
22	Taxilane E7	23	685	Concrete	1175/R/A/W/T	•BTN TWY D and TWY G. •Suitable for up to Code Letter C aircraft.
23	Taxilane E9	23	675	Concrete	1506/R/B/W/T	•Connecting TWY G and TWY D. •Suitable for up to Code letter C Aircraft
24	G	23	2238	Asphalt	943/F/B/W/T	•North of RWY 10/28; Between TWY H2 and TWY G5. •Suitable for up to Code letter E Aircraft
25	G1	23	330	Asphalt	882/F/B/W/T	•BTN RWY 10/28 and TWY G. •Suitable for up to Code Letter E Aircraft.
26	G2	23	330	Asphalt	862/F/A/W/T	•BTN RWY 10/28 up to TWY G. •Suitable for up to Code Letter E Aircraft
27	G3	23	172	Concrete	1765/R/C/W/T	• BTN RWY 10/28 and TWY G. • Suitable for up to Code Letter E Aircraft.
28	G4	23	172	Concrete	1506/R/B/W/T	•BTN RWY 10/28 up to TWY G. •Suitable for up to Code Letter E Aircraft
29	G5	23	172	Concrete	1506/R/B/W/T	•Connecting RWY 10/28 up to TWY G. •Suitable for up to Code Letter E Aircraft.

30	H1	23	460	Asphalt/ Concrete	763/F/A/W/T 1765/R/C/W/T	•From RWY 10/28 up to 203M PCR 1765/R/C/W/T •BTN 203M and 360M PCR 763/F/A/W/T. •From 360M up to TWY D4 PCR 1765/R/C/W/T • Suitable for up to Code Letter C Aircraft
31	H2	23	152	Asphalt	862/F/A/W/T	•BTN RWY 10/28 up to junction of TWY G and TWY D4 •Suitable for up to Code letter E Aircraft.
32	H3	23	318	Concrete/ Asphalt	2037/R/D/W/T 1333/F/D/W/T	•BTN RWY 10/28 up to TWY D. •Suitable for up to Code Letter F Aircraft
33	H6	23	249	Concrete	1175/R/A/W/T	•BTN RWY 10/28 up to TWY D. •Suitable for up to Code Letter F Aircraft.
34	J7	25	204	Concrete	1506/R/B/W/T	•Suitable for up to Code Letter F Aircraft
35	J1	23	429	Asphalt	943/F/B/W/T	Suitable for up to Code Letter F Aircraft
36	J2	23	408	Concrete	2037/R/D/W/T	•Suitable for up to Code Letter F Aircraft
37	J3	25	408	Concrete	1765/R/C/W/T	•Suitable for up to Code Letter F Aircraft
38	J5	25	204	Concrete	1765/R/C/W/T	•Suitable for up to Code Letter F Aircraft
39	J6	23	204	Concrete/ Asphalt	1506/R/B/W/T 788/F/B/W/T	BTN RWY 10/28 and TWY K. • Suitable for up to Code Letter F Aircraft.
40	K1	23	410	Concrete/ Asphalt	2037/R/D/W/T 1181/F/C/W/T	•Suitable for up to Code Letter F Aircraft
41	K2	23	421	Concrete/ Asphalt	1175/R/A/W/T 862/F/B/W/T	•Suitable for up to Code Letter F Aircraft
42	K3	23	230	Asphalt/ Concrete	768/F/A/W/T 1765/R/C/W/T	•Suitable for up to Code Letter F Aircraft
43	K5	23	204	Concrete/ Asphalt	1191/R/B/W/T 943/F/B/W/T	•BTN RWY 10/28 and TWY K. •Suitable for up to Code Letter F Aircraft.
44	K6	23	245	Concrete	1175/R/A/W/T	• Between RWY 28/10 and TWY K. •Suitable for up to Code Letter F Aircraft
45	K	23	3960	Asphalt	688/F/A/W/T	•Suitable for up to Code Letter F Aircraft
46	N7	23	440	Concrete	1506/R/B/W/T	•Between TWY K and TWY N. •Suitable for up to Code Letter F Aircraft
47	N6	23	399	Asphalt	862/F/A/W/T	•Connecting TWYs N and K, east and parallel of TWY N7. •Suitable for up to Code Letter F Aircraft
48	LINK 34	23	298	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft
49	LINK 33	25	298	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft
50	LINK 39	23	298	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter E Aircraft
51	LINK 32	23	298	Concrete	1175/R/A/W/T	•Portion of TWY LINK 32 south of TWY M Suitable for up to Code Letter C Aircraft •Portion of TWY LINK 32between TWY K and TWYMSuitable for up to Code Letter F Aircraft.
52	LINK 31	15	68	Asphalt	943/F/C/W/T	•Suitable for up to Code Letter C Aircraft

53	LINK 30	23	156	Concrete	1175/R/A/W/T	•Portion of TWYLINK 30 between TWY K and TWY M suitable for upto CodeLetterFAircraft,restricted up to max wingspan 68.40m •Portion of TWYLINK 30 south of TWY M suitablefor up to Code letter D Aircraft
54	M	23	504	Asphalt	793/F/A/W/T	•Portion of TWY M between TWYs M1 and TWY A suitable for up to code F Aircraft. •Portion of TWY M between TWYs A and LINK 30 suitable for up to Code Letter F Aircraft restricted up to max wingspan 68.40m
55	M1	23	91	Asphalt	758/F/A/W/T	•Connecting TWY M and TWY K. •Suitable for up to Code letter F Aircraft.
56	Taxilane N	23	2377.5	Concrete	1506/R/B/W/T	•Portion of Taxilane N between TWY C and TWY A Suitable for up toCode Letter F Aircraft . Portion of Taxilane Nbetween TWY C and TWY LINK 34 Suitable for upto Code Letter F Aircraft restricted up to max wingspan73.30m • Portion of Taxilane N between TWY LINK 34 and TWY LINK 33 Suitable for up to Code Letter F Aircraft
57	Taxilane N1	15	450	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter C Aircraft
58	Taxilane N3	23	252	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter E Aircraft
59	Taxilane N4	23	237	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter E Aircraft
60	Taxilane N8	15	626	Concrete	1175/R/A/W/T	•Suitable for up to Code letter C Aircraft having OMGWS up to but not including 9m
61	Taxilane N5	23	370	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter E Aircraft
62	C	25	1386	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft •Portion of TWY C between TWY N and TWY LINK 4
63	C	25	3321	Asphalt	793/F/A/W/T	•Portion of TWY C between TWY LINK 5 and TWY P6. •Suitable for up to Code Letter F Aircraft.
64	C	23	1962.5	Asphalt	862/F/A/W/T	• Portion of TWY C between TWY P6 and LINK 32. •Suitable for up to Code Letter F Aircraft.
65	A	25	1485	Asphalt	862/F/A/W/T	•Portion of TWY A between TWY N and TWY R. •Suitable for up to Code Letter F Aircraft.
66	A	25	3419	Asphalt	805/F/A/W/T	•Portion of TWY A between TWY R4 and TWY P6. •Suitable for up to Code Letter F Aircraft
67	A	23	1881.5	Asphalt	862/F/A/W/T	•Portion of TWY A between TWY P7 and TWY K. •Suitable for up to Code Letter F Aircraft
68	LINK 1	25	98	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft
69	LINK 2	25	98	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft
70	LINK 3	25	98	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft

71	LINK 4	25	98	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft
72	Q	25	671	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft
73	R	25	718	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft
74	LINK 5	25	98	Asphalt	1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft
75	LINK 6	25	98	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft
76	LINK 7	25	98	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft
77	LINK 8	25	98	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft
78	LINK 9	25	98	Asphalt	862/F/A/W/T	
79	LINK 10	18	80	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter C Aircraft
80	Taxi lane LINK 13	15	290	Concrete	1175/R/A/W/T	•Suitable up to Code letter C aircraft •Connecting GA apron to TWY C
81	Taxi lane LINK 14	15	162	Concrete	1175/R/A/W/T	•Suitable up to Code letter C •Inside GA apron parallel to TWY C
82	Taxi lane LINK 15	10.5	218	Concrete	1175/R/A/W/T	•Suitable up to Code letter B aircraft. •Inside GA apron parallel to Taxilane LINK 16 and LINK 13.
83	Taxi lane LINK 16	10.5	279	Concrete	1175/R/A/W/T	•Suitable up to Code letter B aircraft •Connecting GA apron to TWY C
84	TWY LINK 16	25	98	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft •Connecting to TWY C and TWY A
85	LINK 19	25	97.5	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft
86	P1(RET)	23	560	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft.
87	P2	23	210	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft
88	P3(RET)	23	545	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft.
89	P4	23	198	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft.
90	P5	23	196	Concrete	1175/R/A/W/T	•Concrete portionUp to 150M from RWY 29R centreline. •Suitable for up to Code Letter F Aircraft.
91	P6	23	150	Concrete	1031/R/A/W/T	•Concrete portionUp to 150M from RWY 29R centreline. •Suitable for up to Code Letter F Aircraft.
		23	136.5	Asphalt	696/F/A/W/T	
92	P7	23	187.5	Concrete	1139/R/A/W/T	•Suitable for up to Code Letter F Aircraft.
		23	37.5			
93	P8	23	150	Concrete	1175/R/A/W/T	•Concrete portionUp to 150M from RWY 29R centreline. •Suitable for up to Code Letter F Aircraft.
		23	159	Asphalt	732/F/B/W/T	
94	P9	23	316	Asphalt	732/F/B/W/T	•Between TWY C and TWY P8. •Suitable for up to Code Letter F Aircraft.
95	R1	25	228	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft.
96	R2(RET)	23	619	Asphalt	943/F/B/W/T	•Suitable for up to Code Letter F Aircraft.

97	R3(RET)	23	635	Asphalt	873/F/C/W/T	•Suitable for up to Code Letter F Aircraft.
98	R4	25	249	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft.
99	R5	25	150	Concrete	1175/R/A/W/T	•Concrete portionUp to 150M from RWY 29R centreline.
		25	203	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft.
100	R6	25	150	Concrete	1175/R/A/W/T	•Concrete portionUp to 150M from RWY 29R centreline.
		25	207	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft.
101	R7	25	150	Concrete	1175/R/A/W/T	•Concrete portionUp to 150M from RWY 29R centreline.
		25	209	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft.
102	S1	25	190	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft
103	S2	25	190	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft
104	S5	25	190	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft.
105	S6	23	190	Concrete	1506/R/B/W/T	•Suitable for up to Code Letter F Aircraft
106	S7	23	190	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft
107	S8	23	190	Concrete	1506/R/B/W/T	•Suitable for up to Code Letter F Aircraft
108	T4	25	207	Asphalt	700/F/A/W/T	•Suitable for up to Code Letter F Aircraft
109	T5	23	190	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft
110	T6	25	92	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft.
		25	98	Concrete	1765/R/C/W/T	•Asphalt portionFrom RWY 29R edge up to 70 M
111	T7	25	92	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft.
		25	98	Concrete	1506/R/B/W/T	•Asphalt portionFrom RWY 29R edge up to 70 M
112	Z (TWY Z7 up to TWY Z4))	25	385	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft
113	Z (TWY Y5 up to TWY Y8)	25	366	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft
114	Z(TWY Z4 up to TWY Y5)	25	3760	Asphalt	862/F/A/W/T	•Suitable for up to Code Letter F Aircraft
115	Y8	25	318	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft
116	Y7	25	190	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft
117	Y6	25	190	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft
118	Y5	25	190	Asphalt/ Concrete	862/F/A/W/T 1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft
119	Y4	25	531	Asphalt	862/F/A/W/T	•Drain in TWY strip at distance of 33M west to TWY centerline •Suitable up to Code letter F Aircraft
120	Y2	25	531	Asphalt	862/F/A/W/T	
121	Y1	25	531	Asphalt	862/F/A/W/T	
122	Z2	25	531	Asphalt	862/F/A/W/T	
123	Z3	25	531	Asphalt	862/F/A/W/T	
124	Z4	25	190	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft

125	Z5	25	190	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft
126	Z6	25	218	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft
127	Z7	25	269	Concrete	1175/R/A/W/T	•Suitable for up to Code Letter F Aircraft

III. OTHER DETAILS OF TWYS

i. Rapid exit TWY D6: available for RWY27, Angle of turn 30deg, Radius of turn off curve 550m, suitable for aircraft upto code letter E.

ii. Rapid exit TWY D5: available for RWY27, Angle of turn 30deg, Radius of turn off curve 550m, suitable for aircraft up to code letter E with OMGWS up to but not including 15m.

iii. Rapid exit TWY H1: available for RWY28, angle of turn 30deg, radius of turn off curve 550m, suitable for code letter A, B and C aircraft with OMGWS up to but not including 9.6m.

iv. TWY D: From TWY D9 up to abeam ARP is realigned parallel to RWY 09/27, at a distance of 172.5m south of RWY 09/27 centre line.

v. Extended portion of TWY D: From TWY D, abeam ARP, up to TWY D1 at a distance of 168m south of RWY 09/27 centre line.

vi. TWY G2: Connects RWY 10 to TWY G , suitable for code letter C, Angle of turn 30 deg, Radius of Turn off Curve 550m.

vii. Rapid exit TWY J2: Angle of turn 30deg, Radius of turn off curve 550M, suitable for code letter F aircraft.

viii. Rapid exit TWY J3: Angle of turn 30deg, Radius of turn off curve 550M, suitable for code letter F aircraft.

IV. **Exemption Granted by DGCA:**

DGCA India has granted exemption vide letter dated 10.11.2025 for non-compliant width of runway strip for RWY 09/27 against provisions in CAR Section 4, series B, part 1, para 3.4.3 for the period up to 31.12.2026.

V.Aeronautical Ground Light (AGL) team working on watch and ward basis in grassy area both side of RWY 10/28 (75M from Runway centreline). Pilots to exercise caution while Landing or Take-off from RWY 10/28.

VI. Aeronautical Ground Light (AGL) team working on watch and ward basis in grassy area both side of RWY 11L/29R (75M from Runway centreline). Pilots to exercise caution while Landing or Take-off from RWY 11L/29R.

VIDP AD 2.24 CHARTS RELATED TO AN AERODROME

1 Aerodrome Chart (14 May 2026)
2 Aerodrome Chart Hot Spot (14 May 2026)
3 Aircraft Parking/Docking Chart Apron II (19 March 2026)
4 Aircraft Parking/Docking Chart Apron 31(12 June 2025)
5 Aircraft Parking/Docking Chart Apron 32(12 June 2025)
6 Aircraft Parking/Docking Chart Apron 33(12 June 2025)
7 Aircraft Parking/Docking Chart Apron 34(12 June 2025)
8 Aircraft Parking/Docking Chart Apron 35(12 June 2025)
9 Aircraft Parking/Docking Chart Cargo Apron (12 June 2025)
10 Aircraft Parking/Docking Chart Stand 801(12 June 2025)
11 Aircraft Parking/Docking Chart Stand 802(12 June 2025)
12 Aircraft Parking/Docking Chart Apron I Stand 101-137 (12 June 2025)
13 Aircraft Parking/Docking Chart General Aviation Apron Stands: 901-911C & 912-920(12 June 2025)
14 Aircraft Parking/Docking Chart General Aviation Apron Stands: 921-928 & 929-937(12 June 2025)
15 Aircraft Parking/Docking Chart General Aviation Apron Stands: 938-947 & 951-956 (12 June 2025)
16 Aircraft Parking/Docking Chart Stand 803(12 June 2025)
17 Aircraft Parking/Docking Chart Apron I Stand 140-178(12 June 2025)
18 Aerodrome Obstacle Chart Type – A (Operating limitations) RWY 09 (02 October 2025)
19 Aerodrome Obstacle Chart Type – A (Operating limitations) RWY 10

20 Aerodrome Obstacle Chart Type – A (Operating limitations) RWY 11L (23 January 2025)
21 Aerodrome Obstacle Chart Type – A (Operating limitations) RWY 11R
22 Aerodrome Obstacle Chart Type – A (Operating limitations) RWY 27 (21 March 2024)
23 Aerodrome Obstacle Chart Type – A (Operating limitations) RWY 28 (21 March 2024)
24 Aerodrome Obstacle Chart Type – A (Operating limitations) RWY 29L
25 Aerodrome Obstacle Chart Type – A (Operating limitations) RWY 29R (23 January 2025)
26 Aerodrome Obstacle Chart Type – B (23 January 2025)
27 Take Off Run Available from Intersection RWY 09
28 Take Off Run Available from Intersection RWY 10 (08 August 2024)
29 Take Off Run Available from Intersection RWY 11L (23 January 2025)
30 Take Off Run Available from Intersection RWY 11R
31 Take Off Run Available from Intersection RWY 27
32 Take Off Run Available from Intersection RWY 28 (08 August 2024)
33 Take Off Run Available from Intersection RWY 29L
34 Take Off Run Available from Intersection RWY 29R (23 January 2025)
35 Location of Different Exit TWYs RWY 09
36 Location of Different Exit TWYs RWY 10 (08 August 2024)
37 Location of Different Exit TWYs RWY 11L (23 January 2025)
38 Location of Different Exit TWYs RWY 11R(14 May 2026)
39 Location of Different Exit TWYs RWY 27
40 Location of Different Exit TWYs RWY 28 (08 August 2024)
41 Location of Different Exit TWYs RWY 29L(14 May 2026)
42 Location of Different Exit TWYs RWY 29R (23 January 2025)
43 ILS CAT III Critical And Sensitive Areas- RWY 11R
44 ILS CAT III Critical And Sensitive Areas- RWY 28 (22 January 2026)
45 ILS CAT III Critical And Sensitive Areas- RWY 29L
46 ILS CAT III Critical And Sensitive Areas- RWY 11L (23 January 2025)
47 ILS CAT III Critical And Sensitive Areas- RWY 10 (22 January 2026)
48 ILS CAT III Critical And Sensitive Areas- RWY 29R (23 January 2025)
49 LVP Taxi Routing - Easterly Flow - RWY 11L & RWY10 (14 May 2026)
50 LVP Taxi Routing - Easterly Flow - RWY 11R, RWY 11L & RWY 09 (14 May 2026)
51 LVP Taxi Routing - Easterly Flow - RWY 11R, RWY 11L & RWY 10 (14 May 2026)
52 LVP Taxi Routing - Easterly Flow - RWY 11R & RWY10 (14 May 2026)
53 LVP Taxi Routing - Westerly Flow - RWY 29L & RWY28 (14 May 2026)
54 LVP Taxi Routing - Westerly Flow - RWY 29R & RWY28 (14 May 2026)
55 LVP Taxi Routing - Westerly Flow - RWY 29R, RWY 29L & RWY 27 (14 May 2026
56 LVP Taxi Routing - Westerly Flow - RWY 29R, RWY 29L & RWY 28 (14 May 2026)
57 LVP Taxi Routing - Easterly Flow - RWY 09, RWY 10 & RWY 11L,RWY 11R (14 May 2026)
58 Taxiway Compatibility Chart (14 May 2026)
59 Precision Approach Terrain Chart RWY 09
60 Precision Approach Terrain Chart RWY 10(22 January 2026)
61 Precision Approach Terrain Chart RWY 11L
62 Precision Approach Terrain Chart RWY 11R
63 Precision Approach Terrain Chart RWY 27
64 Precision Approach Terrain Chart RWY 28(22 January 2026)
65 Precision Approach Terrain Chart RWY 29L

66 Precision Approach Terrain Chart RWY 29R
67 Instrument Approach Chart – ILS or LOC RWY 11R (27 November 2025)
68 Instrument Approach Chart – ILS Procedure (CAT-II/III) RWY 11R (27 November 2025)
69 Instrument Approach Chart – ILS Procedure (CAT-II/III) RWY 11L
70 Instrument Approach Chart – ILS or LOC RWY 27 (08 August 2024)
71 Instrument Approach Chart – ILS Procedure (CAT-II/III) RWY 28
72 Instrument Approach Chart – ILS or LOC RWY 29R (16 May 2024)
73 Instrument Approach Chart – VOR Procedure RWY 09
74 Instrument Approach Chart – VOR Procedure RWY 10(27 November 2025)
75 Instrument Approach Chart – VOR Procedure RWY 11R (08 August 2024)
76 Instrument Approach Chart – VOR Procedure RWY 28 (27 November 2025)
77 Instrument Approach Chart – ILS or LOC RWY 11L (17 April 2025)
78 Instrument Approach Chart – ILS (CAT-II/III) Procedure RWY 29R (17 April 2025)
79 Instrument Approach Chart – ILS or LOC RWY 09 (17 April 2025)
80 Instrument Approach Chart – RNP Y Procedure RWY 27 (27 November 2025)
81 Coding Table 1 - RNP Y Procedure RWY 27 (27 November 2025)
82 Coding Table 2 - RNP Y Procedure RWY 27 (27 November 2025)
83 Instrument Approach Chart – RNP Y Procedure RWY 10 (17 April 2025)
84 Coding Table 1 - RNP Y Procedure RWY 10 (17 April 2025)
85 Coding Table 2 - RNP Y Procedure RWY 10(17 April 2025)
86 Tactical Air Traffic Flow Procedures (T-ATFP) in Delhi FIR – RNAV STAR RWY 09, 10, 11R and ATS Route Structure in Delhi TMA
87 Tactical Air Traffic Flow Procedures (T-ATFP) in Delhi FIR – RNAV STAR RWY 27, 28, 29L and ATS Route Structure in Delhi TMA
88 Tactical Air Traffic Flow Procedures (T-ATFP) in Delhi FIR – Establishing Arrival Sequence, CTO and RTO computation for RWY 27, 28 and 29L
89 Instrument Approach Chart – RNP Y Procedure RWY 29R
90 Coding Table 1 - RNP Y Procedure RWY 29R
91 Coding Table 2 - RNP Y Procedure RWY 29R
92 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 09, AKRIB & DUDUM, 6E (13 June 2024)
93 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 09, AKRIB & DUDUM, 6E - Tabular Description (13 June 2024)
94 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 09, ALI & ITBAN, 6E (13 June 2024)
95 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 09, ALI & ITBAN, 6E - Tabular Description (13 June 2024)
96 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 09, OROTI & SURAM, 6E (13 June 2024)
97 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 09, OROTI & SURAM, 6E - Tabular Description (13 June 2024)
98 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 09, SP 6U - Conditional (13 June 2024)
99 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 09, SP 6U - Tabular Description (13 June 2024)
100 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 10, AKRIB & DUDUM, 6F (13 June 2024)
101 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 10, AKRIB & DUDUM, 6F - Tabular Description (13 June 2024)
102 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 10, ALI & ITBAN, 6F (13 June 2024)

103 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 10, ALI & ITBAN, 6F - Tabular Description (13 June 2024)

104 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 10, OROTI & SURAM, 6F (13 June 2024)

105 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 10, OROTI & SURAM, 6F - Tabular Description (13 June 2024)

106 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 10, SP 6V - Conditional (13 June 2024)

107 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 10, SP 6V - Tabular Description (13 June 2024)

108 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 27/28, AKRIB & DUDUM, 6A/6B (13 June 2024)

109 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 27, AKRIB 6A & DUDUM 6A - Tabular Description (13 June 2024)

110 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 28, AKRIB 6B & DUDUM 6B - Tabular Description (13 June 2024)

111 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 27/28, ALI & ITBAN, 6A/6B (13 June 2024)

112 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 27, ALI 6A & ITBAN 6A - Tabular Description (13 June 2024)

113 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 28, ALI 6B & ITBAN 6B - Tabular Description (13 June 2024)

114 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 27/28, OROTI & SURAM, 6A/6B (13 June 2024)

115 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 27, OROTI 6A & SURAM 6A - Tabular Description (13 June 2024)

116 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 28, OROTI 6B & SURAM 6B - Tabular Description (13 June 2024)

117 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 27, SP 6J (13 June 2024)

118 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 27, SP 6J - Tabular Description (13 June 2024)

119 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 28, SP 6K (13 June 2024)

120 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 28, SP 6K - Tabular Description (13 June 2024)

121 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 11L, AKRIB, ALI, DUDUM, ITBAN, OROTI & SURAM 6G (13 June 2024)

122 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 11L, AKRIB, ALI, DUDUM & ITBAN 6G - Tabular Description (13 June 2024)

123 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 11L, OROTI & SURAM 6G - Tabular Description (13 June 2024)

124 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 29R, AKRIB, ALI, DUDUM & ITBAN 6C (13 June 2024)

125 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 29R, AKRIB, ALI, DUDUM & ITBAN 6C - Tabular Description (13 June 2024)

126 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 29R, OROTI & SURAM, 6C (13 June 2024)

127 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 29R, OROTI & SURAM, 6C - Tabular Description (13 June 2024)

128 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 11R, AKRIB, ALI, DUDUM, ITBAN, OROTI & SURAM 6H (13 June 2024)

129 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 11R, AKRIB, ALI, DUDUM & ITBAN 6H - Tabular Description (13 June 2024)

130 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 11R, OROTI & SURAM 6H - Tabular Description (13 June 2024)

131 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 29L, AKRIB, ALI, DUDUM & ITBAN 6D (13 June 2024)

132 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 29L, AKRIB, ALI, DUDUM & ITBAN 6D - Tabular Description (13 June 2024)

133 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 29L, OROTI & SURAM, 6D (13 June 2024)

134 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 29L, OROTI & SURAM, 6D - Tabular Description (13 June 2024)

135 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 09/10/11L/11R, ATRIN, LKA & UKASO, 6E/6F/6G/6H (13 June 2024)

136 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 09/10/11L/11R, ATRIN, LKA & UKASO, 6E/6F/6G/6H - Tabular Description (13 June 2024)

137 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 09/10/11L/11R, BAVOX & POSIG, 6E/6F/6G/6H (13 June 2024)

138 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 09/10/11L/11R, BAVOX & POSIG, 6E/6F/6G/6H - Tabular Description (13 June 2024)

139 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 09/10/11L/11R, HW, JAL & PUMOT, 6E/6F/6G/6H - Conditional (13 June 2024)

140 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 09/10/11L/11R, HW, JAL & PUMOT, 6E/6F/6G/6H - Tabular Description (13 June 2024)

141 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 09/10/11L/11R, ELKUX & SP, 6E/6F/6G/6H (13 June 2024)

142 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 09/10/11L/11R, ELKUX & SP, 6E/6F/6G/6H - Tabular Description (13 June 2024)

143 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 09/10/11L/11R, HW, JAL & PUMOT, 6Q/6R/6S/6T (13 June 2024)

144 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 09/10/11L/11R, HW, JAL & PUMOT, 6Q/6R/6S/6T - Tabular Description (13 June 2024)

145 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 09/10/11L/11R, APUDI, MUPID & UPTEX, 6E/6F/6G/6H - Conditional (13 June 2024)

146 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 09/10/11L/11R, APUDI, MUPID & UPTEX, 6E/6F/6G/6H - Tabular Description (13 June 2024)

147 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 27/28/29R/29L, ATRIN, LKA & UKASO, 6A/6B/6C/6D (13 June 2024)

148 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 27/28/29R/29L, ATRIN, LKA & UKASO, 6A/6B/6C/6D - Tabular Description (13 June 2024)

149 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 27/28/29R/29L, BAVOX & POSIG, 6A/6B/6C/6D (13 June 2024)

150 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 27/28/29R/29L, BAVOX & POSIG, 6A/6B/6C/6D - Tabular Description (13 June 2024)

151 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 27/28/29R/29L, HW, JAL & PUMOT, 6A/6B/6C/6D (13 June 2024)

152 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 27/28/29R/29L, HW, JAL & PUMOT, 6A/6B/6C/6D - Tabular Description (13 June 2024)

153 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 27/28/29R/29L, ELKUX & SP, 6A/6B/6C/6D - Conditional (13 June 2024)

154 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 27/28/29R/29L, ELKUX & SP, 6A/6B/6C/6D - Tabular Description (13 June 2024)

155 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 27/28/29R/29L, ELKUX & SP, 6L/6M/6N/6P (13 June 2024)

156 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 27/28/29R/29L, ELKUX & SP, 6L/ 6M/6N/6P - Tabular Description (13 June 2024)

157 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 27/28/29R/29L, APUDI, MUPID & UPTEX, 6A/6B/6C/6D - Conditional (13 June 2024)

158 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 27/28/29R/29L, APUDI, MUPID & UPTEX, 6A/6B/6C/6D - Tabular Description (13 June 2024)

159 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 27/28/29R/29L, APUDI, MUPID & UPTEX, 6L/6M/6N/6P - Conditional (13 June 2024)

160 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 27/28/29R/29L, APUDI, MUPID & UPTEX, 6L/6M/6N/6P - Tabular Description (13 June 2024)

VIDP AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

Procedure	Procedure Minima	VSS Penetration
1	2	3